ALMA MATER STUDIORUM
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Theoretical Physics

Statistical Field Theory

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Part I

Phase Transitions and Critical Phenomena

Introduction

1 Statistical Mechanics Review

In this chapter we will review some basic concepts of statistical mechanics, which will be useful for the rest of the course. We will start with the microcanonical ensemble, which is the simplest ensemble, and then we will move to the canonical ensemble, which is more general and more useful for practical applications. Finally, we will briefly discuss quantum statistical mechanics, which is a generalization of classical statistical mechanics to quantum systems.

1.1 | Equal a Priori Probabilities

We begin recalling the foundamental principles of statistical mechanics, starting from a classical isolated system of N particles, with total energy E and volume V.

The **microstates** of the system are the points in the phase space of the system, which is a $6N$-dimensional space spanned by the coordinates and momenta of the particles. The **macrostates** of the system are defined by the macroscopic variables, such as energy, volume, and number of particles. The **ensemble** is a collection of microstates that are compatible with a given macrostate. The **probability distribution** of the system is a function that assigns a probability to each microstate in the ensemble.

It is useful to recall the Hamiltonian formalism, which is a powerful tool for describing the dynamics of classical systems. The Hamiltonian H is a function of the coordinates and momenta of the particles, and it encodes the energy of the system. The equations of motion for the system can be derived from the Hamiltonian using Hamilton's equations, which describe how the coordinates and momenta evolve in time. For each particle i we have

$$(q_1, q_2, q_3, p_1, p_2, p_3) \in \mathcal{M},$$

where $\mathcal{M}$ is the phase space of the system, which is a $6N$-dimensional manifold. The Hamiltonian H is a function that maps points in the phase space to real numbers, and it encodes the energy of the system

$$H = H(q_1, \ldots, q_{3N}, p_1, \ldots, p_{3N}) : \mathcal{M} \to \mathbb{R}.$$

The equations of motion for the system can be derived from the Hamiltonian using Hamilton's equations, which describe how the coordinates and momenta evolve in time:

$$\begin{cases} \dot{q}_i = \dfrac{\partial H}{\partial p_i}, \\ \dot{p}_i = -\dfrac{\partial H}{\partial q_i}. \end{cases}$$

We can visualize the phase space of the system as a high-dimensional space, where each point represents a microstate of the system. The energy shell defined by the total energy E is a hypersurface in this phase space, and the trajectories of the system are represented by curves that lie on this hypersurface.

TODO: add a picture of the phase space and the energy shell

If we suppose that

1. we know positions and momenta of all particles at a given time t, thus we know the microstate of the system at time t,

2. we have an "infinite-powerful" computer that can solve the equations of motion for the system, thus we can predict the microstate of the system at any time t' given the microstate at time t,

we should be able to predict the future evolution of the system with perfect accuracy. However, in practice, we do not have access to the microstate of the system, and we do not have an infinite-powerful computer that can solve the equations of motion for the system. Specially if we think about a system with $N \sim N_A = 6 \times 10^{23}$ particles, it is impossible to know the microstate of the system, and it is impossible to solve the equations of motion for the system. We need to introduce approximations and assumptions in order to be able to make predictions about a large collection of particles.

A standard way to introduce statistical mechanics is through the so-called **postulate of equal a priori probabilities**:

Postulate 1.1 (Equal A Priori Probabilities)**.** *Macroscopic observables can be computed defining a* ***probability distribution*** *on the phase space of the system, which assigns probabilities to all microstates, and taking the expectation value of the observable with respect to this probability distribution. In an isolated system, all microstates are equally probable, thus the probability distribution is uniform on the energy shell defined by the sub-manifold of the phase space where the energy is equal to E:*

$$\mathcal{M}_E = \{(q_1, \dots, q_{3N}, p_1, \dots, p_{3N}) \in \mathcal{M} : H(q_1, \dots, q_{3N}, p_1, \dots, p_{3N}) = E\}.$$

Let us analyze this postulate in more detail through an exemplification.

Example. Suppose we have a system of N non-interacting particles in a box of volume V, with total energy E. We are interested in computing the average particle velocity $\overline{v}_x$ along some axis x.

We should compute the time dependent function $v_x(t)$ and then take the time average of it:

$$\overline{v}_x(t) = \overline{v}_x(\{q_j(t)\},\, \{p_j(t)\}) = \frac{1}{N} \sum_{j=1}^{N} \frac{v_{x,j}(t)}{m},$$

which is very complicated in general, since we have to solve the equations of motion for the system in order to find the trajectories of the particles in the phase space. However, we can use the postulate of equal a priori probabilities to compute the average particle velocity without solving the equations of motion.

The function $\overline{v}_x(\{q_j\},\, \{p_j\})$ is defined on the phase space of the system, and it assigns a value to each microstate of the system. Given the knowledge on some initial microstate, the system will evolve along a curve in the fixed energy shell defined by E:

$$\mathbf{R}(t) = \{q_j(t),\, p_j(t)\}, \quad \overline{v}_x(t) = \overline{v}_x(\mathbf{R}(t)).$$

We can now assume the system to be at the equilibrium, thus we can take the time average of the function $\overline{v}_x(t) \sim \overline{v}_x$ over a long time interval T as

$$\overline{v}_x = \frac{1}{T} \int_0^T \mathrm{d}t\, \overline{v}_x(\mathbf{R}(t)).$$

Now we can use the postulate of equal a priori probabilities to replace the time average with an average over the energy shell defined by E:

$$\overline{v}_x = \frac{1}{|\mathcal{M}_E|} \int_{\mathcal{M}_E} \mathrm{d}\mathcal{M}\, \overline{v}_x(\{q_j\},\, \{p_j\}),$$

where we are using the previous definition of the energy shell $\mathcal{M}_E$ and $|\mathcal{M}_E|$ is the volume of the energy shell in the phase space

$$|\mathcal{M}_E| = \int_{\mathcal{M}_E} \mathrm{d}\mathcal{M}.$$

Thus we have computed the average particle velocity without solving the equations of motion for the system, but using only the postulate of equal a priori probabilities: it told us that *averaging over the system trajectory is equivalent to averaging over the whole manifold at fixed energy.*

Let us now analyze the postulate of equal a priori probabilities in more detail. We have to stress some important points about it:

- In typical equilibrium situations, macroscopic quantities (such as $\overline{v}_x$ from the example) usually are constant in time only up to small fluctuations

$$\overline{v}_x(t) = \overline{v}_x + \delta v_x(t).$$

  We have previously neglected these fluctuations, but they are actually present in the system, and they can be formally removed by studying the time averages. Indeed we expect those fluctuations to be small and to occur at microscopic time scales, much smaller than the time scales of the macroscopic observables, thus they will not affect the macroscopic properties of the system. Thus measuring (i.e. averaging) over a long time interval T will remove these fluctuations:

$$\frac{1}{\tau}\int_0^\tau \mathrm{d}t\,(\overline{v}_x + \delta v_x(t)) = \frac{1}{\tau}\left(\overline{v}_x\tau + \int_0^\tau \mathrm{d}t\,\delta v_x(t)\right) = \overline{v}_x,$$

  where we have used the fact that the time average of the fluctuations is zero.

- The traditional definition of the postulate of equal a priori probabilities is simpler than the one we have given, since it is usually stated as *in an isolated system, all microstates are equally probable*, without specifying that the microstates are the points in the energy shell defined by E. We have given a more modern and precise definition of the postulate, which is more useful for our purposes, since it allows us to compute the average of macroscopic observables as an average over the energy shell in the phase space.

- In the study of dynamical systems, it is possible to mathematically formalize the problem of proving the postulate of equal a priori probabilities, and to find conditions under which it holds. This is a very active field of research, called **ergodic theory**, and in some cases one can prove its validity, while in other cases one can find counterexamples where it does not hold. There exist complicated systems for which the postulate of equal a priori probabilities does not hold, but these systems are not relevant for our purposes, thus we will assume the validity of the postulate for the rest of the course, and we will use it as a fundamental principle.

- We have assumed above that the system is at equilibrium, thus we can take the time average of the macroscopic observable over a long time interval T. If the system is not at equilibrium, then the time average of the macroscopic observable will not be constant in time, and it will depend on the initial conditions of the system. Studying how a system reaches the equilibrium (or even defining it rigorously) is non-trivial, and it is a very active field of research, called **non-equilibrium statistical mechanics** or **thermalization**. Typically the approach to equilibrium is obtained via a weak coupling among theh system and the environment, which allows the system to exchange energy and information with the environment, so that the system is not completely isolated.

With the above discussion at hand, it is useful to define the concept of **typical behaviour**. Given a function defined on the phase space (an observable) $f(\{q_j, p_j\})$ associated with some macroscopic quantity, we can define a **typical point** $\mathbf{R} = \{q_j, p_j\}$ as the point in which the function f takes a value that is close to the average value of the function over the energy shell defined by E:

$$f(\mathbf{R}) = \frac{1}{|\mathcal{M}_E|} \int_{\mathcal{M}_E} \mathrm{d}\mathcal{M}\, f(\{q_j\},\, \{p_j\}) + \delta f,$$

where δf is a small fluctuation. The postulate of equal a priori probabilities implies that the typical behaviour of the system is close to the average behaviour of the system, thus we can say that the typical point in the phase space of the system is close to the average value of the function f over the energy shell defined by E.

1.2 | Statistical Ensembles

[...]

1.2.1 | Microcanonical Ensemble

The postulate of equal a priori probabilities defines the microcanonical ensemble (in general an ensemble is a collection of all possible microstates that are consistent with a given set of macroscopic constraints), which is the ensemble of microstates that are compatible with a given macrostate defined by the energy E, volume V, and number of particles N. The probability distribution of the system in the microcanonical ensemble is uniform on the energy shell defined by E, thus we can write it as

$$\rho_{mc}(\{q_j\},\,\{p_j\}) = \begin{cases} \dfrac{1}{|\Gamma_\Delta(E)|}, & \text{if } E \leq H(\{q_j\},\,\{p_j\}) \leq E+\Delta, \\ 0, & \text{otherwise}, \end{cases} \tag{1.2.1}$$

where Δ is a small energy interval that defines the width of the energy shell, and $|\Gamma_\Delta(E)|$ is the volume of the energy shell in the phase space, which can be computed by considering the normalization condition for the probability distribution

$$\int \rho_{mc}\,\mathrm{d}^{3N}q\,\mathrm{d}^{3N}p = 1 \implies \rho_{mc} = \frac{1}{\int_{\Gamma_\Delta(E)} \mathrm{d}^{3N}q\,\mathrm{d}^{3N}p} = \frac{1}{|\Gamma_\Delta(E)|}.$$

Given a function f defined on the phase space of the system, we can compute its **ensamble average**, denoted by $\langle f\rangle_{mc}$ or $\mathbb{E}_{mc}[f]$, as the average of the function over the energy shell defined by E with respect to the probability distribution of the microcanonical ensemble:

$$\langle f\rangle_{mc} = \int f(\{q_j\},\,\{p_j\})\rho_{mc}(\{q_j\},\,\{p_j\})\,\mathrm{d}^{3N}q\,\mathrm{d}^{3N}p. \tag{1.2.2}$$

A *consistency condition* for the microcanonical ensemble (or any other ensemble) is that the statistical fluctuations of macroscopic observables are small in the thermodynamic limit $N\to\infty$. This means that the variance of a macroscopic observable f should vanish in the thermodynamic limit, thus we can write

$$\sigma_f^2 = \frac{\langle f^2\rangle_{mc} - \langle f\rangle_{mc}^2}{\langle f^2\rangle_{mc}} \ll 1 \quad \text{for } N\to\infty.$$

Now we can use these conditions to further analyze this ensamble; in order to do so let us introduce the entropy S of the system, which can be used to derive the thermodynamic properties of the system, and to find the most probable configuration of the system.

Entropy

As seen in previous courses, in this framework we are outlining a microscopic basis for the thermodynamic properties of the system, thus we need to introduce a microscopic definition of the entropy S of the system, which is a measure of the number of microstates available to the system at a given energy E. The **Boltzmann formula** for the entropy is given by

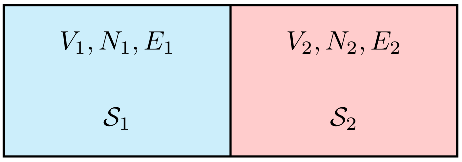
$$S(E) = k_B \log|\Gamma_\Delta(E)|, \tag{1.2.3}$$

where k_B is the Boltzmann constant, and $|\Gamma_\Delta(E)|$ is the volume of the energy shell in the phase space. The Boltzmann formula for the entropy tells us that the entropy of the system is proportional to the logarithm of the number of microstates available to the system at a given energy E. This is a fundamental result in statistical mechanics, and it allows us to derive the thermodynamic properties of the system from its microscopic properties.

We will not derive the Boltzmann formula for the entropy, but we will use it as a fundamental principle to derive the thermodynamic properties of the system. For example, we can derive a definition of temperature T of the system from the entropy S.

Temperature

Let us consider two systems $\mathcal{S}_1$ and $\mathcal{S}_2$ in thermal contact such that they can exchange energy, but they are isolated from the rest of the universe. The total system is isolated, as well as the subsystems, thus we can apply the microcanonical ensemble to them.



$$\begin{cases} N = N_1 + N_2, \\ V = V_1 + V_2, \\ E = E_1 + E_2. \end{cases}$$

Thermal contact means that the two systems can exchange energy, thus the total energy E is fixed, but the energy of each subsystem E_1 and E_2 can fluctuate. The total number of particles N and the total volume V are also fixed, since there is no exchange of matter or volume between the two systems.

The total energy can now be computed from the Hamiltonian of the system, sum of the subsystems' Hamiltonians plus an interaction term

$$H(\{q_j\}, \{p_j\}) = H_1(\{q_{1j}\}, \{p_{1j}\}) + H_2(\{q_{2j}\}, \{p_{2j}\}) + \delta H_{12}(\{q_{1j}\}, \{p_{1j}\}, \{q_{2j}\}, \{p_{2j}\}),$$

where the last term accounts for the interaction between the two systems, which we will neglect in the following. The energy of the total system is fixed, but the energy of each subsystem can fluctuate. The probability of finding the system in a state with energy E_1 is proportional to the number of states available to the second system with energy $E_2 = E - E_1$. If we fix an energy shell of width Δ, we can write

$$E_k^{(1)} = k\Delta, \qquad E_k^{(2)} = E - E_k^{(1)} = E - k\Delta,$$

and with this "discretization" we can count the number of states available to the total system at a fixed energy E as

$$\Gamma(E) = \sum_{k=0}^{E/\Delta} \Gamma_1(E_k^{(1)})\Gamma_2(E - E_k^{(1)}),$$

where $\Gamma_k(E) = \int_{E<H_k<E+\Delta} \mathrm{d}\{q_{kf}\}\mathrm{d}\{p_{kf}\}$ is the number of states available to the k-th subsystem at energy E.

Thus we are saying that *in order to count the number of microstates at energy E we have to multiply the number of microstates of the subsystem S_1 with energy ϵ_1 to the number of microstates of S_2 at energy $\epsilon_2 = E - \epsilon_1$, then sum over all possible values of ϵ_1.*

The **most probable configuration** of the system is the one that maximizes the number of states available to the total system, which is equivalent to **maximizing the entropy** of the total system. Thus we can compute the entropy of the total system as

$$S(E) = k_B \log \left[\sum_{k=0}^{E/\Delta} \Gamma_1(E_k^{(1)}) \Gamma_2(E - E_k^{(1)}) \right].$$

Now, in order to maximize it, we can take the derivative of the entropy with respect to E_1 and set it to zero: let us denote $\bar{E}_1$ the energy of the subsystem S_1 that maximizes the entropy, then $\bar{E}_2 = E - \bar{E}_1$ is the energy of the subsystem S_2 that maximizes the entropy. We have

$$\Gamma(\bar{E}^{(1)})\Gamma(\bar{E}^{(2)}) \leq \Gamma(E) \leq \frac{E}{\Delta}\Gamma(\bar{E}^{(1)})\Gamma(\bar{E}^{(2)}),$$

where the first inequality comes from the fact that $\Gamma(E)$ is a sum over all the values of E_1, thus it is greater than or equal to the single term with $E_1 = \bar{E}_1$, while the second inequality comes from the fact that there are at most E/Δ terms in the sum, thus we can upper bound it by multiplying the maximum term by the number of terms. Taking the logarithm of these inequalities we arrive at

$$S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2) \leq S(E, V) \leq S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2) + k_B \log \left(\frac{E}{\Delta} \right),$$

where both inequalities are saturated in the thermodynamic limit $N \to \infty$: entropy and energy are extensive quantities, thus $S(E, V) \sim N$ and $E \sim N$, so the last term in the upper bound is negligible in the thermodynamic limit

$$k_B \log \left(\frac{E}{\Delta} \right) \sim k_B \log(N) \ll N \sim S(E, V).$$

Thus we can write, in the thermodynamic limit, the entropy of the total system as the sum of the entropies of the subsystems, computed at the most probable values of the energy of the subsystems:

$$S(E,\, V) = S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2).$$

The first consequence of this result is that the **entropy is an extensive quantity**, since it scales with the number of particles N in the system. The second consequence is that the *most probable configuration of the system is the one that maximizes the entropy*, which is equivalent to maximizing the number of states available to the total system. Thus we can write

$$\Gamma(E) = \Gamma_1(\bar{E}^{(1)})\Gamma_2(\bar{E}^{(2)}).$$

Therefore terms with different values of E_1 are negligible compared to the term with $E_1 = \bar{E}_1$, which is the one that maximizes the number of states available to the total system. This is a consequence of the fact that the number of states available to the total system grows exponentially with the number of particles N, thus even a small difference in energy between different configurations can lead to a huge difference in the number of states available to the total system. $\bar{E}^{(1)}$ and $\bar{E}^{(2)}$ are the most probable values of the energy of the subsystems, which are the ones that maximize the entropy of the total system: the *equilibrium energy values*. Mathematically, we can obtain $\bar{E}^{(1)}$ from the saddle-point condition:

$$\frac{\partial}{\partial \epsilon} \left[\Gamma_1(\epsilon)\Gamma_2(E - \epsilon) \right]_{\epsilon = \bar{E}^{(1)}} = 0,$$

which, expanding the derivative, can be written as

$$\left[\frac{\partial}{\partial \epsilon}\Gamma_1(\epsilon) \right]_{\epsilon = \bar{E}^{(1)}} \Gamma_2(E - \epsilon) + \Gamma_1(\epsilon) \left[-\frac{\partial}{\partial \epsilon}\Gamma_2(E - \epsilon) \right]_{\epsilon = \bar{E}^{(2)}} = 0,$$

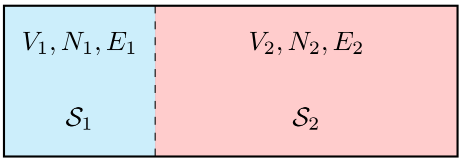
which now, dividing by $\Gamma_1(\bar{E}^{(1)})\Gamma_2(\bar{E}^{(2)})$ (in order to build the logarithmic derivative), and using the definition of entropy we arrive at

$$\frac{\partial}{\partial \epsilon} S_1(\epsilon)\bigg|_{\epsilon=\bar{E}^{(1)}} = \frac{\partial}{\partial \epsilon} S_2(\epsilon)\bigg|_{\epsilon=\bar{E}^{(2)}}.$$

This is a fundamental result, which tells us that at equilibrium the derivative of the entropy with respect to the energy is the same for both subsystems. This allows us to define a thermodynamic quantity which is equal in systems in thermal equilibrium, called **temperature** T (recovering the first principle of thermodynamics). The temperature is defined as the inverse of the derivative of the entropy with respect to the energy, thus we can write

$$\frac{1}{T} = \frac{\partial S}{\partial E}(E, V).$$

1.2.2 | Canonical Ensemble



$$\begin{cases} N = N_1 + N_2, \\ V = V_1 + V_2, \\ E = E_1 + E_2, \end{cases} \quad \text{and} \quad \begin{cases} N_2 \gg N_1, \\ V_2 \gg V_1, \\ E_2 \gg E_1. \end{cases}$$

The second system S_2 is called the *heat bath*, and it is so large that it can exchange energy with the first system S_1 without changing its thermodynamic properties. The total system is isolated, thus we can apply the microcanonical ensemble to it. The probability of finding the system in a state with energy E_1 is proportional to the number of states available to the second system with energy $E_2 = E - E_1$, which is given by $\Omega_2(E - E_1)$.

Technically speaking, we want to find the probability distribution of the subsystem S_1 when it is in contact with a heat bath S_2, knowing the total probability distribution: the total system is isolated, thus we can apply the microcanonical ensemble to it, and we know that it will be evolving with a trajectory in the phase space that is uniformly distributed on the energy shell defined by the total energy E. This means that if we analyze the evolution of the subsystem S_1 in its phase space, we will find that it will not be limited to fixed energy states, but it will be able to explore all the energy states available to it, with a probability that is proportional to the number of states available to the heat bath S_2 at the corresponding energy. we are interested in finding the **induced probability distribution** of the subsystem S_1.

Let us visualize the phase space of the total system as a high-dimensional space, where each point represents a microstate of the total system. The energy shell defined by the total energy E is a hypersurface in this phase space, and the trajectory of the total system is uniformly distributed on this hypersurface. The subsystem S_1 can be thought of as a projection of the total system onto a lower-dimensional subspace, where each point represents a microstate of the subsystem S_1. The probability distribution of the subsystem S_1 is then given by the projection of the uniform distribution on the energy shell onto this lower-dimensional subspace. Differents probabilities correspond to different energy states of the subsystem S_1, and the probability of finding the subsystem S_1 in a state with energy E_1 is proportional to the number of states available to the heat bath S_2 at energy $E - E_1$.

The idea is to fix a point in the phase space of the subsystem S_1 with energy E_1, and then count the number of points in the phase space of the heat bath S_2 that are compatible with this point, i.e. that have energy $E - E_1$. This will give us the number of states available to the heat bath S_2 at energy $E - E_1$, which is proportional to the probability of finding the subsystem S_1 in a state with energy E_1.

So we fix an energy ϵ for the subsystem S_1, and we want to find the number of states available to the heat bath S_2 at energy $E - \epsilon$. *The number of points in the phase space of S that corresponds to P is equal to the number of points in the phase space of S_2 that corresponds to $E - \epsilon$.* Thus we can write

$$\rho(\{q_f\}, \{p_f\}) \propto \frac{\Gamma_2(E - \epsilon)}{\Gamma(E)}.$$

We know that the entropy of the heat bath S_2 is related to the number of states available to it by the Boltzmann formula (we are using $k_B = 1$ for simplicity), such that

$$\Gamma_2(E - \epsilon) = e^{S_2(E-\epsilon)} \sim e^{S_2(E) - \epsilon \frac{\partial S_2}{\partial E} \epsilon},$$

where we have used the fact that the entropy is an extensive quantity, thus we can write it as a function of the energy per particle, and we have expanded it around the total energy E of the heat bath S_2. We can now recall the temperature as a thermodynamic quantity defined as the inverse of the derivative of the entropy with respect to the energy, while reinserting the Boltzmann constant k_B for clarity, we have

$$\rho(\{q_f\}, \{p_f\}) \propto e^{-\beta\epsilon} = e^{-\frac{\epsilon}{k_B T}},$$

where $\beta = \frac{1}{k_B T}$ is the inverse temperature. Thus we have derived the Boltzmann distribution for the subsystem S_1 in contact with a heat bath S_2. The probability of finding the subsystem S_1 in a state with energy ϵ is proportional to the exponential of the negative of the energy divided by the temperature of the heat bath S_2. This is the canonical ensemble, where the subsystem S_1 is in thermal equilibrium with a heat bath S_2 at temperature T.

We can now define the partition function Z as the normalization constant for the probability distribution of the subsystem S_1, such that

$$Z = \int \frac{\mathrm{d}^{3N}q \, \mathrm{d}^{3N}p}{h^{3N} N!} \, e^{-\beta H(\{q\}, \{p\})},$$

which can be used as a generating function for the thermodynamic properties of the subsystem S_1. For example, the free energy F can be computed from the partition function as

$$F = -k_B T \log Z = -\frac{1}{\beta} \log Z,$$

and the internal energy U can be computed as

$$U = -\frac{\partial}{\partial \beta} \log Z,$$

while the entropy S can be computed as

$$S = -\frac{\partial F}{\partial T} = k_B \beta^2 \frac{\partial F}{\partial \beta}.$$

Discrete System

A discrete system is a system that can only take a finite number of states, such as a spin system or a lattice gas. In this case, let us imagine a collection of N particles that can only occupy a finite number of states, let us imagine two values of spin, up and down, thus we have 2^N possible configurations of the system. The phase space of the system is now a discrete set of points, configurations (a set of sets) $\{c_j\}_{j=1}^N$, and the probability distribution of the system is defined on this discrete set of points. The partition function can be written as a sum over all possible configurations of the system, such that

$$Z = \sum_{\{c_j\}} e^{-\beta H(\{c_j\})}.$$

1.3 | Quantum Statistical Mechanics

Considering a general quantum mechanical system, we can study it with different tools, such as the hamiltonian operator, the density matrix, the partition function, etc. The Hamiltonian operator H is a self-adjoint operator that acts on the Hilbert space of the system, and it encodes the energy of the system. The density matrix ρ is a positive semi-definite operator that acts on the Hilbert space of the system, and it encodes the state of the system. The partition function Z is a scalar quantity that encodes the thermodynamic properties of the system.

$$
\begin{aligned}
\hat{H}\left|n\right\rangle &= E_n \left|n\right\rangle, \\
\left|\psi\right\rangle &\in \mathcal{H}, \\
\{\left|E_n\right\rangle\}_{n=1}^{D} \implies \left|\psi\right\rangle &= \sum_{n=1}^{D} c_n \left|E_n\right\rangle, \\
\left\langle\psi\right| \hat{A} \left|\psi\right\rangle &= \sum_{n,m} c_n^* c_m \left\langle E_n\right| \hat{A} \left|E_m\right\rangle, \\
\left|\psi(t)\right\rangle = \sum_n c_n(t) \left|E_n\right\rangle, &\quad c_n(t) = c_n e^{-i\frac{E_n}{\hbar}t} \\
\left\langle\psi(t)\right| \hat{A} \left|\psi(t)\right\rangle &= \sum_{n,m} c_n^* c_m e^{i\frac{E_n - E_m}{\hbar}t} \left\langle E_n\right| \hat{A} \left|E_m\right\rangle, \\
\implies = \frac{1}{T}\int_0^T \mathrm{d}t \, \left\langle\psi(t)\right| \hat{A} \left|\psi(t)\right\rangle &= \sum_n |c_n|^2 \left\langle E_n\right| \hat{A} \left|E_n\right\rangle.
\end{aligned}
$$

Let us report some postulates of quantum statistical mechanics:

- Postulate of equal a priori probabilities: in an isolated system, all microstates are equally probable,
$$
|\bar{c_n}|^2 = \begin{cases} \frac{1}{D}, & \text{if } E_n \in [E, E+\Delta], \\ 0, & \text{otherwise.} \end{cases}
$$
- postulate of random phases: the phases of the coefficients c_n are random variables uniformly distributed in the interval $[0, 2\pi]$, thus
$$
\frac{1}{T}\int_0^T \mathrm{d}t \, c_n^*(t) c_m(t) = \frac{1}{T}\int_0^T \mathrm{d}t \, c_n^* c_m \, e^{i\frac{E_n - E_m}{\hbar}t} = \begin{cases} |c_n|^2, & \text{if } n = m, \\ 0, & \text{otherwise.} \end{cases}
$$

The time average of an observable $\hat{A}$ is given by

$$
\overline{\left\langle\psi(t)\right| \hat{A} \left|\psi(t)\right\rangle} = \frac{1}{T}\int_0^T \mathrm{d}t \, \left\langle\psi(t)\right| \hat{A} \left|\psi(t)\right\rangle = \sum_n |c_n|^2 \left\langle E_n\right| \hat{A} \left|E_n\right\rangle.
$$

[...]

We can define the density matrix ρ as

$$
\rho = \frac{1}{Z} \sum_{E \le E_n \le E+\Delta} \left|E_n\right\rangle \left\langle E_n\right|,
$$

which is a positive semi-definite operator that encodes the state of the system. The expectation value of an observable $\hat{A}$ can be then computed as

$$
\langle \hat{A} \rangle = \mathrm{Tr}(\rho \hat{A}).
$$

We have to note that dealing with quantum statistical mechanics is more complicated than dealing with classical statistical mechanics, since we have to deal with operators instead of functions, and we have to deal with the non-commutativity of operators. However, the general principles of statistical mechanics still apply, and we can still define ensembles, partition functions, and thermodynamic quantities in the quantum case. The main difference is that we have to use the language of operators and Hilbert spaces instead of the language of functions and phase space.

Defining an ensemble in quantum statistical mechanics automatically defines a density matrix, which encodes the state of the system, and from which we can compute the expectation values of observables and the thermodynamic properties of the system. We are governing our set of eigenstates with a probability distribution, which is encoded in the density matrix, and we can compute the expectation values of observables and the thermodynamic properties of the system from this density matrix.

For a quantum system at thermal equilibrium, we have the following identification of the thermodynamic free energy F with the partition function Z:

$$F = -k_B T \log Z,$$

which is the same relation that we have in classical statistical mechanics. These postulates are like operational tools: justify them as they are can be rally difficult and it is an actual field of research with a lot of literature, but they are very useful for computing the thermodynamic properties of quantum systems at thermal equilibrium. For us, justifying why a precise postulate workd in the actual setting will be enough.

2 Phase Transitions and Mean Field Theory

Phase Transitions are fundamental phenomena in statistical physics, where a system undergoes a qualitative change in its macroscopic properties as external parameters (like temperature or pressure) are varied. These transitions can be classified into different types based on their characteristics and the nature of the change in the system. Let us start from the definition of a phase of a many-particle system, which is crucial for understanding phase transitions.

Definition 2.1 (Phase of Many-Particle System)**.** A phase of a many-particle system is a region in the parameter space (e.g. temperature, pressure) where the system exhibits distinct physical properties and behavior: it is a "type" of collective state of matter. They are defined in terms of macroscopic properties such as density, magnetization, or symmetry.

A phase transition occurs when a system undergoes an *abrupt change from one phase to another* (states) *as we smoothly vary an external parameter.* Often PTs are accompanied by a discontinuity or singularity in some thermodynamic quantity, which serves as an order parameter to characterize the transition.

A phase transition is always a result of a competition between "order" and "disorder" in the system. For example, in a ferromagnetic material, the ordered phase corresponds to a state where the magnetic moments are aligned, while the disordered phase corresponds to a state where the magnetic moments are randomly oriented. The transition between these two phases can be driven by temperature, where thermal fluctuations disrupt the order at high temperatures.

We are now going to define PTs in a mathematical framework, which will allow us to classify them into different types. To start we have to highlight how PTs are related to **singularities in the partition function** of the system at the thermodynamic limit, or equivalently in the free energy. The **free energy** is a thermodynamic potential that encodes the equilibrium properties of a system, and its behavior near a phase transition can reveal important information about the nature of the transition:

$$F = -\frac{1}{\beta}\ln Z = U - TS,$$

where Z is the partition function of the system and $\beta = 1/(k_B T)$ is the inverse temperature. A phase transition is typically associated with a non-analytic behavior of the free energy as a function of the external parameters, such as temperature or pressure. Thus PTs are defined as singularities in the thermodynamic limit of the partition function Z or equivalently of the free energy F.

Definition 2.2 (Phase Transition)**.** A phase transition is a non-analytic behavior of the free energy F as a function of external parameters (e.g., temperature, pressure) in the thermodynamic limit.

This non-analyticity manifests as a discontinuity or singularity in some thermodynamic quantity, which serves as an order parameter to characterize the transition.

Example (Finite system and phase transitions). If we consider a finite set of spins, the partition function Z is a finite sum of exponentials, which is an analytic function of the parameters:

$$Z = \sum_{\{\text{conf}\}} e^{-\beta E[\text{conf}]}.$$

Therefore, in a finite system, there can be no true phase transition, as the free energy will always be analytic. However, as we take the thermodynamic limit (i.e., let the number of spins go to infinity), the partition function can develop singularities, leading to non-analytic behavior in the free energy and thus allowing for phase transitions to occur.

In physics we never encounter real infinite systems, but we can approximate them by considering large systems and taking the thermodynamic limit. This is a common approach in statistical mechanics to study phase transitions, as it allows us to capture the essential features of the transition while avoiding the complications that arise from finite-size effects.

2.1 | Classification of Phase Transitions

Phase transitions can be classified into different types based on their characteristics and the nature of the change in the system. One of the first classifications was proposed by *Paul Ehrenfest*, who categorized phase transitions based on the order of the derivative of the free energy that exhibits a discontinuity at the transition point. Today, we have a more refined classification, which is based on the concept of order parameters and symmetry breaking.

Phase transitions can be classified depending on the type of singularity in the free energy. We can distinguish between first-order and continuous (second-order) phase transitions:

- **First-order phase transitions** are characterized by a discontinuity in the first derivative of the free energy with respect to some external parameter (e.g., temperature). This means that there is a **latent heat** associated with the transition, and the order parameter changes discontinuously.[1] Examples include the liquid-gas transition and the melting of a solid.

- **Continuous (second-order) phase transitions** are characterized by a continuous first derivative of the free energy, but a discontinuity in the second or higher derivatives. In this case, there is **no latent heat**, and the order parameter changes continuously. Examples include the ferromagnetic transition in the Ising model and the superconducting transition in certain materials. These are the most interesting PTs and will drive most of our discussion in the following chapters.

The latter group of PTs is also called **critical phenomena**, and they are associated with the emergence of long-range correlations and *scale invariance* at the critical point. The study of critical phenomena has led to the development of powerful theoretical tools, such as the *renormalization group*, which allows us to understand the universal behavior of systems near criticality. Another important phenomenon associated with continuous phase transitions is the concept of **spontaneous symmetry breaking**. In many cases, the ordered phase of a system exhibits a lower symmetry than the disordered phase. This spontaneous symmetry breaking is a key feature of many continuous phase transitions and plays a crucial role in determining the properties of the system near criticality.

2.1.1 | Phase Diagram of Water

Looking at the phase diagram of water, we can see that there are three main phases: solid (ice), liquid (water), and gas (vapor). The **phase boundaries** between these phases are represented by lines in the pressure-temperature (p-T) diagram, also called coexistence lines. The point where the liquid-gas phase boundary ends is called the **critical point**, which is a second-order phase transition (only "crossing" the specific point). Over this point, the properties of the liquid and gas phases become indistinguishable, suggesting that the two phases are not fundamentally different, and the system exhibits critical phenomena such as large fluctuations and long-range correlations.

[1] This happens because we can express order parameters as derivatives of the free energy, which is continuous but has discontinuous first derivative.

Moving along the phase boundaries, we can observe first-order phase transitions, where the system undergoes a discontinuous change in density and other properties. For example, when water freezes into ice or boils into vapor, there is a latent heat associated with the transition, and the order parameter (density) changes discontinuously. The first derivative of the free energy with respect to temperature (which is related to the entropy) and pressure (which is related to the volume) will show a discontinuity at these transitions, confirming their first-order nature.

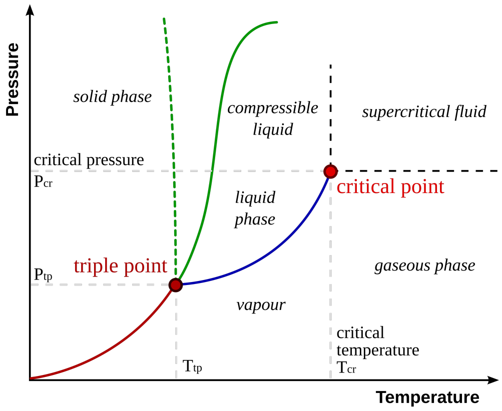
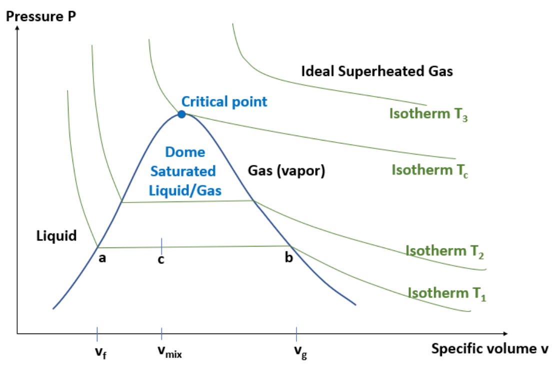


Figure 2.1: Phase diagram of water in the pressure-temperature (p-T) plane. The lines represent the phase boundaries between solid, liquid, and gas phases, with the critical point marking the end of the liquid-gas phase boundary.

Decreasing the pressure at a fixed temperature can also lead to phase transitions. For example, if we decrease the pressure of liquid water at a constant temperature, we can reach the point where it transitions into vapor. This is another example of a first-order phase transition, as there is a discontinuity in the density and other properties of the system.

Each point in the (p-T) phase diagram corresponds to a specific value of $v = \frac{V}{N}$ specific volume, except for the points on the phase boundaries (I order transition lines), where we have a coexistence of phases and the specific volume can take on different values depending on the relative proportions of the coexisting phases. For example, along the liquid-gas phase boundary, we can have a mixture of liquid and gas phases, and the specific volume will depend on the ratio of liquid to gas in the mixture. This can be seen in the following phase diagram.

The phase diagram of water in the pressure-volume (p-v) plane shows indeed the different phases and the transitions between them, with the critical point marking the end of the liquid-gas phase boundary.



If we are below the critical temperature, at a fixed T we can observe all three phases by varying the pressure or volume: this isothermal lines form a mesoscopic domain where we can observe the coexistence of phases and the transitions between them. This is a rich area of study in statistical mechanics, as it allows us to explore the behavior of systems near criticality and understand the underlying mechanisms driving phase transitions.

Figure 2.2: Phase diagram of a pure substance in the pressure-volume (p-v) plane. The lines represent the phase boundaries between different phases, with the critical point marking the end of the liquid-gas phase boundary.

The isotherms in the previous diagram are not to be confused with the coexistence lines in the p-T diagram, which represent the phase boundaries. In the p-v diagram, the isotherms represent the behavior of the system at a fixed temperature as we vary the pressure and volume, and we encounter phase I order phase transitions as we cross the coexistence region ("dome" of saturated liquid/gas). This region is characterized by a coexistence of liquid and gas phases, where the specific volume can take on different values depending on the relative proportions of the coexisting phases

$$\rho_g \sim \frac{1}{v_g}, \quad \rho_l \sim \frac{1}{v_l}, \quad \text{at } T < T_c.$$

The critical point marks the end of this coexistence region, where the properties of the liquid and gas phases become indistinguishable.

The full phase diagram of water in the pressure-volume-temperature (p-v-T) space provides a comprehensive view of the different phases and the transitions between them. It shows how the phases of water change as we vary both temperature and pressure, with the critical point marking the end of the liquid-gas phase boundary.

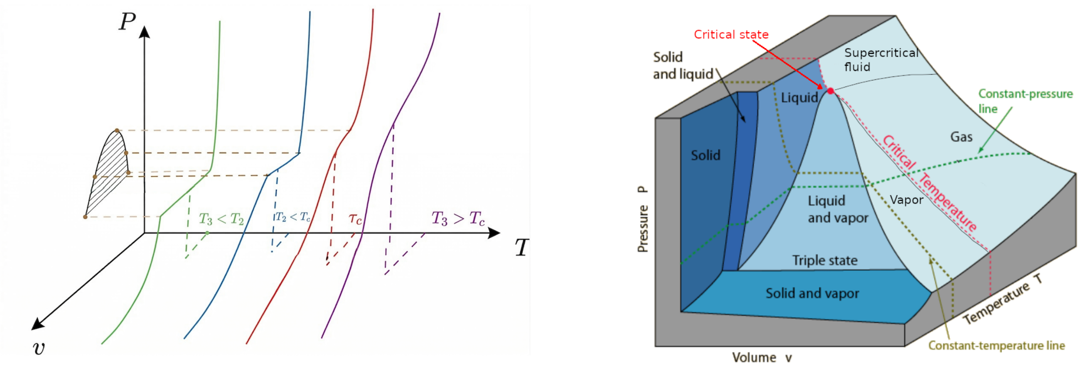


Figure 2.3: Phase diagram of water in the pressure-volume-temperature (p-v-T) space. On the right, the surfaces represent the different phases of water, with the critical point marking the end of the liquid-gas phase boundary. On the left, different isotherms are shown, illustrating how the phases change as we vary temperature and pressure (only near the critical point).

Transitions may be direction dependent in the phase space, meaning that the nature of the transition can change depending on the path taken in the parameter space. At the critical point, the system exhibits critical phenomena such as large fluctuations and long-range correlations. The properties of the system near the critical point can be described by critical exponents, which characterize how various physical quantities diverge or vanish as the critical point is approached. These critical exponents are universal, meaning that they depend only on the dimensionality of the system and the symmetry of the order parameter, rather than on the specific details of the microscopic interactions. This universality is a key feature of critical phenomena and allows us to classify different systems into universality classes based on their critical behavior.

Phase transitions are well defined when described with a curve in the parameter space, but we can also have transitions that occur at a single point, such as the critical point in the phase diagram of water. In this case, the transition is not associated with a curve but rather with a specific set of conditions (temperature and pressure) where the properties of the system change dramatically.

We are only going to discuss PTs which occurs with a smooth variation of the parameters, but there are also PTs that can occur with a sudden change in the parameters, such as a quench or a rapid cooling. These non-equilibrium phase transitions can lead to interesting phenomena such as the formation of topological defects and the emergence of new phases that are not present in equilibrium conditions. The study of non-equilibrium phase transitions is an active area of research in statistical mechanics and condensed matter physics.

Remark (Critical Opalescence). *The **milky appearence** (or critical opalescence) of a system near the critical point is a direct consequence of the divergence of the correlation length ξ (for example in a liquid-gas mixture). As the correlation length increases, fluctuations in the density or order parameter occur over larger and larger distances, leading to the scattering of light and giving rise to a milky or cloudy appearance. This phenomenon is a visual manifestation of the critical behavior of the system and is often observed in fluids near their critical point, such as in the case of water at its critical temperature and pressure.*

2.1.2 | Magnetic System

In this setting, we are considering a universal magnetic system, which can be described by a Hamiltonian that includes interactions between magnetic moments (spins) and an external magnetic field. The behavior of the system can be analyzed in terms of its phase transitions, which are characterized by changes in the order parameter (magnetization) as we vary the temperature and the external magnetic field.

We have few control parameters over the system: the temperature T and the external magnetic field h. The phase diagram of the magnetic system can be represented in the $T - h$ plane, where we can identify different phases based on the behavior of the magnetization.

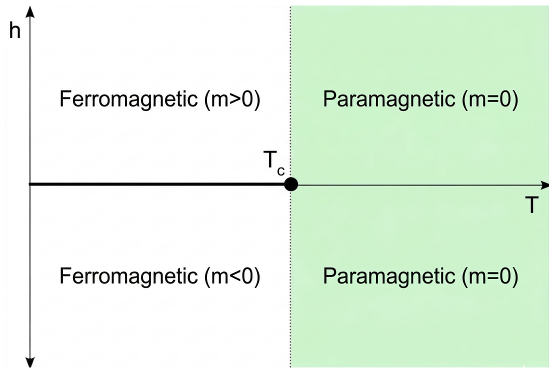


Figure 2.4: Phase diagram of a magnetic system in the (h-T) plane, showing the ferromagnetic phase at low temperatures, cut in half by the first-order phase transition line, which ends at the critical point. Over the critical point, the system is in the paramagnetic phase.

- At high temperatures and low magnetic fields, the system is in a disordered phase where the magnetization is zero: **paramagnetic phase**.
- At low temperatures and high magnetic fields, the system is in an ordered phase where the magnetization is non-zero: **ferromagnetic phase**.
- The transition between these two phases can be either first-order or second-order, depending on the specific interactions in the system. If we cross the phase buondary under the critical point, we observe a first-order phase transition, while if we cross the critical point, we observe a second-order phase transition. Over it we have only the paramagnetic phase, which is a disordered phase with zero magnetization.
- At the critical point, the system exhibits critical phenomena such as large fluctuations in magnetization and long-range correlations. The properties of the system near the critical point

can be described by critical exponents, which characterize how various physical quantities diverge or vanish as the critical point is approached.

Introducing the **magnetization** as the order parameter of the system, we can analyze how it changes across the phase transitions.

Definition 2.3 (Order Parameter)**.** An order parameter is a quantity that serves as a measure of the degree of order in a system and can be used to characterize different phases. It typically takes on a non-zero value in the ordered phase and vanishes in the disordered phase. It is expressed generally as

$$m(T) = \frac{1}{V} \lim_{h \to 0} \langle M(h, T) \rangle,$$

where the notation is deliberately describing the case of magnetization, but we can interpret more generally as the order parameter of a system, which can be defined in terms of the expectation value of some observable that characterizes the order in the system.

Thus we interpret in general as

$$\phi(T) = \frac{1}{V} \lim_{h \to 0} \langle O(h, T) \rangle,$$

where $O(h, T)$ is an observable that characterizes the order in the system, and V is the volume of the system, h is an external field that couples to the order parameter, and the limit $h \to 0$ ensures that we are measuring the spontaneous magnetization (or order parameter) in the absence of an external field.

In the paramagnetic phase, the magnetization is zero, while in the ferromagnetic phase, it takes on a non-zero value. The behavior of the magnetization near the critical point can be described by a power law, with a critical exponent that characterizes how the magnetization vanishes as we approach the critical point from the ordered phase.

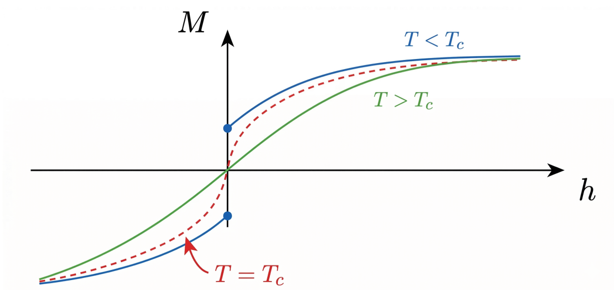


Figure 2.5: Phase diagram of a magnetic system in the $(M\text{-}h)$ plane, showing the behavior of the magnetization as an order parameter as a function of the external magnetic field at different temperatures. The coexistence line at the critical temperature is shown.

This is equivalent to the jump of the volume in the liquid-gas transition (p-v water diagram), which is the order parameter of that system. Indeed, the red line in the previous figure represents the coexistence line, where the system can exist in both phases. The magnetization serves as a measure of the degree of order in the magnetic system, and it is associated to a first derivative of the free energy, which is discontinuous at a first-order phase transition and continuous at a second-order phase transition.

At any temperature below the critical temperature, we can observe a first-order phase transition by varying the external magnetic field. As we increase the magnetic field from zero, the system transitions from the paramagnetic phase to the ferromagnetic phase, with a discontinuous change in magnetization. This is a first-order phase transition, as there is a latent heat associated with the transition and the order parameter (magnetization) changes discontinuously.

The critical temperature separates two phases with different symmetries: the paramagnetic phase has a higher symmetry (rotational symmetry) than the ferromagnetic phase (which has a lower symmetry due to the alignment of spins). This is an example of spontaneous symmetry breaking, where the ordered phase exhibits a lower symmetry than the disordered phase. The study of this magnetic system and its phase transitions provides valuable insights into the behavior of many other systems that exhibit similar phenomena, such as superconductors and liquid crystals.

2.2 | Critical Behaviour

When the order parameter approaches zero continuously at the critical point, we can describe its behavior using a power law:

$$m(T) = \begin{cases} 0 & \text{for } T > T_c, \\ (T_c - T)^\beta & \text{for } T < T_c, \end{cases}$$

where β is the critical exponent that characterizes how the magnetization vanishes as we approach the critical point from the ordered phase. The value of β depends on the universality class of the system, which is determined by factors such as the dimensionality of the system and the symmetry of the order parameter. For example, in the three-dimensional Ising model, which is a prototypical model for ferromagnetic phase transitions, the critical exponent β is approximately 0.326.

The surprising thing is that if we compute the same exponent for a completely different system, such as the liquid-gas transition, using some difference in densities as the order parameter, we find the same value of β:

$$\begin{cases} |\rho_l - \rho_c| \sim (T - T_c)^\beta, \\ |\rho_c - \rho_g| \sim (T - T_c)^\beta, \end{cases} \quad \beta \approx 0.326,$$

where ρ_l and ρ_g are the densities of the liquid and gas phases, respectively, ρ_c at the critical point, and the critical exponent β is exactely the same as in the magnetic system, despite the fact that these systems have different microscopic details and interactions.

This is a manifestation of the **universality of critical phenomena**, which states that systems with different microscopic details can exhibit the same critical behavior near the phase transition, as long as they belong to the same universality class. This universality allows us to classify different systems based on their critical exponents and understand their behavior near criticality without needing to know the specific details of their interactions.

2.2.1 | Critical Exponents and Universality Classes

The critical exponents are a set of numbers that characterize the behavior of physical quantities near the critical point. They describe how various properties of the system diverge or vanish as the critical point is approached. Some of the most common critical exponents include:

- β: describes the behavior of the **order parameter** *as a function of temperature* near the critical point, $m(T) \sim (T_c - T)^\beta$ for $T < T_c$. We have already discussed this exponent in the context of the magnetization and the liquid-gas transition, where it characterizes how the order parameter vanishes as we approach the critical point from the ordered phase.

- α: describes the divergence of the specific heat (heat capacity) at constant volume and zero external field as a function of temperature near the critical point,

  $$C_V \sim |T - T_c|^{-\alpha},$$

  and the heat capacity is defined as the second derivative of the free energy with respect to temperature:

  $$C_V = \frac{\partial U}{\partial T} = \frac{\partial}{\partial T}(F + TS) = T\frac{\partial S}{\partial T} = -T\frac{\partial^2 F}{\partial T^2}.$$

If we denote by $c_{\pm}(T, h = 0)$ the heat capacity at zero external field above/below the critical point ($t = \frac{T-T_c}{T_c}$), we can express the critical behavior of the heat capacity as

$$c_{\pm}(t, h = 0) \sim |t|^{-\alpha_{\pm}},$$

where $\alpha_{\pm}$ are the critical exponents for the heat capacity, in principle different above and below the critical point.

- δ: describes the behavior of the **order parameter** *as a function of the external field* at the critical temperature,

$$m(h) \sim h^{1/\delta} \quad \text{at } T = T_c,$$

  where h is the external magnetic field. This means that the magnetization, along the critical isotherm, behaves as a power law in the external field, going to zero as the field goes to zero with the previus dependence on δ.

- γ: describes the divergence of the *zero-field* **susceptibility** *as a function of temperature*:

$$\chi \sim |T - T_c|^{-\gamma},$$

  where the susceptibility χ is defined as the derivative of the order parameter with respect to the external field:

$$\chi_{\pm} = \chi(t \to 0^{\pm}) = \left(\frac{\partial m}{\partial h}\right)_{h=0}, \quad \chi_{\pm} \sim |t|^{-\gamma_{\pm}},$$

  with $t = (T - T_c)/T_c$ being again the reduced temperature. The susceptibility measures how responsive the system is to changes in the external field, and its divergence at the critical point indicates that even a small change in the magnetic field can lead to a large change in the magnetization. This is a hallmark of criticality and is associated with the emergence of long-range correlations in the system.

2.2.2 | Correlation Length and Universality

Before continuing with the definitions of ν and η, we have to introduce the concept of **correlation length**, which is a measure of how far correlations in the system extend. Near the critical point, the correlation length diverges, indicating that fluctuations occur over increasingly large distances. Let us introduce it through a little model: let us consider a system of spins on a lattice, where each spin s_i can take on values of +1 or -1. The magnetization m is defined as the average spin per site:

$$m = \frac{1}{N}\sum_{i=1}^{N}\langle s_i\rangle,$$

where the angle brackets denote the thermal average:

$$\langle s_i\rangle = \mathbb{E}_c\left[s_i\right] = \frac{1}{Z}\sum_{\{s_k\}} s_i e^{-\beta H(\{s_k\},h)},$$

with Z being the partition function and H the Hamiltonian of the system. The correlation function $G(r)$ is defined as the average relative alignment of spins, i.e. the average product of spins at a distance r:

$$G_{ij}^{(2)} = \langle s_i s_j\rangle = \frac{1}{Z}\sum_{\{s_k\}} s_i s_j e^{-\beta H(\{s_k\},h)} = G^{(2)}(\mathbf{r}), \quad \text{where } \mathbf{r} = |\mathbf{r}_i - \mathbf{r}_j|.$$

Since $G_{ij}^{(2)}$ depends only on the distance between spins for translational invariance, it depends only on the distance r. We can then define the **connected correlation function**, which measures the correlations between fluctuations in the system:

$$G_c^{(2)}(\mathbf{r}) = \langle (s_i - \langle s_i \rangle)(s_j - \langle s_j \rangle) \rangle = \langle s_i \rangle \langle s_j \rangle - \langle s_i \rangle \langle s_j \rangle.$$

The correlation length ξ is defined as the characteristic length scale over which correlations decay in the system, away from the critical point:

$$G_c^{(2)}(\mathbf{r}) \sim e^{-r/\xi} \quad \text{for } r \gg \xi.$$

Near the critical point, the correlation function typically decays as a power law, which can be described by the critical exponent η. Let us introduce two more critical exponents describing the behavior of the correlation function and the correlation length near the critical point:

- ν: describes the divergence of the correlation length, $\xi_\pm \sim |T - T_c|^{-\nu_\pm}$, with $\nu_\pm > 0$.
- η: describes the behavior of the correlation function at the critical point, $G(r) \sim r^{-(d-2+\eta)}$ at $T = T_c$, where d is the dimensionality of the system.

Remark (Critical Opalescence and Correlation Length). *Going back to the phenomenon of critical opalescence, we can understand it better after having introduced the correlation length. Since the wavelength of visible light is much larger than the microscopic length scales of the system (i.e. atomic distances), the scattering of light must be related to density fluctuations that occur over large distances. Near the critical point, the correlation length diverges, which means that fluctuations occur over increasingly large distances. This leads to a significant scattering of light, giving rise to the milky or cloudy appearance known as critical opalescence. When the correlation length diverges, the system becomes increasingly sensitive to fluctuations, and exhibits macroscopic collective behavior that allow us to ignore the microscopic details of the system and focus on the universal properties that emerge near criticality.*

Now we are ready to define and discuss the concept of **universality classes**. Systems that share the same critical exponents and scaling behavior near the critical point are said to belong to the same universality class. The universality class of a system is determined by factors such as the dimensionality of the system, the symmetry of the order parameter, and the range of interactions between particles.

Definition 2.4 (Universality Class)**.** A universality class is a category of systems that share the same critical exponents and scaling behavior near the critical point, regardless of their microscopic details. Systems that belong to the same universality class exhibit the same critical behavior and can be described by the same effective field theory near the critical point. The universality class of a system is determined by factors such as the dimensionality of the system, the symmetry of the order parameter, and the range of interactions between particles.

Let us now make some observations about the critical behavior of systems near phase transitions, which will be important for our understanding of critical phenomena and the renormalization group.

Remark. *There is a connection between the divergence of the correlation length the response funcion. The connection relies on the fact that near the critical point, the system exhibits long-range correlations, which means that fluctuations occur over increasingly large distances. This leads to a divergence in the correlation length ξ, which in turn causes the response function (such as susceptibility) to diverge as well.*

TODO: *add reference to Kardar - stat ph of fields, 1.4*

The response function measures indeed how the system responds to external perturbations, and its divergence at the critical point indicates that even a small change in the external field can lead to a large change in the order parameter, which is a hallmark of criticality.

TODO: *add reference to J. Cardy - scaling and renormalization in statistical physics, 1.2*

Remark. *Second order phase transitions are also called* ***continuous phase transitions*** *because the order parameter changes continuously across the transition, and there is no latent heat associated with the transition. As we will see, these PTs are associated with the* ***spontanoeus symmetry breaking****. In some cases, the symmetry is only* ***emergent****, meaning that it is not present in the microscopic description of the system but emerges as a result of the collective behavior of the system near the critical point (approximate symmetry which becomes exact only near the critical point). This emergent symmetry can play a crucial role in determining the properties of the system and its critical behavior.*

2.3 | Mean Field Theory

Mean field theory is a powerful approximation method used to study phase transitions and critical phenomena in many-body systems. The basic idea behind mean field theory is to replace the complex interactions between particles in the system with an average or "mean" field that each particle experiences. This simplification allows us to derive self-consistent equations for the order parameter and analyze the behavior of the system near the critical point.

TODO: reference: D. Tong - statistical field theory, 1 | D. Tong - statistical physics, 5

Let us concentrate on a classical model, the Ising model, which is a prototypical model for studying ferromagnetic phase transitions.

2.3.1 | Ising Model

The Ising model consists of a lattice of spins, where each spin s_i can take on values of +1 or -1. The Hamiltonian of the Ising model is given by:

TODO: insert images of spins on a lattice

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i,$$

where J is the coupling constant that determines the strength of the interaction between neighboring spins, h is the external magnetic field, and the sum $\langle i, j \rangle$ runs over pairs of **nearest-neighbor** spins. The first term in the Hamiltonian represents the interaction between neighboring spins, which favors alignment (ferromagnetic order) when $J > 0$. The second term represents the coupling of the spins to the external magnetic field, which tends to align the spins in the direction of the field.

Definition 2.5 (Coordination Number)**.** The coordination number z of a lattice is the number of nearest neighbors that each site has. For example, in a two-dimensional square lattice, each site has 4 nearest neighbors, so the coordination number is $z = 4$. In a three-dimensional cubic lattice, each site has 6 nearest neighbors, so the coordination number is $z = 6$. The coordination number plays an important role in determining the behavior of the system and the nature of the phase transition.

Our aim is to study the equilibrium properties (where observables can effectively be calculated through thermal averaging) of the Ising model and understand the nature of the phase transition it exhibits. To do this, we will apply mean field theory to derive self-consistent equations for the magnetization (order parameter) and analyze the behavior of the system near the critical point.

Each configuration of the system is specified by the set of spin values $\{s_i\}_{i=1}^N$, and it is associated to a probability given by the Boltzmann distribution:

$$P(\{s_i\}) = \frac{1}{Z} e^{-\beta H(\{s_i\})},$$

where Z is the **partition function**, defined as the sum over all possible configurations:

$$Z = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})}.$$

Example (Pair of spins). When we have only two spins, the partition function can be calculated explicitly. The Hamiltonian for two spins s_1 and s_2 is given by:

$$H = -J s_1 s_2 - h(s_1 + s_2),$$

where the sum is removed since we only have two spins. We can now compute the energy for each of the four possible configurations of the spins:

- $s_1 = +1$, $s_2 = +1$: $H = -J - 2h$.
- $s_1 = +1$, $s_2 = -1$: $H = J$.
- $s_1 = -1$, $s_2 = +1$: $H = J$.
- $s_1 = -1$, $s_2 = -1$: $H = -J + 2h$.

The partition function is then given by:

$$Z = e^{\beta J + 2\beta h} + 2e^{-\beta J} + e^{\beta J - 2\beta h}.$$

If we now wanted to compute the probability of finding the system in the configuration $s_1 = +1$, $s_2 = +1$, we would use the Boltzmann distribution:

$$P(s_1 = +1, s_2 = +1) = \frac{e^{\beta J + 2\beta h}}{Z}.$$

For the magnetization, we can compute the average spin per site:

$$M = \mathbb{E}_c[\sum_i s_i], \quad m = \frac{M}{N} = \frac{1}{N}\mathbb{E}_c[s_1 + s_2],$$

where the summations run over the configuration space.

$$m = \sum_{\langle s_i \rangle} p(\{s_i\}) \left(\frac{1}{N} \sum_i s_i \right),$$

where the sum runs over all possible configurations of the spins. In this case, we can explicitly compute the magnetization by summing over the four configurations.

If we were now to consider a system with $N = 100$, we would have 2^{100} possible configurations for a chain with $z = 2$, which is an astronomically large number. This makes it impossible to compute the partition function and the magnetization by explicitly summing over all configurations. This is where mean field theory comes into play, as it allows us to approximate the behavior of the system without needing to consider every possible configuration.

2.3.2 | Mean Field Approach - Euristic

Let us now apply the mean field approximation to the Ising model. The key idea is to replace the interaction term in the Hamiltonian with an average or "mean" field that each spin experiences. This allows us to derive a self-consistent equation for the magnetization.

It is a *euristic* approach, which is not exact but provides a good approximation, remaining not systematic, not rigorous and hard to fully justify. We will see how it can be derived from a more rigorous approach, which is the **Landau-Ginzburg effective field theory**.

There are many versions of this approach (different implementations of the approximation), but the idea is almost the same: we want to replace some microscopic degrees of freedom with an average field, which is the mean field. The mean field is defined as the average value of the spin at a given site:

Let us rewrite the Hamiltonian of the Ising model

$$H = -J \sum_{\langle s_i, s_j \rangle} s_i s_j - h \sum_i s_i.$$

If we now consider the case in which the coupling constant J is positive ($J > 0$ which corresponds to a ferromagnetic interaction), and a null external magnetic field ($h = 0$), the Hamiltonian simplifies to:

$$H = -J \sum_{\langle s_i, s_j \rangle} s_i s_j.$$

In this case the system exhibits a phase transition at a critical temperature T_c, below which the spins tend to align and the system enters an ordered ferromagnetic phase, while above T_c the spins are disordered and the system is in a paramagnetic phase.

Being interested in the behavior of the magnetization, we can express it as a thermal average in the canonical ensemble:

$$\mathbb{E}_c \left[\frac{1}{N} \sum_i s_i \right] = \frac{1}{N} \sum_{\{s_i\}} \left(\sum_i s_i \right) \frac{e^{-\beta H(\{s_i\})}}{Z},$$

where the sum runs over all possible configurations of the spins. The mean field approximation consists in building a term which encodes the average effect of the neighboring spins on a given spin: it is natural to think about building this average field from the magnetization, so let us analyze it in more detail. The magnetization can assume the following values:

$$M \in [-N, \, -N+2, \, -N+4, \, \ldots, \, +N],$$

where the values change by 2 because each spin can flip from +1 to -1 or vice versa. The magnetization per site is then given by:

$$m = \frac{M}{N} \in \left[-1, \, -1 + \frac{2}{N}, \, -1 + \frac{4}{N}, \, \ldots, \, +1 \right].$$

The average field we are building has to be proportional to the magnetization m, which serves as a measure of the degree of order in the system. Thus we can express the sum over configurations of the spins as a sum over the magnetization M and then a sum over all configurations that correspond to that magnetization:

$$\sum_{\{s_i\}_{i=1}^N} [\ldots] = \sum_{M=-N}^{N} \sum_{\{s_i\}_{i=1}^N} \delta \left(\sum_i s_i - M \right) [\ldots],$$

where we have introduced a delta function to enforce the constraint that the sum of the spins equals the magnetization M. This allows us to rewrite the partition function as a sum over the magnetization, but let's understand better the meaning of this delta function through an example.

Example ($N = 2$). If we have only two spins, the possible configurations and their corresponding magnetizations are:

- A: $s_1 = +1$, $s_2 = +1$: $M = 2$.
- B: $s_1 = +1$, $s_2 = -1$: $M = 0$.
- C: $s_1 = -1$, $s_2 = +1$: $M = 0$.
- D: $s_1 = -1$, $s_2 = -1$: $M = -2$.

The delta function $\delta\left(\sum_i s_i - M\right)$ ensures that we only sum over configurations that correspond to a specific magnetization M.

Let us analyze the contribution of each configuration to the partition function: starting from a sum over all configurations, we get

$$Z = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})} = e^{-\beta H(A)} + e^{-\beta H(B)} + e^{-\beta H(C)} + e^{-\beta H(D)},$$

where $H(A)$, $H(B)$, $H(C)$, and $H(D)$ are the Hamiltonian evaluated at each configuration: considering the case $h = 0$ for simplicity, we have

$$H(A) = -J, \quad H(B) = J, \quad H(C) = J, \quad H(D) = -J.$$

Thus the partition function can be rewritten as

$$Z = 2(e^{\beta J} + e^{-\beta J}).$$

We now want to get the same result using the sum over magnetization, which is given by

$$Z = \sum_{M=-2}^{2} \sum_{\{s_i\}} \delta\left(\sum_i s_i - M\right) e^{-\beta H(\{s_i\})} = \sum_{M=-2}^{2} e^{-\beta H(M)} \sum_{\{s_i\}} \delta\left(\sum_i s_i - M\right) = 2(e^{\beta J} + e^{-\beta J}),$$

where we have used the fact that the Hamiltonian depends only on the magnetization M and not on the specific configuration of the spins, and that the sum over configurations for each magnetization value is just the number of configurations with that value of magnetization. This example illustrates how the sum over configurations can be rewritten as a sum over magnetization, with the delta function ensuring that we only consider configurations that correspond to a specific magnetization.

Now we can implement the mean field approximation by replacing the interaction term in the Hamiltonian with an average field that each spin experiences. This average field is proportional to the magnetization m, which serves as a measure of the degree of order in the system: in a fixed magnetization sector (m-sector) we can replace the interaction term with an effective field that depends on the magnetization (reinserting the external magnetic field h):

$$H_{mf}(\{s_i\}, m) = -J \sum_{\langle i,j \rangle} s_i\, m - h \sum_i s_i,$$

where we are summing over single sites (only s_i appears in the sum), but we kept the notation of the sum over nearest neighbors to indicate that the effective field is proportional to the number of nearest neighbors (coordination number z) and the magnetization m. This is the point where we have to accept that this method works even if it is hard to justify: there exist some ad-hoc argument, but it would be meaningless to use those since we will anyway present the Landau-Ginzburg effective field theory in the following sections, which then will make us able to justify this approximation in a systematic way.

Remark. *The hamiltonian H_{mf} is independent of the specific configuration of the spins $\{s_i\}$, as long as*

$$\sum_i s_i = M,$$

So that we have efficiently replaced the interaction term with an average field that depends only on the magnetization m. This allows us to derive a self-consistent equation for the magnetization and analyze the behavior of the system near the critical point without needing to consider every possible configuration of the spins.

If we continue the computation of the mean field hamiltonian we get

$$H_{mf}(\{s_i\}, m) = -Jm \sum_{\langle i,j \rangle} s_i - h \sum_i s_i = -(Jzm + h) \sum_i s_i = -(Jzm + h)Nm,$$

where we have used the fact that the sum over the spins is equal to the magnetization $M = Nm$ and the sum over the nearest neighbors gives a factor of the coordination number $\frac{Nz}{2}$.[2] This mean field Hamiltonian describes a system of non-interacting spins in an *effective magnetic field* that depends on the magnetization m and the external field h.

The average magnetization per spin in the canonical ensamble can now be performed easily applying the mean field approximation:

$$\begin{aligned} \mathbb{E}_c\left[\frac{1}{N}\sum_i s_i\right] &= \sum_{m\in[-1,\dots,+1]} m \sum_{\{s_i\}} \delta\left(\sum_i s_i - Nm\right) \frac{e^{-\beta H_{mf}(m)}}{Z} \\ &= \frac{1}{Z}\sum_m m e^{-\beta H_{mf}(m)} \left[\sum_{\{s_i\}} \delta\left(\sum_i s_i - Nm\right)\right], \end{aligned}$$

where the last term identifieas a sum over all the configurations of the spins that correspond to a given magnetization m. Note that a configuration of $N_\uparrow$ of up spins and $N_\downarrow$ of down spins corresponds to a magnetization

$$m = \frac{N_\uparrow - N_\downarrow}{N} = \frac{2N_\uparrow - N}{N} = 2\frac{N_\uparrow}{N} - 1.$$

The number of such configurations is given by the binomial coefficient:

$$\binom{N}{N_\uparrow} = \frac{N!}{N_\uparrow!(N - N_\uparrow)!} = \Omega(N_\uparrow),$$

which we can approximate using Stirling's formula for large N:

$$\log(\Omega(N_\uparrow)) \approx N\log N - N_\uparrow \log N_\uparrow - (N - N_\uparrow)\log(N - N_\uparrow),$$

which after some algebraic manipulations can be rewritten as

$$\log(\Omega(N_\uparrow)) \approx N\left[\log(2) - \frac{(1+m)}{2}\log(1+m) - \frac{(1-m)}{2}\log(1-m)\right].$$

Thus we have found an expression for the number of configurations that correspond to a given magnetization m, which can be used to compute the final form of the magnetization:

$$\mathbb{E}_c\left[\frac{1}{N}\sum_i s_i\right] = \frac{1}{Z}\sum_m m e^{-\beta N f(m)} = \frac{1}{Z}\sum_m m e^{-\beta H_{mf}(m)}\Omega(m),$$

where we have defined the function $f(m)$ as

$$f(m) = -hm - \frac{Jzm^2}{2} - T\left[\log(2) - \frac{(1+m)}{2}\log(1+m) - \frac{(1-m)}{2}\log(1-m)\right],$$

which can be interpreted as a **free energy density** (per site) that depends on the magnetization m. Indeed we have the last sum over the magnetization m getting dominated by the minimum of

[2]When we count the same site for each of its n.n., we are counting each s_i a number of z times, but we have to divide by 2 to avoid double-counting. Thus the total is $N \cdot \frac{z}{2}$.

the function $f(m)$ in the thermodynamic limit $N \to \infty$, while the other terms get suppressed by the factor $e^{-\beta N f(m)}$.[3] Thus the average magnetization can be approximated as

$$\mathbb{E}_c\left[\frac{1}{N}\sum_i s_i\right] \approx \frac{1}{e^{-\beta N f(m^*)}}\left(m^* e^{-\beta N f(m^*)}\right) = m^*, \qquad \left.\frac{\partial f(m)}{\partial m}\right|_{m^*} = 0,$$

TODO: Add a figure of a parabula with the minimum at m-star.

where m^* is the value of m that minimizes the function $f(m)$. This is known as a *saddle point approximation*, which is a common technique used in statistical mechanics to evaluate integrals or sums that are dominated by the contribution of a single point (the minimum of the free energy in this case). We can use the same approximation to compute the partition function Z as well, which is given by

$$Z = \sum_m e^{-\beta N f(m)} \approx e^{-\beta N f(m^*)}.$$

Self-consistency relation. Now computing the derivative of $f(m)$ with respect to m and setting it to zero, we get the self-consistent equation for the magnetization:

$$\beta(h + Jzm) = \frac{1}{2}\log\left(\frac{1+m}{1-m}\right) = \tanh^{-1}(m),$$

where the last step is obtaines after some algebraic manipulations. This equation can be rewritten as

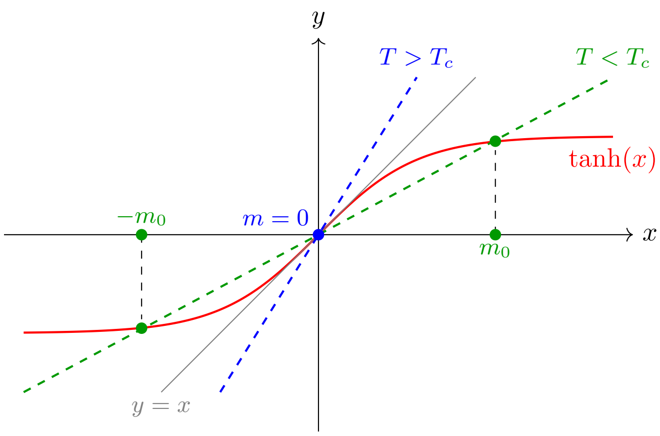
$$m = \tanh\left(\beta h_{eff}\right), \quad h_{eff} = h + Jzm,$$

where h_{eff} is the effective magnetic field that each spin experiences due to the mean field approximation. This self-consistent equation can be solved numerically to find the magnetization m as a function of temperature T and external field h, allowing us to analyze the behavior of the system near the critical point and understand the nature of the phase transition. The effective field h_{eff} captures the influence of the neighboring spins on a given spin, and the self-consistent equation reflects the fact that the magnetization m depends on itself through the effective field. This is a hallmark of mean field theory, where the order parameter (magnetization) is determined by a self-consistent equation that takes into account the average effect of the interactions in the system.

[3] Since we always have to consider an enormously large N (the real world is at the thermodynamic limit), even small variations in $f(m)$ will be exponentially suppressed.

2.4 | Self-Consistency and Phase Transitions

In order to solve the self-consistent equation, we can analyze it graphically by plotting the function m on the left-hand side and the function $\tanh(\beta h_{eff})$ on the right-hand side as functions of m. The points where these two functions intersect correspond to the solutions of the self-consistent equation. By varying the temperature T and the external field h, we can observe how the number and nature of the solutions change, which provides insights into the phase transition and critical behavior of the system. Let us start from the case $h = 0$: **TODO:** fix the plot



The graphical solution shows that for $T > T_c$ the only intersection is at the origin, yielding $m = 0$: this corresponds to the **paramagnetic phase**. For $T < T_c$, instead, the line intersects the curve at two symmetric non-zero points, corresponding to two stable ordered states with opposite magnetization: this is the **ferromagnetic phase**.

- For $T > T_c$, there is only one solution at $m = 0$, which corresponds to the paramagnetic phase where the spins are disordered and there is no net magnetization.
- For $T < T_c$, there are three solutions: one at $m = 0$ and two symmetric solutions at $m = \pm m^*$, which correspond to the ferromagnetic phase where the spins are aligned and there is a non-zero magnetization. The solution at $m = 0$ becomes unstable (not a minimum) in this regime, while the two symmetric solutions represent the stable ordered phases.
- At $T = T_c$, the solution at $m = 0$ becomes marginally stable, and the system undergoes a second-order phase transition from the paramagnetic phase to the ferromagnetic phase. The magnetization m changes continuously across the transition, and there is no latent heat associated with the transition.

The critical temperature T_c is determined by the condition that the slope of the tanh function at $m = 0$ equals 1, which gives

$$\beta_c J z = 1 \quad \Rightarrow \quad T_c = \frac{Jz}{k_B},$$

where k_B is the Boltzmann constant. Thus the condition on the critical temperature can be expressed as $\beta_c J z = 1$.

We can now study the beaviour of the magnetization as a function of temperature. As we decreas the temperature T below T_c, the magnetization m increases continuously from zero, indicating the onset of ferromagnetic order. We got two symmetric solutions for the magnetization, which correspond to the two possible orientations of the spins in the ferromagnetic phase. The magnetization m serves as an order parameter that characterizes the phase transition: it is zero in the paramagnetic phase and becomes non-zero in the ferromagnetic phase, indicating the spontaneous breaking of symmetry. **TODO:** add plot of m vs T

We can obtain the plot of the free energy as a function of the magnetization m for different temperatures to visualize the phase transition. For $T > T_c$, the free energy has a single minimum at $m = 0$, corresponding to the stable paramagnetic phase. As we decrease the temperature below T_c, the free energy develops two symmetric minima at $m = \pm m^*$, corresponding to the stable ferromagnetic phases, while the minimum at $m = 0$ becomes a local maximum, indicating that it is no longer a stable solution. This can be seen studying the expression for the free energy density $f(m)$ that we derived earlier as $h = 0$:

$$f(m) = \ldots$$

TODO: add plot of f(m) vs m for different T

and we get then two different expressions for $T > T_c$ and $T < T_c$:

- For $T > T_c$, the free energy has a single minimum at $m = 0$, which corresponds to the stable paramagnetic phase: we have a parabula with a minimum at the origin, which is the only stable solution in this regime.
- For $T < T_c$, the free energy develops two symmetric minima at $m = \pm m^*$, which correspond to the stable ferromagnetic phases, while the minimum at $m = 0$ becomes a local maximum, indicating that it is no longer a stable solution: we have a double-well potential with two minima at $m = \pm m^*$ and a local maximum at $m = 0$. As the temperature decreases further below T_c, the minima at $m = \pm m^*$ become deeper, indicating that the ferromagnetic phases become more stable.

If we were to plot the free energy as a function of the magnetization m for different temperatures, we would see how the shape of the free energy changes across the phase transition, with the emergence of two minima below T_c and the disappearance of the minimum at $m = 0$. This visualization helps to understand the nature of the phase transition and the role of the magnetization as an order parameter.

If we now imagine to "turn on" the external magnetic field h, we would see that the symmetry between the two minima at $m = \pm m^*$ is broken, and one of the minima becomes deeper than the other, depending on the sign of h. The easier way is to consider an external field which is positive but very small in comparison to the coupling constant J, so that we can treat it as a perturbation.

TODO: add plot of f(m) vs m for different h

- For $h > 0$, the minimum at $m = +m^*$ becomes deeper than the minimum at $m = -m^*$, indicating that the ferromagnetic phase with positive magnetization is more stable than the one with negative magnetization. The system will preferentially align its spins in the direction of the external field, leading to a net positive magnetization.
- For $h < 0$, the minimum at $m = -m^*$ becomes deeper than the minimum at $m = +m^*$, indicating that the ferromagnetic phase with negative magnetization is more stable than the one with positive magnetization. The system will preferentially align its spins in the opposite direction of the external field, leading to a net negative magnetization.

This means that the external magnetic field h breaks the $\mathbb{Z}_2$ symmetry between the two ferromagnetic phases and selects one of them as the stable phase, depending on the sign of h. This is a manifestation of the fact that the external field acts as a symmetry-breaking field, which can influence the behavior of the system near the critical point and affect the nature of the phase transition, especially in the presence of a small external field where the system can exhibit critical behavior and scaling properties.

Remark (On Symmetries and Spontaneous Symmetry Breaking). *A symmetry of a system is a transformation that leaves the Hamiltonian (and so the physical properties and evolution of the system) invariant.*

In the case of the Ising model with $h = 0$, there is a $\mathbb{Z}_2$ symmetry corresponding to the transformation $s_i \to -s_i$ for all spins, which leaves the Hamiltonian unchanged:

$$H(\{s_1, s_2, \ldots, s_n\}) = H(\{-s_1, -s_2, \ldots, -s_n\}).$$

When we flip all spins, is like changing the sign of the magnetization m, but the Hamiltonian remains the same; in the graph of the free energy, this corresponds to the symmetry of the double-well potential around $m = 0$. This symmetry is a fundamental property of the system and plays a crucial role in determining the nature of the phase transition and the behavior of the system near the critical point.

*This symmetry is **spontaneously broken** as $T > T_c$, since ...*

Once the system is in one of the two ferromagnetic phases (with $m = +m^*$ or $m = -m^*$), we could expect the system to be able to transition between these two phases by flipping all the spins, since the states are completely symmetric: same energy, entropy etc. However, in the thermodynamic limit $N \to \infty$, the system becomes trapped in one of the two minima of the free energy, and it cannot transition to the other minimum due to the presence of an infinite energy barrier between them: the transition probability between the two minima approches zero as a power law, and the system remains in one of the two ferromagnetic phases.

2.4.1 | Critical Exponents

The mean field theory allows us to compute the critical exponents that characterize the behavior of the system near the critical point. By analyzing the self-consistent equation for the magnetization and the free energy, we can derive expressions for the critical exponents α, β, γ, and δ in terms of the parameters of the model. The mean field theory predicts specific values for these critical exponents, which can be compared to experimental results and more sophisticated theoretical approaches to understand the limitations and applicability of mean field theory in describing critical phenomena.

We start from the analysis of the **free energy** $f(m)$ and the magnetization m as a function of temperature T near the critical point T_c. We can expand the the free energy in a Taylor series around $m = 0$ to analyze the behavior of the system near the critical point. The expansion takes the form:

TODO: correct the expansion

$$f(m) \sim T\left[\frac{m^2}{2}\left(1 - \frac{T}{T_c}\right) + \frac{m^4}{12}T_c^3 + O(m^6)\right],$$

where the interesting feature is that the coefficient of the m^2 term changes sign at $T = T_c$, which indicates the onset of the phase transition. We can now find an analytic expression for the derivative of the free energy with respect to m and set it to zero to find the equilibrium magnetization m as a function of temperature T:

$$\frac{\partial f}{\partial m} \propto m(T - T_c) + Bm^3 + O(m^5) = 0, \quad B \sim O(1) > 0,$$

which can be solved to find the behavior of the magnetization near the critical point. For $T > T_c$, the only solution is $m = 0$, while for $T < T_c$, we have two symmetric solutions given by

$$m = \pm\sqrt{\frac{T_c - T}{B}} \sim (T_c - T)^{1/2},$$

which indicates that the magnetization m behaves as a power law near the critical point with a critical exponent

$$m \propto (T_c - T)^\beta, \quad \beta = \frac{1}{2}.$$

This critical exponent β characterizes the behavior of the magnetization as the system approaches the critical temperature from below, and it is a key quantity in understanding the nature of the phase transition.

We can now analyze the behavior of the **magnetic susceptibility** χ near the critical point. The magnetic susceptibility is defined as the derivative of the magnetization with respect to the external magnetic field h at $h = 0$:

$$\chi = N \frac{\partial m}{\partial h}\bigg|_{h=0}.$$

Recovering the expression for the magnetization

$$m = \tanh\left(\beta(h + Jzm)\right),$$

we know that near the critical point (approached from above)T_c, the magnetization m is small, so we can expand the tanh function to first order in m:

$$m \approx \beta(h + Jzm).$$

Thus the magnetic susceptibility can be computed as

$$\chi = \frac{N\beta}{1 - \beta Jz},$$

which diverges as T approaches T_c from above, since $\beta Jz = 1$ at $T = T_c$. The critical exponent γ characterizes the divergence of the magnetic susceptibility near the critical point, and it can be defined as

$$\chi \propto |T - T_c|^{-\gamma}, \quad \gamma = 1.$$

These critical exponents helps us to understand the behavior of the system near the critical point and to classify the universality class of the phase transition. Usually when quantities diverge at the critical point, we can characterize their behavior by a power law with a critical exponent, which provides insights into the nature of the fluctuations and correlations in the system as it undergoes a phase transition. Indeed in an experiments, we can measure also **finite size effect**: all of our computations are true in the thermodynamic limit $N \to \infty$, but in a real experiment we have a finite number of particles, so we can observe how the critical behavior is affected by the finite size of the system, which can lead to rounding and shifting of the critical point, as well as modifications to the scaling behavior of physical quantities near the transition. All of these is heavily connected to the concept of **correlation length**, which is a measure of how far correlations between particles extend in the system, and it diverges at the critical point, leading to long-range correlations and critical fluctuations that dominate the behavior of the system near the transition.

How can we concile the mean field predictions for the critical exponents with experimental results? The mean field theory provides is just an approximation, and it is known to give incorrect predictions for the critical exponents in low-dimensional systems (e.g., 2D and 3D), where fluctuations play a significant role (speaking about the Ising model). In higher dimensions (e.g., 4D and above), the mean field theory becomes more accurate, and the critical exponents predicted by mean field theory are closer to experimental results. For Ising

- In 1D, the Ising model does not exhibit a phase transition at finite temperature, and the critical exponents are not defined in the usual sense. The mean field theory incorrectly predicts a phase transition in 1D, which is a clear indication of its limitations in low-dimensionality.
- In 2D, the exact solution of the Ising model gives $\beta = 1/8$ and $\gamma = 7/4$, which are different from the mean field predictions.
- In 3D, numerical simulations and experiments suggest that $\beta \approx 0.326$ and $\gamma \approx 1.237$, which again differ from the mean field values.
- In 4D and above, the critical exponents approach the mean field values of $\beta = 1/2$ and $\gamma = 1$.

Here we can see that the mean field theory provides a qualitative understanding of the phase transition and critical behavior, but it fails to capture the correct critical exponents in low-dimensional systems due to the neglect of fluctuations, which are crucial in determining the behavior near the critical point. More sophisticated approaches, such as renormalization group theory, are needed to accurately describe critical phenomena and compute critical exponents in low-dimensional systems.

We will solve the Ising model in 1D in the next section, and we will give some arguments about the behavior of the model in 2D, which will allow us to understand better the limitations of mean field theory and the importance of fluctuations in low-dimensional systems.

3 | Exact Solution of the Ising Model

[...]

3.1 | Ising Model in One Dimension

The hamiltonian of the 1D Ising model is given by

$$H = -J\sum_{i=1}^{N} s_i s_{i+1} - h\sum_{i=1}^{N} s_i,$$

where we have periodic boundary conditions $s_{N+1} = s_1$. The partition function depends upon the temperature T, the external magnetic field h and the number of spins N, and it can be written as

$$Z_N(\beta, h) = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})}.$$

It is important since we can derive an expression for the ensemble average of the magnetization per site as a function of the partition function:

$$\mathbb{E}_c\left[\frac{1}{N}\sum_{i=1}^{N} s_i\right] = -\frac{1}{N\beta}\frac{\partial \log Z_N}{\partial h}.$$

If we manipulate the expression of the hamiltonian, to express two symmetric sums, we get

$$H = -J\sum_{i=1}^{N} s_i s_{i+1} - \frac{h}{2}\sum_{i=1}^{N} s_i s_{i+1}$$

$$Z = \sum_{\{s_i\}}\prod_{i=1}^{N}(e^{\beta J s_i s_{i+1} + \frac{\beta h}{2} s_i s_{i+1}})\ldots$$

Now we can introduce the **transfer matrix** T, which is a 2×2 matrix that encodes the interactions between neighboring spins and the effect of the external magnetic field. The transfer matrix is defined as

$$T_{s_i, s_{i+1}} = e^{\beta J s_i s_{i+1} + \frac{\beta h}{2}(s_i + s_{i+1})},$$

where s_i and s_{i+1} can take values of ± 1. The partition function can then be expressed as a product of transfer matrices:

$$Z = \sum_{\{s_i\}} T_{s_1,s_2} T_{s_2,s_3}\cdots T_{s_N,s_1} = \mathrm{Tr}(T^N),$$

where the trace is taken over the possible spin states. The transfer matrix method allows us to compute the partition function and other thermodynamic quantities by diagonalizing the transfer matrix and finding its eigenvalues. The largest eigenvalue of the transfer matrix dominates the behavior of the partition function in the thermodynamic limit, and it can be used to derive expressions for the free energy, magnetization, and other physical quantities of interest. This method is particularly powerful for solving one-dimensional models like the 1D Ising model, where it provides an exact solution for the partition function and allows us to analyze the behavior of the system at different temperatures and external fields.

Transfer Matrix

Expressing the partition function in terms of the transfer matrix, we can notice how the computational complexity of the problem is reduced from summing over 2^N configurations to diagonalizing a 2×2 matrix multiplied N times with itself, which is a much more tractable problem. The eigenvalues of the transfer matrix can be computed explicitly, and they determine the thermodynamic properties of the system. We can write the transfer matrix explicitly as

$$T = \begin{pmatrix} e^{\beta J + \beta h} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J - \beta h} \end{pmatrix},$$

with the idea of diagonalizing this matrix to find its eigenvalues λ_1 and λ_2. Let us note that $T = T^\dagger$, thus we know there exist an orthogonal matrix $O = O^T$ such that $T = ODO^T$, where D is a diagonal matrix containing the eigenvalues of T. The diagonal maatrix D can be expressed as

$$D = \begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix},$$

and we can compute the the Nth power of the transfer matrix as

$$T^N = (ODO^T)^N = (ODO^T)(ODO^T)\cdots(ODO^T) = OD^N O^T,$$

where we have used the fact that $O^T O = I$ to simplify the expression. The partition function can then be expressed as

$$Z = \mathrm{Tr}(T^N) = \mathrm{Tr}(OD^N O^T) = \mathrm{Tr}(D^N) = \lambda_+^N + \lambda_-^N,$$

where λ_+ and λ_- are the eigenvalues of the transfer matrix T and we have used the cyclic property of the trace.

Now we can also compute the determinant of the transfer matrix, which is given by

$$\det(T) = \lambda_+ \lambda_- = e^{2\beta J} - e^{-2\beta J} = 2\sinh(2\beta J),$$

while for the trace we had

$$\mathrm{Tr}(T) = \lambda_+ + \lambda_- = e^{\beta J + \beta h} + e^{\beta J - \beta h} + 2e^{-\beta J} = 2e^{\beta J}\cosh(\beta h) + 2e^{-\beta J}.$$

The eigenvalues of the transfer matrix can be computed using the characteristic polynomial, which is given by

$$x^2 - \mathrm{Tr}(T)x + \det(T) = 0,$$

which, inserting the expressions for the trace and determinant, can be rewritten as

$$\ldots$$

and in the end we get the eigenvalues as

$$\lambda_\pm = e^{\beta J}\left(\cosh(\beta h) \pm \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right),$$

which gives

$$|\frac{\lambda_-}{\lambda_+}| < 1, \quad |\frac{\lambda_-}{\lambda_+}|^N \ll 1.$$

thus we can write the partition function in the thermodynamic limit as

$$Z = \lambda_+^N + \lambda_-^N = \lambda_+^N \left(1 + \left(\frac{\lambda_-}{\lambda_+}\right)^N\right) \approx \lambda_+^N,$$

which allows us to compute the free energy per site as

$$f = -\frac{1}{\beta} \lim_{N\to\infty} \frac{1}{N} \log Z,$$

where $f = \lim_{N\to\infty} \frac{F}{N}$; thus we have

$$f = -\frac{1}{\beta} \log \lambda_+ = -J - \frac{1}{\beta} \log \left(\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}} \right).$$

from this we can understand that there are no singularities in the free energy as a function of temperature T and external field h, which indicates that there is no phase transition in the 1D Ising model at finite temperature. This is consistent with the known result that the 1D Ising model does not exhibit a phase transition at finite temperature, and it highlights the importance of dimensionality and fluctuations in determining the behavior of the system near critical points.

3.1.1 | Magnetization

Let us now compute the magnetization per site as a function of temperature T and external field h. We can use the expression for the partition function to compute the magnetization as

$$\mathbb{E}_c\left[s_i\right] = m = -\frac{1}{N} \frac{\partial}{\partial h} \log Z = -\frac{1}{\beta} \frac{\partial}{\partial h} (\frac{F}{N}).$$

Inserting the expression for the free energy per site, we get

$$m = \frac{\sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}}.$$

This expression shows that the magnetization m is a smooth function of temperature T and external field h, and it does not exhibit any singularities or discontinuities, which is consistent with the absence of a phase transition in the 1D Ising model at finite temperature.

TODO: add plot of m vs h for different T.

As T increases, the curve becomes more steep, but it never exihibit a discontinuity or a singularity, which confirms that there is no phase transition in the 1D Ising model at finite temperature. Only in the limit $T \to 0$ we have a discontinuity in the magnetization at $h = 0$, where the system transitions from a fully magnetized state with $m = +1$ for $h > 0$ to a fully magnetized state with $m = -1$ for $h < 0$:

$$\lim_{T\to 0} m(T, h) = \operatorname{sgn}(\sinh(\beta h)) = \operatorname{sgn}(h).$$

We could repeat the computation for χ the magnetic susceptibility, which is given by

$$\chi = \frac{\partial}{\partial h} m(T, h) = \dots$$

and it can be shown that χ does not diverge at any finite temperature, which is consistent with the absence of a phase transition in the 1D Ising model at finite temperature. Only in the limit $T \to 0$ we have a divergence in the magnetic susceptibility at $h = 0$, which indicates a critical point at zero temperature.

The euristic derivation of mean field theory has to have some assumptions which do not hold at low dimensions, and this is the reason why it fails to capture the correct critical behavior in 1D and 2D systems. Only in higher dimensions, where fluctuations are less important, the mean field theory provides a more accurate description of the critical behavior and the critical exponents.

The exact solution of the 1D Ising model using the transfer matrix method allows us to understand the limitations of mean field theory and the importance of fluctuations in determining the behavior of the system near critical points.

3.1.2 | Two Point Correlation Function

It is now time to compute the two-point correlation function for the 1D Ising model: the correlation function is defined as

$$\mathbb{E}_c\left[s_i s_{i+r}\right] = \frac{1}{Z}\sum_{\{s_k\}} s_i s_{i+r} e^{-\beta H(\{s_k\})},$$

which can become a connected correlation function if we subtract the product of the magnetizations:

$$\mathbb{E}_c\left[s_i s_{i+r}\right] - \mathbb{E}_c\left[s_i\right]\mathbb{E}_c\left[s_{i+r}\right] = \frac{1}{Z}\sum_{\{s_k\}} s_i s_{i+r} e^{-\beta H(\{s_k\})} - m^2.$$

To compute the correlation function, we can use the transfer matrix method again. We can express the correlation function in terms of the transfer matrix as follows:

$$\mathbb{E}_c\left[s_i s_{i+r}\right] = \frac{1}{Z}\sum_{\alpha_1=1}^{2}\sum_{\alpha_2=1}^{2}\cdots\sum_{\alpha_N=1}^{2} \sigma_{\alpha_j}\sigma_{\alpha_{j+r}} T_{\alpha_1,\alpha_2} T_{\alpha_2,\alpha_3}\cdots T_{\alpha_N,\alpha_1},$$

where σ_{α_j} is the spin value corresponding to the state α_j (i.e., $\sigma_1 = +1$ and $\sigma_2 = -1$). We can rewrite this expression in a more compact form using matrix notation. Let us define a diagonal matrix D such that

$$D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \text{ and } D_{\alpha,\beta} = \delta_{\alpha,\beta}\cdot\sigma_\alpha,$$

meaning that D is a diagonal matrix with entries corresponding to the spin values. Then we can express the correlation function as

$$\mathbb{E}_c\left[s_j s_{j+r}\right] = \frac{1}{Z}\sum_{\alpha_1=1}^{2}\sum_{\alpha_2=1}^{2}\cdots\sum_{\alpha_N=1}^{2} T_{\alpha_1,\alpha_2} T_{\alpha_2,\alpha_3}\cdots T_{\alpha_{j-1},\alpha_j} D_{\alpha_j,\beta_1} T_{\beta_1,\alpha_{j+1}}\cdots T_{\alpha_{j+r-1},\alpha_{j+r}} D_{\alpha_{j+r},\beta_2} T_{\beta_2,\alpha_{j+r+1}}\cdots T_{\alpha_N,\alpha_1},$$

where we have inserted the diagonal matrices D at the positions corresponding to the spins s_j and s_{j+r}. This expression can be further simplified by recognizing that the sums over the indices α_k correspond to matrix multiplications, and we can express

$$\mathbb{E}_c\left[s_j s_{j+r}\right] = \frac{1}{Z}\mathrm{Tr}(T^{j-1} D T^r D T^{N-r+1-j}),$$

where we have used the cyclic property of the trace to rearrange the order of the matrices. This expression allows us to compute the two-point correlation function in terms of the transfer matrix and the diagonal matrix D, which encodes the spin values. By diagonalizing the transfer matrix with a unitary matrix U:

$$T = U\begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix}U^\dagger,$$

where U can be computed explicitly, we can express the correlation function in terms of the eigenvalues and eigenvectors of the transfer matrix, which allows us to analyze its behavior as a function of the distance r and the temperature T. Let us focus on the case without external field: the unitary transformation becomes

$$\cdots$$

and so we can write the transfer matrix in the diagonal basis as

$$\cdots$$

Now putting everything together, we can express the correlation function as

$$\cdots$$

In the end we find a close analytical form for the correlation function, which can be expressed as

$$\mathbb{E}_c\left[s_j s_{j+r}\right]_{h=0} = \frac{\lambda_+^{N-r}\lambda_-^r + \lambda_-^{N-r}\lambda_+^r}{\lambda_+^N + \lambda_-^N},$$

and again if we use tha fact that $|\frac{\lambda_-}{\lambda_+}| < 1$, we can simplify this expression in the thermodynamic limit as

$$\mathbb{E}_c\left[s_j s_{j+r}\right]_{h=0} = \cdots \sim e^{-\frac{r}{\xi}},$$

effectively defining the correlation length ξ as

$$\xi^{-1} = \log\left(\frac{\lambda_+}{\lambda_-}\right) > 0,$$

which is a measure of how far correlations between spins extend in the system. Now considering the absence of an external field, we can express the eigenvalues of the transfer matrix as ...

and in the end we get the correlation length in the $T \to 0$ limit without external field as

$$\xi \sim e^{2\beta J},$$

so that it diveres exponentially as $T \to 0$, which indicates that the system exhibits long-range correlations at low temperatures, and it confirms the presence of a critical point at zero temperature in the 1D Ising model. This behavior of the correlation length is consistent with the absence of a phase transition at finite temperature, as the correlation length remains finite for all $T > 0$ and only diverges as $T \to 0$.

The correlation lenght correspond to the distance after which we can consider the spins as independent, since the correlation function decays exponentially with distance r as $e^{-r/\xi}$. As the temperature approaches zero, the correlation length diverges, indicating that the spins become correlated over longer distances, which is a signature of critical behavior at zero temperature. This analysis of the two-point correlation function and the correlation length provides insights into the nature of correlations in the 1D Ising model and helps to understand the absence of a phase transition at finite temperature. But since this divergence occur at $T = 0$ then everything is in accord to the fact that there is no phase transition at finite temperature, and the critical point is located at zero temperature.

3.2 | Ising Model in Two Dimensions

We will not solve exactly the 2D Ising model, but we will give some arguments about its behavior and the nature of the phase transition it exhibits, and then in the end we will present the proof of the existence of a phase transition in the 2D Ising model at finite temperature, which is a non-trivial result that highlights the importance of dimensionality and fluctuations in determining the behavior of the system near critical points.

3.2.1 | Phase Transition - Euristic

Consider the 2D Ising model on a square lattice, where each spin interacts with its four nearest neighbors. Take the limit for a very low temperature $T \to 0$, and suppose that there is a region in the first 1D raw where the system is in a fully magnetized state, with all l spins aligned (e.g., $m = +1$). We want to ask ourselves if this state is stable against the formation of a droplet of flipped spins (e.g., $m = -1$) of size $l \times l$. The energy cost of creating such a droplet can be estimated as follows:

- Case A: $\uparrow\uparrow \dots \uparrow$, where we have a single domain wall of length l separating the two regions of opposite magnetization. The energy cost of creating this domain wall is proportional to its length, which gives
$$E = -Jl,$$
where J is the coupling constant. The energy cost is negative, which means that it is energetically favorable to create such a droplet, and the system will tend to form domains of flipped spins, leading to a disordered state with zero magnetization.

- Case B: $\uparrow \dots \uparrow\downarrow \dots \downarrow$, where we have two domain walls of length l separating the three regions of magnetization. The energy cost of creating these two domain walls is proportional to their total length, which gives
$$E = -Jl + 2Jl = Jl,$$
where the first term corresponds to the energy cost of creating the first domain wall, and the second term corresponds to the energy cost of creating the second domain wall. The energy cost is positive, which means that it is energetically unfavorable to create such a droplet.

Remark (entropy of domains). *...*

If we now compute the free energy cost of creating such a droplet, we need to take into account both the energy cost and the entropy gain associated with the formation of the droplet. The free energy cost can be expressed as

$$F_B - F_A = \Delta E - T\Delta S = 2Jl - Tk_B \log(l-1),$$

since the energy cost of creating the droplet in case B is $2Jl$ and the entropy gain is proportional to the logarithm of the number of ways to arrange the droplet, which is approximately $\log(l-1)$. As l increases, the energy cost dominates over the entropy gain, and the free energy cost becomes positive, which means that it is energetically unfavorable to create such a droplet. This indicates that the system will tend to remain in a magnetized state with non-zero magnetization, and it suggests that there is a phase transition at a finite temperature T_c below which the system exhibits spontaneous magnetization.

This argument provides an intuitive understanding of why the 2D Ising model exhibits a phase transition at finite temperature, and it highlights the importance of dimensionality and fluctuations in determining the behavior of the system near critical points.

It seems that breaking a domain wall is energetically unfavorable, but there is a maximum value of l for which the entropy does not make unfavorable the creation of the droplet, and this maximum value of l can be estimated by setting the free energy cost to zero:

$$\Delta F = 0 \implies 2Jl - Tk_B \log(l-1) = 0 \implies l \sim e^{\frac{2J}{Tk_B}} = \xi,$$

which indicates that there is a characteristic length scale l above which the domain wall are not favorable anymore, and this length scale can be identified with the correlation length ξ of the system. As the temperature approaches the critical temperature T_c from below, the correlation length diverges, indicating that the system exhibits long-range correlations and critical behavior at the phase transition.

Euristic arguments in statistical physics can provide valuable insights into the behavior of complex systems and help to understand the underlying mechanisms driving phase transitions and critical phenomena. Sometimes, when a problem is mathematically intractable, we can use physical intuition and qualitative reasoning to make predictions about the behavior of the system, which can then be used to address the problem in new ways and in the end find the analutical solution.

3.2.2 | Peierls's argument - Euristic

TODO: add picture of the two configurations

Let us compare the free energy of the two configurations:

$$\dots$$

We ask what is the difference in energy of the two 2D configurations described. Each set of nearest neighbours of the subdomain considered in the configuration contributes to the change in energy from one configuration to the other. In particular

$$\Delta U = U_B - U_A = 2Jl,$$

where now in 2D l represents the lenght of the perimeter of the flipped subdomain, which is proportional to the number of nearest neighbor pairs that are affected by the change in configuration. The energy cost of creating a droplet of flipped spins is proportional to the length of the domain wall, which is $2Jl$ in this case. This energy cost is positive, which means that it is energetically unfavorable to create such a droplet, and it suggests that the system will tend to remain in a magnetized state with non-zero magnetization, indicating the presence of a phase transition at finite temperature.

But we have always to take into account the entropy gain associated with the formation of the droplet, which can be estimated after counting the number of ways to arrange the droplet of flipped spins. The number of ways to arrange a droplet of size l can be approximated as the path we can take to create the domain wall, which is given by three choices of direction at each step (since we cannot go back),[1] thus we can estimate the number of configurations as

$$\#DM \sim 3^l,$$

thus if the number of domains wall of length l is given by $\#DM$, then the entropy gain associated with the formation of the droplet can be estimated as

$$\Delta S = k_B \log(\#DM) = k_B l \log(3),$$

which is proportional to the length of the domain wall.

Thus in the end, the free energy cost of creating such a droplet can then be expressed as

$$\Delta F = \Delta U - T\Delta S = 2Jl - Tk_B l \log(3) = l(2J - Tk_B \log(3)),$$

which indicates that there is a critical temperature T_c at finite temperature, at which we observe a phase transition.

3.2.3 | Rigorous Proof of Phase Transition

TODO: Add reference to 14.3 Huang's book stat mech.

Let us consider a system with **fixed open boundary conditions**, where we have an Ising model defined on a square lattice of size $N \times N$, and we fix the spins on the boundary to be all up (i.e., $s_i = +1$ for all spins on the boundary). We want to show that there is a phase transition at finite temperature, which can be done by comparing the free energy of two configurations: one with all spins up (the ordered phase) and one with a droplet of flipped spins (the disordered phase).

The hamiltonian of the system is unchanged, and it is given by

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i,$$

where the first sum is taken over nearest neighbor pairs of spins, and the second sum is taken over all spins in the lattice.

We can fix the temperature and compute the partition function and magnetization for both configurations. Then in the end we will find: if we denote by N_- the average number of down spins, we will have only two cases:

- $\lim_{N\to\infty} \frac{N_-}{N} = \frac{1}{2}$: in this case the system is in a disordered phase with zero magnetization, since the number of up spins and down spins are equal in the thermodynamic limit.
- $\lim_{N\to\infty} \frac{N_-}{N} < \frac{1}{2}$: in this case the system is in an ordered phase with non-zero magnetization, since the number of up spins is greater than the number of down spins in the thermodynamic limit.

We used a preferred direction setting the boundary conditions, but we could have done the same analysis with all spins down at the boundary, and we would have found the same result. Exact numbers will fluctuate, but these ratios will converge to the same limit, and in the TD limit we will have a well defined phase transition at finite temperature.

[1] Extimating like this is exagerated, since we are counting also “snakes” that do not close on a region, but let us do this anyay and then reason on it.

4 | Landau-Ginzburg Effective Field Theory

4.1 | Mesoscopic Description of Phase Transitions

4.1.1 | Partition Function

4.2 | Principles and Free Energy Functional

- **Locality and uniformity**:

- **Analiricity**:

- **Symmetry**:

4.2.1 | Application on One Dimensional Ising Model

4.3 | Saddle Point Approximation

4.3.1 | Lower and UpperCritical Dimension

4.3.2 | Non Uniform Boundary Conditions: Domain Walls

4.4 | Fluctuations

4.4.1 | Heat Capacity

4.4.2 | Correlation Functions

4.4.3 | The Ginzburg Criterion

4.5 | The Scaling Hypothesis

Discussing the realations between critical exponents is a foundamental step in the study and comprehension of critical phenomena. The scaling hypothesis is a powerful tool that allows us to derive these relations and understand the behavior of physical quantities near the critical point.

...

$$f(t,h) = \text{min}_m \left[\frac{t}{2}m^2 + um^4 - h \cdot m\right] = \begin{cases} \dots \\ \dots \end{cases}$$

...

Nom taking a generalized homogeneous function of two variables $f(x, y)$ such that:

$$f(\lambda^{d_1}x, \lambda^{d_2}y) = \lambda f(x, y),$$

if we set

$$\lambda = \left(\frac{1}{x}\right)^{\frac{1}{d_1}},$$

we can notice how the function f can be rewritten as

$$f(1, \frac{1}{x^{d_2/d_1}}y) = \left(\frac{1}{x}\right)^{\frac{1}{d_1}} f(x, y).$$

Now if we define $g(z) = f(1, z)$, we can invert the relation above to obtain

$$f(x, y) = x^{d_1} g\left(\frac{y}{x^{d_2/d_1}}\right),$$

which therefore shows how the function f can be expressed in terms of a single variable, which is the ratio $\frac{y}{x^\Delta}$ (where $\Delta = \frac{d_2}{d_1}$) of the two original variables, up to an overall prefactor x^{d_1}. This is the essence of the scaling hypothesis, which states that near the critical point, physical quantities can be expressed as homogeneous functions of the relevant variables, leading to scaling relations between critical exponents; we will see this more explicitly in the following sections.

If we assume that the true free energy density f is a generalized homogeneous function of the reduced temperature $|t|$ and the external field h when we study it close to the critical point T_c, we can write

$$f(|t|, h) = |t|^{d_1} g\left(\frac{h}{|t|^{\Delta}}\right),$$

for some d_1, Δ and g (notice how we are now treaing d_1 and Δ as indipendent parameters). Knowing the singularities of the free energy density $f(t, h)$, we can fix the values of d_1 and Δ and therefore derive the scaling relations between critical exponents.

For $h = 0$, we must have

$$f(t, 0) \sim \frac{|t|^2}{u}, \qquad \Longrightarrow \qquad d_1 = 2, \quad \lim_{x \to 0} g(x) \sim \frac{1}{u}.$$

Furthermore, for $t \to 0$, we must have

$$\lim_{t \to 0} f(t, h) \sim \frac{h^{4/3}}{u^{1/3}},$$

which can be verified by setting

$$\lim_{x \to \infty} g(x) \sim \frac{x^{4/3}}{u^{1/3}}.$$

Now, using the last relation, we can fix the value of Δ as follows:

$$\lim_{t \to 0} f(t, h) \sim |t|^2 \frac{h^{4/3}}{u^{1/3}\, t^{4\Delta/3}}, \qquad \Longrightarrow \qquad \Delta = \frac{3}{2},$$

concluding that the homogeneity assumption in the free energy density f is consistent with the singularities predicted by the saddle point approximation, with

$$d_1 = 2, \quad \Delta = \frac{3}{2}.$$

Next we are going to make the same assumptions for the true free energy density f and see how we can derive the scaling relations between critical exponents from it.

TODO: there were some errors in the notes, check them and fix them

We assume that the true singular part of the free energy density f is a generalized homogeneous function of the reduced temperature $|t|$ and the external field h when we study it close to the critical point T_c, we can write

$$f(|t|, h) = |t|^{d} g\left(\frac{h}{|t|^{\Delta}}\right),$$

for some d, Δ and g. This does not imply the free energy of the mean field theory, which is a particular approximation of the true free energy (which can even have this form in general, but with different exponents), but it is a more general assumption that we make on the true free energy itself. Let us now find a way to relate the exponents d and Δ to the critical exponents.

We start by relating d with the critical exponent α that describes the singularity of the heat capacity C at the critical point. We know that

$$C = -T \frac{\partial^2 f}{\partial T^2},$$

so that we have to compute the second derivative of the free energy density f with respect to the temperature T. Computing the first derivative of f with respect to T:

$$\begin{aligned} \frac{\partial f}{\partial T} &\sim (2 - \alpha) \dots \\ &=: |t|^{1-\alpha} \tilde{g}\left(\frac{h}{|t|^{\Delta}}\right), \end{aligned}$$

since in the previous parrenthesis h appeared always divided by $|t|^{\Delta}$.

Remark. *More generally, partial derivative of an homogeneus generalized function will again be an homogeneus generalized function, with different prefactors and a different function $\tilde{g}$ (which is related to the original function g by differentiation).*

Repeating the same procedure to compute the second derivative of f with respect to T, we obtain

$$C \propto -\frac{\partial^2}{\partial t^2} f \sim |t|^{-\alpha} \tilde{\tilde{g}} \left(\frac{h}{|t|^\Delta} \right),$$

where $\tilde{\tilde{g}}$ is another appropritely defined function. Therefore, we can conclude that the critical exponent α is related to the exponent d by the following relation:

$$\lim_{x \to 0} \tilde{\tilde{g}}(x) = \text{const} \implies \alpha = 2 - d,$$

and thus we found the relation between the critical exponent α and the exponent d that appears in the homogeneity assumption of the free energy density f: $d = 2 - \alpha$. The logic behind this is that we need to now how this function behaves at the critical point, where it could go to 0, diverge to infinity or go to a constant; since

$$\lim_{t \to 0} C(t, h = 0) \sim |t|^{-\alpha},$$

then we need to have $\lim_{x \to 0} \tilde{\tilde{g}}(x) = \text{const}$ in order to have the correct singularity of the heat capacity C at the critical point.

Now the only unknown parameter is Δ, which we call the **gap exponent**: it is related to the critical exponent β that describes the singularity of the order parameter m at the critical point. We are saying that it is not important the theory in which we compute the critical exponent β (mean field, Landau-Ginzburg, etc), but it is important that the true free energy density f has the homogeneity property that we assumed above, with some value of Δ. Thus let us relate Δ to β by computing the order parameter m as a function of the reduced temperature t at zero external field $h = 0$: remember that

$$m(t, h = 0) \sim |t|^\beta,$$

then we can compute m by taking the derivative of the free energy density f with respect to the external field h:

$$m = -\frac{\partial f}{\partial h} \sim |t|^{d-\Delta} g' \left(\frac{h}{|t|^\Delta} \right),$$

and therefore we can conclude that the critical exponent β is related to the gap exponent Δ by the following relation:

$$\lim_{x \to 0} g'(x) = \text{const} \implies \beta = d - \Delta,$$

which is the second scaling relation that we were looking for

$$\beta = d - \Delta = 2 - \alpha - \Delta.$$

now we can consider the critical exponent δ, which describes the singularity of the order parameter m at the critical point as a function of the external field h:

$$m(t = 0, h) \sim h^{1/\delta},$$

then we can assume that the function g behaves as

$$\lim_{x \to \infty} g(x) \sim x^{p+1}, \quad \implies \quad \lim_{x \to \infty} g'(x) \sim x^p,$$

which is the only reasonable behavior that we can assume for the function g in order to have a power law singularity of the order parameter m at the critical point (indeed we need $\lim_{x\to\infty} g'(x) =$ const). Therefore, we can compute the order parameter m at the critical point $t = 0$ as a function of the external field h:

$$\lim_{t\to 0} m(t,h) \sim |t|^{2-\alpha-\delta} \left(\frac{h}{|p|^{\Delta}}\right)^{p},$$

which is well-defined only if $p\Delta = 2 - \alpha - \delta$ leading us to

$$\lim_{t\to 0} m(t,h) \sim h^{\frac{2-\alpha-\delta}{\Delta}},$$

which recalling the definition of the critical exponent δ leads us to the following scaling relation:

$$\delta = \frac{\Delta}{2-\alpha-\Delta} = \frac{\Delta}{\beta}.$$

For the susceptibility χ, we can use the same arguments and procedures, computing it as the derivative of the magnetization m with respect to the external field h:

$$\chi(t,h) \propto \frac{\partial m}{\partial h} \sim |t|^{2-\alpha-2\Delta} g'\left(\frac{h}{|t|^{\Delta}}\right),$$

[...]

obtaining

$$\gamma = 2\Delta + \alpha - 2.$$

After these computations, we have found the following scaling relations between critical exponents:

$$\begin{cases} \alpha = 2 - d, \\ \beta = d - \Delta, \\ \delta = \dfrac{\Delta}{2-\alpha-\Delta}, \\ \gamma = 2\Delta + \alpha - 2, \end{cases}$$

showing how the gap exponent Δ appear in all the expressions for the critical exponents; now from these relations we can derive the following identities:

$$\begin{cases} \alpha + 2\beta + \gamma = 2, \text{ Rushbrooke's identity,} \\ \delta - 1 = \dfrac{\gamma}{\beta}, \text{ Widom's identity,} \end{cases}$$

[...]

$$\delta - 1 = \frac{\Delta}{2-\alpha-\Delta} - 1,$$

$$\frac{\gamma}{\beta} = \frac{\gamma\delta}{\Delta} = \frac{2\Delta+\alpha-2}{\Delta}\left(\frac{\Delta}{2-\alpha-\Delta}\right),$$

[...]

These relations can be tested experimentally and numerically, and they are indeed satisfied with good precision, despite the fact that the individual exponents are different than the one predicted by the mean field theory.

[*values of exponents*]

So it seems that even in systems with very different exponents, the scaling relations are still satisfied, which is a strong evidence in favor of the scaling hypothesis and the homogeneity assumption that we made on the true free energy density f.

4.5.1 | Correlation Functions

The homogeneity assumption do not apply only to the free energy density f, but also to other physical quantities derived from it. But it says nothing about the correlation functions. In order to obtain identities related to exponents of the correlation functions, we need to make two additional assumptions which will seem to be more foundamental and more general than the homogeneity assumption on the free energy density f:

- The correlation length ξ is a generalized homogeneous function of the reduced temperature $|t|$ and the external field h when we study it close to the critical point T_c:

$$\xi(t,h) = |t|^{-\nu} g\left(\frac{h}{|t|^{\Delta}}\right).$$

- Close to criticality, ξ is the most important lenght scale in the system and is solely responsible for the singular contributions to thermodynamic quantities.

[...]

4.5.2 | Anomalous Dimension

[...]

5 Renormalization Group

The renormalization group (RG) is a powerful theoretical framework that allows us to understand the behavior of physical systems at different scales, and in particular to analyze critical phenomena and phase transitions. It provides a unified perspective on the many facets of critical phenomena.

The name "renormalization group" is rather misleading: it is a framework in which one applies transformations on a system, but such transformations do not form a group in the mathematical sense (since they do not have an inverse). However, the name has been kept for historical reasons, since it was first introduced in the context of quantum field theory, where it was used to understand the behavior of quantum fields at different energy scales; now it is used to refer to a more general framework that can be applied to a wide range of systems, and it is associated with a wide set of ideas and techniques that are used to understand critical phenomena in statistical mechanics.

TODO: Add reference to the John Cardy book, chapter 3.

The renormalization group is based upon the observation that critical phenomena are dominated by fluctuations at a scale ξ which diverges at the critical point, and thus the system is effectively scale-invariant at criticality. The idea is then to iteratively modify our description of the system, eliminating details about the short range correlations at each step, and thus to understand how the system behaves at different scales. In this way, one is eventually left with a simplified description of the system that captures the essential features of the critical behavior through correlations at large scales.

This iterative procedure is called a **coarse-graining transformation**, and it allows us to understand how the parameters of the system change as we look at it at different scales. It is indeed equivalent to a flow in the space of parameters (space of all possible free energies or actions), which is called the **action space**, and the fixed points of this flow correspond to critical points of phase transitions.

The RG flow allows us to understand the behavior of the system near criticality, and in particular to identify the relevant and irrelevant directions in the parameter space, which are associated with the critical exponents that describe the behavior of physical quantities near the critical point.

The conceptual framework derived from the RG analysis is very general, and can be implemented in different forms and with different techniques; traditionally, the main distinction is between the **real space RG** and the **momentum space RG**:

- The first one is very useful to introduce the framework, as it involves a more intuitive approach.

- The latter provides more powerful tools for quantitative computations: it encodes a systematic approach to derive critical exponents below the upper critical dimension for istance.

However, as we said, real space RG is more intuitive and thus we will start by using this approach to revisit the 1D Ising model, which we have already solved, in order to understand how the RG works in a simple case, and then we will introduce the more general framework that can be applied to more complicated systems.

5.1 | Coarse-Graining in Real Space

[...]

Starting from the Hamiltonian

$$\ldots$$

and having introduced the coefficients in such a way that they respect the following identity:

$$\ldots$$

It is convenient to parametrize as

$$\begin{cases} x = e^k, \\ y = e^h, \\ z = e^g, \end{cases} \quad \begin{cases} x' = e^{k'}, \\ y' = e^{h'}, \\ z' = e^{g'}, \end{cases}$$

then we have four choices for $(s_1, s_2) = (\pm 1, \pm 1)$:

$$\begin{aligned}
(+1, +1) \quad &\implies \quad x'y'z' = z^2 y(x^2 y + x^2 y^{-1}), \\
(+1, -1) \quad &\implies \quad (x')^{-1} z' = z^2 (y + y^{-1}), \\
(-1, +1) \quad &\implies \quad (x')^{-1} z' = z^2 (y + y^{-1}), \\
(-1, -1) \quad &\implies \quad x'(y')^{-1} z' = z^2 y^{-1}(x^{-2} y + x^2 y^{-1}).
\end{aligned}$$

With simple algebra we can find the solution

$$\begin{cases} (z')^4 = \ldots \\ (y')^2 = \ldots \\ (x')^4 = \ldots \end{cases}$$

which provides, after taking the logarithm, the RG flow equations for the parameters k and h:

$$\begin{cases} g' = \ldots \\ h' = \ldots \\ k' = \ldots \end{cases}$$

where δg, δh and $k'(k, h)$ are implicitly defined by the above equations. This flow equations describe how the parameters of the Hamiltonian change in the **action space** (or *free energy space*) as we perform a **coarse-graining transformation**. The fixed points of this flow correspond to scale-invariant theories, which are often associated with critical points of phase transitions.

Remark. *When we write the Hamiltonian, if we assume directly that $N \to \infty$, the system has an infinite number of degrees of freedom, and the RG transformation is from an infinite-dimensional space to itself, which is not very useful. Instead, we can consider a finite system with N sites, and then take the limit $N \to \infty$ at the end of the calculation. We are speaking about a space of parameters, not a space of configurations, so the RG transformation is from a finite-dimensional space to itself, which is more manageable.*

TODO: *check if this is correct*

5.2 | Recursion Relations and Flow Diagrams

The previous set of equations is also called the **RG recursion relations**

[...]

In general one can proceed as follows:

1. **Coarse-graining** or “thinning” of the degrees of freedom: ...
2. **Lenght rescaling**: ...
3. Derivation of new **renormalized Hamiltonian**: ...

These steps...

Now let us set $h = 0$, which implies $y = 1$, and see how the RG flow looks like in the (k, h) plane. The flow equations give us

$$y' = 1 \implies h' = 0,$$

and we can write

$$e^{4k'} = \cdots \qquad \implies \qquad k' = \frac{1}{2} \ln\left(\cosh(2k)\right).$$

This recursion relation has two fixed points: $k = 0$ and $k = \infty$. The fixed points are the only candidates for critical points (where the system can undergo a phase transition), since they are the only points where the system is scale-invariant. In this case, the only critical point is at $k = 0$, which corresponds to the infinite temperature limit, where the system is disordered. The other fixed point at $k = \infty$ corresponds to the zero temperature limit, where the system is ordered.

The first one is unstable, while the second one is stable:

- At $k = 0$, we have $\beta = 0$, and thus an infinite temperature $T \to \infty$: it is a **stable** fixed point. If we perturb it slightly, for a small k, we have

  $$k' = \frac{1}{2} \ln\left(\cosh(2k)\right) \approx \frac{1}{2} \ln\left(1 + \tfrac{(2k)^2}{2}\right) \approx k^2 < k,$$

  which means that the flow is towards $k = 0$.

- At $k = \infty$, we have $\beta \to \infty$, and thus a zero temperature $T \to 0$: it is an **unstable** fixed point. If we perturb it slightly, for a large k, we have

  $$k' = \frac{1}{2} \ln\left(\cosh(2k)\right) \approx \frac{1}{2} \ln\left(\frac{2^{2k}}{2}\right) = k - \ln 2 < k,$$

  which means that the flow is towards $k = 0$ and an infinite temperature.

TODO: add drawing of flows on k and T line

[*Drawing of flows on k and T line*]

Thus the initial problem $\mathcal{Z}(\beta, h = 0)$ is recursively mapped onto new ones $\mathcal{Z}'(\beta', h = 0)$ and $\mathcal{Z}''(\beta'', h = 0)$, which by construction have to be numerically equal to the original one, but with different parameters. The flow of the parameters under the RG transformation allows us to understand the behavior of the system at different scales, and in particular to identify the critical points and the nature of the phase transitions. In this case, we see that there is no phase transition at finite temperature, since the only critical point is at $T = \infty$, which is not physically relevant.

[...]

Going to $h = 0$ we can also derive a **flow diagram**...

[*Drawing of flow diagram*]

TODO: add drawing of flow diagram

[...]

Thus if we are close to a fixed point, an RG step will move us away, but the ...

5.2.1 | Linearization and Scaling Form

Thus we can linearze the flow equations around the fixed point. Let us consider in particular $T \sim 0$ and $h \sim 0$, corresponding to $y \sim 1$ and $x \to \infty$. We can write

$$\begin{cases} (x')^4 = \dfrac{x^4}{4}, \\ (y')^2 = y^4, \end{cases} \text{and} \begin{cases} e^{-k'} = \sqrt{2}e^{-k}, \\ h' = 2h. \end{cases}$$

After each RG iteration, the correlation length gets divided by two, and therefore, we can write ξ as a function of e^{-k} and h:

$$\xi = \xi[k, h] \to \dots$$

where the initial form will be

$$\frac{1}{2}\xi(e^{-k}, h) = \xi(\sqrt{2}e^{-k}, 2h),$$

which after ℓ RG iterations becomes

$$\dots$$

... the **scaling form**...

$$\dots$$

This is the most non trivial result of the RG analysis: we are rewriting a new hamiltonian with new parameters, but the partition function is the same, and thus the correlation length is the same. This allows us to derive the scaling form of the correlation length.

Now for $h \to 0$ we have

$$\xi(e^{-k}, 0) \sim e^{2k} \sim e^{2\beta J} \sim e^{\frac{2J}{T}},$$

which is the well-known result for the correlation length of the 1D Ising model at zero magnetic field. This shows how the RG analysis allows us to derive the scaling behavior of physical quantities near critical points, and in this case, it confirms that there is no phase transition at finite temperature, since the correlation length diverges only as $T \to 0$. We had to postulate the scaling form of the correlation length, but now we have an hint the RG ideas are adequate to derive the scaling form of physical quantities near critical points, and in particular to understand the behavior of the correlation length as a function of the parameters of the system.

5.3 | Conceptual Framework

[...]

1. **Coarse-Grain**: ...
$$m'(x) = \frac{1}{b^D}\int_{|y|<b} m(x')dx',$$
2. **Prescale**: ...
3. **Renormalize**: By the way we define m we want to preserve the originale domain for $m(x')$; we need to divide by some scale factor
$$m \to \frac{m}{\zeta},$$
which is equivalent to a variable change
$$m(\mathbf{x}) \xrightarrow{\text{RG step}} m_{\text{new}}(\mathbf{x}_{\text{new}}) = \frac{1}{\zeta b^D}\int_{\mathcal{C}} \mathrm{d}^D\mathbf{x}' m(\mathbf{x}'),$$
where $\mathcal{C}$ is again the cell centered around $b\mathbf{x}_{\text{new}}$.

This reflect on the partition function as
$$\mathcal{Z} \mapsto \mathcal{Z} = \int \mathcal{D}m_{\text{new}} e^{-F_{\text{new}}[m_{\text{new}}(\mathbf{x}_{\text{new}})]},$$
where the functional form of the free energy is preserved, but the parameters are changed. The RG transformation is thus a mapping in the space of parameters, which allows us to understand how the system behaves at different scales.

[...]

We assume that close to the critical temperature T_c we can neglect all the parameters of F except for the most relevant ones, which are t and h:
$$F[m(\mathbf{x})] = F[m(\mathbf{x}); t, h],$$
where t is the reduced temperature defined as $t = \frac{T-T_c}{T_c}$. The RG transformation will then give us the flow equations for t and h, which will allow us to understand the behavior of the system near the critical point.

Thus if we are close to criticality, when applying an RG transformation we will get
$$\implies F_{\text{new}}[m_{\text{new}}(\mathbf{x}_{\text{new}}); t_{\text{new}}, h_{\text{new}}] = F[m_{\text{new}}(\mathbf{x}_{\text{new}}); t_{\text{new}}, h_{\text{new}}],$$
where the functional form of the free energy is preserved, but the parameters t and h are changed to t_{new} and h_{new}. This again translates into a transfomation of the field ...
$$\begin{cases} t_{\text{new}} = \dots \\ h_{\text{new}} = \dots \end{cases}$$
which will give us the flow equations for t and h.

First, since RG steps involve changes at short lenght scales, then it cannot generate singularities; therefore, close to criticality $[(t,h) = (0,0)]$, we can expand the flow equations in Taylor series:
$$\begin{cases} t_b(t,h) \sim A(b)t + B(b)h + \mathcal{O}(t^2, h^2), \\ h_b(t,h) \sim C(b)t + D(b)h + \mathcal{O}(t^2, h^2), \end{cases}$$

from which we will have to deduce symmetry relations for the coefficients $A(b)$, $B(b)$, $C(b)$ and $D(b)$. The free energy is indeed invariant under the combined transformations

$$\begin{cases} m(\mathbf{x}) \to -m(\mathbf{x}), \\ h \to -h, t \to t, \end{cases}$$

then the hamiltonian will also be invariant under the same transformations; now using this transformation we can write

$$\begin{cases} t_b(t,h) = A(b)t, \\ h_b(t,h) = D(b)h, \end{cases}$$

and fixing $A(1) = D(1) = 1$ then

$$\begin{cases} A(b_1 b_2) = A(b_1)A(b_2), \\ D(b_1 b_2) = D(b_1)D(b_2), \end{cases}$$

which suggest the existence of a group structure for the RG transformations, and thus we can write

$$\begin{cases} A(b) = b^{y_t}, \\ D(b) = b^{y_h}, \end{cases}$$

which will give us a solution for certain values of the exponents y_t, $y_h \in \mathbb{R}$. Now we want to understand the sign of thee exponents y_t and y_h, which will give us the stability of the fixed point.

Since after an RG step the correlation length gets divided by b, it seems that we are looking at a less critical system: it moves us away from criticality, which should correspond to

$$y_t,\ y_h > 0.$$

Thus teh parameters t and h transforms under the flow equations near criticality as

$$\begin{cases} t' = t_{\text{new}} = b^{y_t} t, \\ h' = h_{\text{new}} = b^{y_h} h, \end{cases}$$

which means that the flow is away from the critical point, which is thus an unstable fixed point.

Now the idea is to use those to demonstarte the scaling and hyperscaling relations, which are the relations between the critical exponents that describe the behavior of physical quantities near the critical point.

5.3.1 | Scaling Relation for Free Energy

In the path integral formulation, the partition function is given by

$$\mathcal{Z} = \int \mathcal{D}m e^{-F[m;t,h]} = \int \mathcal{D}m_{\text{new}} e^{-F}[m_{\text{new}}; t_{\text{new}}, h_{\text{new}}],$$

but since the integration variable is a dummy variable, we can write

$$\mathcal{Z} = \int \mathcal{D}m e^{-F[m;t_{\text{new}},h_{\text{new}}]},$$

and since the free energy is the logarithm of the partition function, we can write

$$f(t,h) = -\frac{1}{V} \ln \mathcal{Z}.$$

We have to take into account that the volume of the system is also changed under the RG transformation, since we are rescaling the lenghts by a factor b, thus we have to write

$$Vf(t,h) = V_{\text{new}} f(t_{\text{new}}, h_{\text{new}}),$$

and using the fact that $V_{\text{new}}/V = b^{-D}$, we can write

$$f(t,h) = b^{-D} f(b^{y_t} t, b^{y_h} h) \implies f(t,h) = t^{\frac{D}{y_t}} f(1, t^{-\frac{y_h}{y_t}} h) = t^{\frac{D}{y_t}} g\left(\frac{h}{t^{\frac{y_h}{y_t}}}\right),$$

which is the scaling relation for the free energy, where g is a scaling function that depends on the ratio $h/t^{y_h/y_t}$. This scaling relation allows us to understand how the free energy behaves near the critical point, and in particular how it depends on the reduced temperature t and the magnetic field h.

5.3.2 | Scaling Relation for Correlation Length

The correlation length

$$\xi = \Gamma(t,h)$$

is defined as the length scale over which the correlations between the spins decay exponentially. Near the critical point, the correlation length diverges, and its behavior can be described by a scaling relation. Under an RG transformation, again the idea is to write the correlation length in terms of the new parameters:

$$\begin{cases} \xi_{\text{new}} = \Gamma(t_{\text{new}}, h_{\text{new}}), \\ \xi_{\text{new}} = \dfrac{\xi}{b} = \dfrac{\Gamma(t,h)}{b}, \end{cases}$$

and this implies that

$$\Gamma(t,h) = b\Gamma(b^{y_t} t, b^{y_h} h), \quad b = t^{-\frac{1}{y_t}},$$

which again gives us the scaling relation for the correlation length:

$$\Gamma(t,h) = t^{-\frac{1}{y_t}} \Gamma(1, t^{-\frac{y_h}{y_t}} h),$$

and now we can identify the critical exponent ν that describes the divergence of the correlation length as

$$\nu = \frac{1}{y_t}, \text{ since } \xi \sim t^{-\nu} \text{ for } h = 0.$$

The fact that the functional form of the correlation length is preserved under the RG transformation does not mean that this procedure is valid only for simple cases, but we assumed it for simplicity in the derivation of the scaling relations, but we will see the general implications.

In the end the idea will again be to linearize the flow equations around the fixed point and...

5.4 | Formal Framework

[...] In the most general case, we can write the free energy as a functional of the order parameter $m(\mathbf{x})$ and a set of parameters $\mathbf{s}$:

$$F[m] = \int \mathrm{d}^D\mathbf{x} \left[\frac{t}{2}m^2 + um^4 + v_1 m^6 + v_2 m^8 + \cdots + \frac{k}{2}(\nabla m)^2 + \frac{L}{2}(\nabla m)^2 + \cdots + hm + hm^{5/7} + \cdots \right],$$

where this functional form is the most general one that we can write, and the RG transformation will give us the flow equations for all the parameters t, u, v_1, v_2, ..., k, L and h, which will allow us to understand the behavior of the system near the critical point.

When we apply an RG iteration, we are basically performing a change of variables

$$m'(\mathbf{x}') = \frac{1}{\zeta b^D} \int_{\mathcal{C}} \mathrm{d}^D[m(\mathbf{x}')],$$

...

But we can already see that the functional form do not change after a parameter change, which is the key point of the RG transformation:

$$F[m, \mathbf{s}] = F[m, (t, u, v_1, v_2, \ldots, k, L, h, \ldots)] \xrightarrow{\text{RG step}} F[m, (t_{\text{new}}, u_{\text{new}}, v_{1,\text{new}}, v_{2,\text{new}}, \ldots, k_{\text{new}}, \ldots, h_{\text{new}}, \ldots)],$$

and

$$F[m, \mathbf{s}] \xrightarrow{RG} F_{\text{new}}[m, \mathbf{s}_{\text{new}}] = F[m_{\text{new}}, \mathbf{s}_{\text{new}}],$$

since in the parameter space, with the most general functional form, the RG transformation is a mapping from the parameter space to itself

$$\mathbf{s}_1 \mapsto \mathbf{s}_2 \mapsto \mathbf{s}_3 \mapsto \cdots \in \text{Action Space}, \quad \mathbf{s}' = \mathbb{R}_b(\mathbf{s}),$$

which allows us to understand how the system behaves at different scales, and in particular to identify the critical points and the nature of the phase transitions.

We are visualizing the RG transformation as a flow in the parameters space, which is called the **action space**, and the fixed points of this flow correspond to scale-invariant theories, since we are rescaling effectively the system, thus they have to be associated with critical points of phase transitions (at zero or infinite temperature). The flow of the parameters under the RG transformation allows us to understand the behavior of the system at different scales, and in particular to identify the critical points and the nature of the phase transitions.

5.4.1 | Scaling Directions and Anomalous Dimensions

Let us imagine a complicated infinite dimensional parameter function

$$S^* = \ldots$$

we can taylor expand it around the fixed point as

$$\ldots\ldots\ldots$$

and at least we get a formal expression...

commutation property of functions

Hence, we can find a common eigenbasis $\{O_i\}$ for the linearized RG transformation, and we can write

$$\mathcal{R}_b^L O_i = \lambda_i(b) O_i,$$

such that

$$RRO\ldots$$

and we obtain the following constraints on the eigenvalues $\lambda_i(b)$:

$$\lambda(b)\lambda(b') = \lambda(bb'),$$

and recalling that $\lambda(1) = 1$ (since for $b = 1$ the RG transformation is the identity), we can write

$$\lambda(b) = b^{y_i},$$

which let us understand

$$\begin{cases} O_i \text{ are called } \textbf{scaling directions}, \\ y_i \text{ are called } \textbf{anomalous dimensions}. \end{cases}$$

Since the eigenvectors O_i form a complete basis, we can write any Hamiltonian in the vicinity of S^* as

$$S = S^* + \sum_i g_i O_i,$$

which after one RG step becomes

$$\mathcal{R}_b[S] = S^* + \sum_i g_i b^{y_i} O_i,$$

and thus we can classify the scaling directions as

- **Relevant operators** if $y_i > 0$: the flow is away from the fixed point, and thus they are associated with unstable directions.
- **Irrelevant operators** if $y_i < 0$: the flow is towards the fixed point, and thus they are associated with stable directions.
- **Marginal operators** if $y_i = 0$: the flow is neither towards nor away from the fixed point, and thus they are associated with marginal directions.

We call them "operators" even if they are just directions in the parameter space, because they are associated with physical operators in the theory and historically they were called operators in the context of quantum field theory, where the RG was first developed. There is a natural way to associate to each scaling direction a physical operator, since the parameters of the Hamiltonian are associated with physical operators. We accept it as a name for now.

Relevant operators are those for which g_i becomes larger and larger under the RG flow, and thus they are associated with unstable directions, since they move us away from the fixed point. Irrelevant operators are those for which g_i becomes smaller and smaller under the RG flow, and thus they are associated with stable directions, since they move us towards the fixed point. Marginal operators are those for which g_i does not change under the RG flow, and thus they are associated with marginal directions, since they do not move us towards or away from the fixed point.

The saddle point calculation give us exact result when concerned about criticality, because neglecting irrelevant operators not only we make a small error, but we are actually neglecting terms that are going to zero under the RG flow, after looking at an infinite scale of the system. This is the key point of the RG analysis: we are not neglecting terms that are small, but we are neglecting terms that are going to zero.

Let us consider again the scale lenght ξ; if we repeat the same procedure as before, we can write

$$g_i \to b^{y_i} g_i \implies \xi(g_1, g_2, \dots) = b\xi(b^{y_1} g_1, b^{y_2} g_2, \dots),$$

we can order the operators in such a way that $y_1 > y_2 > \cdots$, and thus we can write

$$b = g^{-\frac{1}{y_1}}, \quad \xi(g_1, g_2, \dots) = g_1^{-\frac{1}{y_1}} \xi(1, g_2 g_1^{-\frac{y_2}{y_1}}, \dots),$$

namely we have derived the scaling form of the correlation length, which is a function of the relevant operators, and the irrelevant operators only appear in the scaling function as corrections to scaling.

If we have a fixed points we have two options: correlation to 0 or correlation to infinity, and thus if we start from a point, if the RG moves me away from the fixed point, then I am in the basin of attraction of the other fixed point, and thus I will end up to an infinite temperature fixed point, while if the RG moves me towards the fixed point, then I am in the basin of attraction of the zero temperature fixed point, and thus I will end up to a zero temperature fixed point.

To reach the repulsive point (infinite correlation) we need to fine-tune the parameters, and thus we need to set $t = 0$ and $h = 0$, which means that we need to set the temperature to the critical temperature and the magnetic field to zero, which is a very special point in the parameter space, and thus it is not surprising that we have a phase transition only at this point.

Historically, the RG was developed in the context of quantum field theory, where it was used to understand the behavior of quantum fields at different energy scales. The idea was coming from Kardanoff, but only after the work of Wilson, who developed the RG in a more systematic way, it was understood that the RG is a powerful tool to understand critical phenomena in statistical mechanics, and it has been applied to a wide range of systems, from classical to quantum systems, and from equilibrium to non-equilibrium systems. **TODO:** Check literally everything.

The epsilon expansion ...

5.5 | The Gaussian Model

The Gaussian model is a simple yet powerful statistical model that assumes that the data follows a normal distribution. It is widely used in various fields, including physics, engineering, and machine learning, due to its mathematical tractability and the central limit theorem.

It is a very limited case of application of RG methods, but it is a good starting point to understand the basic concepts and techniques of RG. In this section, we will explore the Gaussian model in detail, including its formulation, properties, and applications.

The gaussian model is defined by keeping only the quadratic term in the LG partition function:

$$\mathcal{Z} = \int \mathcal{D}m(\mathbf{x}) \exp\left\{ -\int \mathrm{d}^D\mathbf{x}\, [\cdots] \right\},$$

and it is a bit unphysical since it is only defined for $t \geq 0$, but it is a good starting point to understand the basic concepts and techniques of RG. We would not even need RG to solve this model, but it is a good example to illustrate the RG procedure.

We will provide a **momentum space RG solution** for this model.

[...]

5.5.1 | Exact Solution

The Gaussian model is exactly solvable, and we can find the exact solution by performing a Fourier transform of the field $m(\mathbf{x})$. The Fourier transform allows us to express the field in terms of its momentum components, which simplifies the analysis of the partition function. The Fourier transform of the field $m(\mathbf{x})$ is defined as:

$$\tilde{m}(q) = \int \mathrm{d}^D\mathbf{x}\, m(\mathbf{x}) e^{i\mathbf{q}\cdot\mathbf{x}}, \iff m(\mathbf{x}) = \sum_{\mathbf{q}} e^{-i\mathbf{q}\cdot\mathbf{x}} \tilde{m}(\mathbf{q}) \sim \int \mathrm{d}^D\mathbf{q} (2\pi)\, e^{-i\mathbf{q}\cdot\mathbf{x}} \tilde{m}(\mathbf{q}),$$

where the sum is over the discrete momenta allowed by the finite size of the system, and the integral is over the continuous momenta in the thermodynamic limit. The partition function can be rewritten plugging the Fourier transform of $m(\mathbf{x})$ into the original expression:

$$\mathcal{Z} = e^{\frac{V\hbar^2}{2t}} \dots$$

If we take the logarithm, the product becomes a sum which we can express as an integral over the Brillouin zone: since the singularities of the integrand are at $q = 0$, we can replace the Brillouin zone with a sphere of radius Λ (the UV cutoff): ...

And in the limit for $t \to 0$ ...

Therefore we can identify

$$\begin{cases} \alpha_+ = 2 - \dfrac{D}{2}, \\ \Delta = \dfrac{1}{2} + \dfrac{D}{4}. \end{cases}$$

[...]

1. **Coarse Grain**

   [...]

2. **Rescale**

   [...]

3. **Renormalize**

   [...]

Relevant and Irrelevant Operators

[...]

Remark (Dangerous Irrelevant Operators).

Infinitesimal Scale Transformations

[...]

Part II

Conformal Invariance and Minimal Models

Introduction

1 Quantum Field Theory and Statistical Mechanics

A deep connection exists between quantum field theory (QFT) and statistical mechanics. This relation becomes particularly powerful in the study of critical phenomena, as one can see from the **Landau-Ginzburg formulation** and by the observation that systems at phase transitions are described by field theories with scale invariance. At criticality the theory is invariant under dilatations of coordinates

$$x_\mu \to \lambda x_\mu.$$

Order parameters (fields) transform as

$$\Phi_i \to \lambda^{\Delta_i} \Phi_i,$$

where the Δ_i determine critical exponents, so their calculation is crucial. This implies power-law behavior for correlation functions at criticality:

$$\langle \Phi_i(x) \Phi_i(0) \rangle \sim \frac{1}{|x|^{2\Delta_i}},$$

so that the Δ_i can be extracted if one is able to compute the correlation functions exactly.

This suggested Polyakov to formulate a strategy to solve scale invariant QFT known as the **Polyakov programme**:

1. show that scale invariance implies, under very general assumptions, a larger **conformal invariance**;

2. conformal invariance and operator product expansions of local fields lead to **conformal bootstrap**, which is a set of conditions to be satisfied by the 4-point correlation functions;

3. conformal bootstrap constraints the correlation functions and in some cases is able to determine them exactly: we can extract the critical exponents from the obtained conformal blocks;

4. correlation functions allow to compute β-functions, renormalization group eigenvalues, scale dimensions and ultimately critical exponents.

The classification of fixed points of Renormalization Group, in practice of the critical points and therefore the possible universality classes is thus equivalent to classify all the conformal field theories (CFT).

A fundamental theorem states that, unless a theory is particularly pathologically defined, the combination of Poincaré and scale invariance automatically implies a larger symmetry: **conformal symmetry**. This symmetry enhancement is the cornerstone of the whole approach.

Once conformal symmetry is established, it can be used to set up the **conformal bootstrap**. This powerful framework leverages the symmetries to impose rigid mathematical conditions that force the exact form of the correlation functions. In particular, conformal invariance heavily constrains their spatial dependence:

- **2-point functions** are fixed exactly, up to a overall normalization;
- **3-point functions** are determined completely, up to a single constant known as the structure constant;
- **4-point functions** (which initially depend on four coordinate variables x_1, x_2, x_3, x_4) are reduced to a fixed kinematic prefactor multiplied by an unknown function of a single invariant variable, called the **anharmonic ratio** (or cross-ratio).

This reduction is particularly intuitive in two dimensions: using a general conformal transformation, it is always possible to map three of the four points to 0, 1, and ∞. The position of the single remaining point corresponds exactly to this anharmonic ratio.

Conformal invariance further dictates that this remaining function must satisfy a specific set of consistency identities. Solving these *bootstrap identities* allows one to decompose the correlation functions into their fundamental building components, the so-called **conformal blocks**.

However, it is not guaranteed that this exact analytical procedure can be carried out completely for every theory. For most theories in generic dimensions $D > 2$, an exact analytical solution is out of reach. Nevertheless, the conformal symmetry still allows us to strongly constrain the properties of these conformal blocks. In higher dimensions, the bootstrap identities yield very complicated functional equations that can be approached *numerically*. By truncating and solving them recursively, it is possible to build highly accurate estimates of the theory's observables. This numerical bootstrap approach has been phenomenally successful for critical theories up to four dimensions, with the 3D Ising model being one of its most celebrated applications.

1.1 | Path Integral Formulation

The path integral formulation of quantum field theory (QFT) shows deep links with statistical mechanics. It is indeed possible, thanks to a Wick rotation on the time axis, to write the generating functional of Green functions in a way that makes evident its equivalence to a partition function of a statistical mechanical system with one more spatial dimension. If the Hamiltonian density of the QFT is quadratic in the conjugated momenta, this statistical system will be purely configurational. In this section we quickly recall the basics of the path integral formulation of a Wick rotated QFT.

1.1.1 | Quantum Mechanics

A system with degrees of freedom $q(t)$ and Lagrangian $L(q,\dot{q})$ has quantum transition **amplitude between** an initial state $|q_i\rangle$ and a final one $|q_f\rangle$ given by:

$$Z(q_f,t_f;q_i,t_i) = \langle q_f|\, e^{-iH(t_f-t_i)}\,|q_i\rangle\,,$$

which represents the probability amplitude for the system to evolve from the state $|q_i\rangle$ at time t_i to the state $|q_f\rangle$ at time t_f. By discretizing the time interval in n small intervals of size ϵ:

$$t_i = t_0 < t_1 < \ldots < t_n = t_f, \quad t_k - t_{k-1} = \epsilon,$$

and inserting complete sets of states (resolution of the identity) at each time slice:

$$\mathbb{I} = \int \mathrm{d}q\,|q\rangle\,\langle q|\,,$$

one gets:

$$Z = \int \prod_{j=1}^{n-1} \mathrm{d}q_j \prod_{j=0}^{n-1} \langle q_{j+1}|\, e^{-i\epsilon H}\,|q_j\rangle\,.$$

Expanding for small ϵ and inserting momentum eigenstates, one obtains:

$$Z = \int \mathcal{D}q\, e^{iS[q]}, \tag{1.1.1}$$

where

$$S[q] = \int_{t_i}^{t_f} \mathrm{d}t\, L(q,\dot{q})$$

is the action.

In this way, the quantum transition amplitude is expressed as a sum over all possible paths $q(t)$ connecting the initial and final states, weighted by the exponential of the action evaluated on those paths. This formulation is particularly useful for systems where the Hamiltonian is not easily diagonalizable, and it provides a natural framework for perturbation theory.

1.1.2 | Quantum Field Theory

The same construction illustrated above can be done also in QFT. From the Hamiltonian point of view, actually, a field theory is a system with an infinite number of degrees of freedom. To the dynamical variables $q_i(t)$ with $i = 1,\ldots,N$ one substitutes a field $\phi(x,t)$, where the continuum index x plays the role of the discrete index i; in other words, the field $\phi(x,t)$ represents a dynamical variable for any point x of space. Here, we do not better specify the field, which can have a scalar, vector, tensor or spinor nature, with indices in Minkowski spacetime with metric $g = \mathrm{diag}(1, -1, ..., -1)$ and/or other possible indices for internal degrees of freedom.

The same formulae obtained in quantum mechanics for the time evolution amplitude can be applied to the case of a system with Lagrangian density $\mathcal{L}(\phi, \partial_\mu\phi)$ and Hamiltonian density

$$\mathcal{H} = \pi \cdot \dot{\phi} - \mathcal{L},$$

where π represents the conjugate momentum density

$$\pi = \frac{\delta\mathcal{L}}{\delta\dot{\phi}}.$$

It is important to notice that the Lagrangian density contain all of the information about the dynamics of the field, giving us potentially everything we need to know about the theory, including the spectrum of excitations, the interactions and the symmetries. It literally builds the quantum field theory, and it is the starting point for any calculation in the path integral formalism.

For example, the Lagrangian density of a scalar field of mass m is given by

$$\mathcal{L}(\phi, \partial_\mu\phi) = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi),$$

where the first term is called *kinetic term*, while the term describing the **self interaction** of the field

$$V(\phi) = \sum_{k=1}^{\infty} g_k\phi^k,$$

is called *potential*.

The term quadratic in ϕ usually has, in QFT, the meaning of a mass term and $g_2 = \frac{m^2}{2}$ but, for example, in the LG formulation it has a meaning as (dimensionless) temperature $t = (T - T_c)/T_c$. However, it is important to stress that m here represents the *bare* mass of the field. Whether this corresponds to the actual physical mass depends on the presence of other terms in the potential: if the field is interacting (or even self-interacting), quantum fluctuations will shift this initial value. Consequently, the actual physical mass is obtained only after a renormalization procedure, which systematically redefines these bare parameters to absorb the non-physical infinities that arise in the calculations.

The **conjugated momentum density** $\pi = \dot{\phi}$ (through the Legendre transformation) allows to write the Hamiltonian density as

$$\mathcal{H}(\pi, \phi) = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi).$$

Notice that the kinetic and spatial gradient terms are strictly positive, being sums of squares. Provided that the potential $V(\phi)$ is also bounded from below (which is guaranteed if we only include even higher powers of the field, such as the ϕ^4 term in the Ising model, making it the simplest physically viable non-trivial theory), the total Hamiltonian—the spatial integral of this density—is bounded from below. This implies that the operator whose eigenvalues represent the total energy of the system has a well-defined lower bound, meaning the theory is mathematically and physically sensible.

With analogy of what is done in the quantum mechanics case, one obtains

$$\langle\phi_f(x)|\, e^{-i\hat{H}(t_f-t_0)}\, |\phi_0(x)\rangle = \int_{\phi(x,t_0)=\phi_0(x)}^{\phi(x,t_f)=\phi_f(x)} \mathcal{D}\phi\mathcal{D}\pi\, e^{i\hbar\int_{t_0}^{t_f}\mathrm{d}t\int \mathrm{d}^{D-1}x[\pi\cdot\dot{\phi}-\mathcal{H}]}.$$

Physically, this object represents the probability amplitude that an initial configuration of fields at time t_0 will evolve into a final configuration of fields at time t_f.

In QFT one often considers the limit $t_0 \to -\infty$ and $t_f \to +\infty$, therefore considering an interaction running on the whole spacetime. For Hamiltonians quadratic in $\pi(x)$, it is possible to apply the same considerations of the quantum mechanical case and Gauss-integrate on momenta, getting the functional Z:

$$Z = \int \mathcal{D}\phi \, e^{\frac{i}{\hbar}S[\phi]}, \tag{1.1.2}$$

where the action is now the integral over the whole spacetime of the Lagrangian density

$$S[\phi] = \int \mathrm{d}t\mathrm{d}^{D-1}x \, \mathcal{L}(\phi, \partial_\mu\phi).$$

We can identify Z as the generating functional of Green's functions, but it is mathematically hard to give a rigorous meaning to this object due to the strongly oscillating behavior of the complex exponential integrand. To resolve this normalization problem, one typically has to regularize the integral by introducing a small parameter $\epsilon \to 0^+$. A more powerful way to deal with this is to perform a **Wick rotation**:[1] by rotating to imaginary time, the oscillating phase becomes a real damping exponential, making the integral strictly analogous to a statistical mechanics partition function.

Once this setup is generalized, we can express any n-point correlation function using this path integral formulation:

$$\langle\phi(x_1)\cdots\phi(x_n)\rangle = \frac{\int \mathcal{D}\phi \, \phi(x_1)\cdots\phi(x_n)e^{iS[\phi]}}{\int \mathcal{D}\phi \, e^{iS[\phi]}}. \tag{1.1.3}$$

1.1.3 | Wick Rotation

The oscillatory nature of the path integral makes it difficult to define rigorously. For this reason, it is usual to perform a Wick rotation:

$$t \to -i\tau, \implies \begin{cases} \mathrm{d}t \to -i\mathrm{d}\tau, \\ \partial_t \to i\partial_\tau. \end{cases} \tag{1.1.4}$$

In the example of a scalar field considered above, the Lagranian density becomes, after the Wick rotation

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi) = -\mathcal{H},$$

where now the metric is Euclidean $g_{\mu\nu} = \delta_{\mu\nu}$ with the index μ running from 1 to D and setting $\tau = x_D$. Thus, at this point, the action is conveniently substituted by the so called **Euclidean action**

$$S = iS_E, \quad S_E[\phi] = \int \mathrm{d}^D x \left[\frac{1}{2}(\partial_\mu\phi)^2 + V(\phi)\right],$$

or, more generally

$$S_E[\phi] = \int \mathrm{d}^D x \, \mathcal{H}(\phi, \nabla\phi). \tag{1.1.5}$$

The possibility of doing this analytic continuation to imaginary times can easily be established in perturbative QFT. Indeed in the perturbative regime the most general Green function

$$G(x_1, \ldots, x_n) = \int \prod_{i=1}^{n} \frac{\mathrm{d}^D p_i}{(2\pi)^D} e^{i\sum_{i=1}^n p_i\cdot x_i}\tilde{G}(p_1, \ldots, p_n),$$

[1] Named after Gian Carlo Wick, an Italian physicist of the historical Via Panisperna group.

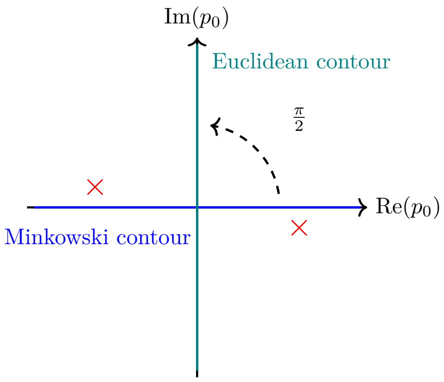
is formed by sums of Feynman diagrams, built with vertices and propagators of the free theory. This latter can be obtained by putting $V(\phi) = 0$. In Minkowski spacetime, giving meaning to the functional in (1.1.2), we add a term of the form

$$\frac{i\epsilon}{2}\phi^2$$

to the action (with ϵ arbitrarily small) and we obtain for the free propagator in momentum space (using units with $\hbar = 1$)

$$\frac{i}{p_0^2 - |\mathbf{p}|^2 - m^2 + i\epsilon}.$$

In the complex plane of $p_0 = \frac{E}{c}$, such propagator has a pole just above the negative part and another just below the positive part of the real axis.



Because of this specific pole structure, we can simultaneously rotate the integration contours of all propagators within a Feynman diagram by an angle of $\pi/2$ counter-clockwise. This procedure safely avoids the singularities, allowing for a rigorous analytic continuation of the Green's functions from the real (Minkowski) axis to the imaginary (Euclidean) axis.

Figure 1.1: Representation of the Wick rotation in the complex p_0 plane.

Thus the analytic continuation of the Green function to imaginary times builds corresponding Euclidean Green function:

$$G(x_1, \ldots, x_n) \to G(\xi t_1, \mathbf{x}_1, \ldots, \xi t_n, \mathbf{x}_n), \quad \xi = \rho e^{i\theta},$$

given by the usual Green's function of the Euclidean theory, with $\rho > 0$ and $\theta \in [0, \pi]$ and all the times t_i replaced by ρt_i. This function is analytical in the half plane $\mathrm{Im}(\rho) < 0$ and the original Green functions are recovered for $\rho \to 1$. Introducing the new integration variables $p_i^0 \to \frac{p_i^0}{\rho}$ in the particular case of $\rho = -1$ we get

$$G(-it_1, \mathbf{x}_1, \ldots, -it_n, \mathbf{x}_n) \propto \int \prod_{i=1}^{n} \frac{\mathrm{d}^D p_i}{(2\pi)^D} e^{-\sum_{i=1}^{n} p_i \cdot x_i} \tilde{G}(p_1, \ldots, p_n),$$

Therefore, in the perturbative expansion the results of the euclidean theory are analytically connected to those in the Minkowski spacetime. Practically, in many cases the results obtained in the euclidean space can be directly used to extract physical information and it is pointless to explicitly continue analytically to the Minkowski spacetime.

There are rigorous theorems as the Osterwalder-Schrader positivity condition, on the validity of such analytic continuation even in non-perturbative regime for almost all QFT of physical interest (with the remarkable exception of Einstein gravity). In the following we shall always assume that the theories we deal with admit the analytic continuation of Wick to an euclidean metric.

From now on, we will always refer to the Euclidean version of QFT, which is much more mathematically well defined and easier to handle, especially in the non-perturbative regime. We will drop the subscript E for simplicity, but it is important to keep in mind that we are working in Euclidean space.

1.2 | Connection with Statistical Mechanics

The great advantage of Wick rotation and of the redefinition of QFT on euclidean space consists in the link we can establish between field theory and statistical mechanics. The Euclidean path integral has the same structure as a partition function:

$$Z = \sum_{\text{conf.}} e^{-\beta H}.$$

This single powerful observation leads to the following correspondence:

QFT	**Statistical Mechanics**
$\phi(x)$	LG Field or microscopical conf.
$S_E[\phi]$	Hamiltonian H
Z	Partition function
$\langle \phi(x_1)\cdots\phi(x_n)\rangle$	Correlation functions

1.2.1 | Generating Functionals

The generating fuctional of Green functions is defined from the partition function Z by introducing a source $J(x)$, similar to coupling a statistical model to an external magnetic field:

$$Z[J] = \int \mathcal{D}\phi\, e^{-S_E[\phi]+\int \mathrm{d}^D x J(x)\phi(x)}. \tag{1.2.1}$$

It is called generating functional since all of the n-point correlation functions are generated by its successive functional derivatives

$$\langle \phi(x_1)\cdots\phi(x_n)\rangle = \left.\frac{\delta^n Z[J]}{\delta J(x_1)\cdots\delta J(x_n)}\right|_{J=0}.$$

This generation includes also disconnected functions. What is physically relevant is the set of **connected n-point functions**, generated by the functional:

$$W[J] = \log\, Z[J]. \tag{1.2.2}$$

Being the logarithm of a partition function, it looks like the free energy in Statistical Mechanics: it is not a surprise that it generates connected correlation functions, since the free energy is the generating function of connected correlation functions in Statistical Mechanics.

1.2.2 | Effective Action

In particular the one point function, or expectation value, of a field, is called classical field in QFT. The name is inspired by **Ehrenfest theorem**: *the average of an operator evoleves in time with the classical equations of motion*, i.e.

$$\phi_{cl}(x) = \frac{\delta W}{\delta J(x)} = \langle \phi(x)\rangle.$$

This definition of classical field has, as analog is Statistical Mechanics: the **order parameter**. The effective action is the Legendre transform of the generating functional of connected correlation functions, defined as

$$\Gamma[\phi_c] = W[J] - \int \mathrm{d}^D x J(x)\phi_c(x).$$

It plays a role analogous to the **Gibbs free energy** in statistical mechanics. In QFT $\Gamma[\phi_c]$ is the generating function of the so-called *1-particle irreducible graphs*, which are very important in the Feynman diagrams perturbative expansion and in its partial resummation through the Dyson equation. The conclusion is that the equivalence of Statistical Mechanics and (Wick rotated) QFT is on a solid ground and shows the wide generality of QFT as a mathematical tool for the systems with an infinite numebr of degrees of freedom, being able to describe a large range of models in many areas of physics.

1.2.3 | Classical Dimensions and Relevance of Operators

Consider a scalar field theory, whose Euclidean action is given by the integral of the Lagrangian density

$$S = \int \mathrm{d}^D x \left[\frac{1}{2}(\partial_\mu \phi)^2 + \sum_n g_n \phi^n \right]. \tag{1.2.3}$$

If we perform dimensional analysis, knowing that with our units the action has dimension $[S] = 0$, we can infer the dimension of the field ϕ from the kinetic term:

$$[\phi] = \frac{D-2}{2},$$

while the dimension of the coupling can be inferred from the potential term:

$$[g_n] = D - n\frac{D-2}{2}.$$

From the **renormalization group analysis**, we know that the fields in the potential are:

- $[g_n] > 0$ *relevant*: they drive the theory out of the fixed point of RG (which corresponds to a critical point). They do not converge in general so we need to renormalize them.
- $[g_n] = 0$ *marginal*: they can be relevant or irrelevant depending on the sign of their β-function.
- $[g_n] < 0$ *irrelevant*: they are suppressed by the renormalization group flow. They originate non-renormalizable theories, which are not predictive and therefore not physically viable.

Thinking of the potential as perturbation of a Gaussian model (free massless boson), we can see that the irrelevant perturbations leave us on the same critical surface and the fluctuations become, in most cases, uncontrollably wild. The perturbation series by these sort of operators turns out to be non-renormalizable, i.e. plugged by infinities that cannot be cured by some regularization procedure.

When an operator is classified as *marginal* ($[g_n] = 0$), it does not induce a Renormalization Group (RG) flow at first order in perturbation theory. Consequently, the system appears to remain precisely at the critical fixed point. One might then ask: does adding such a marginal perturbation change the physical properties of the theory at all? While it might induce finite shifts in parameters such as the mass (up to a suitable renormalization procedure), understanding its true dynamical effect requires going beyond the linear approximation.

To do so, one must examine higher-order quantum corrections by computing the β-function. If the second-order (or higher) contributions cause the coupling to grow, the operator dynamically breaks scale invariance and is classified as *marginally relevant*.

However, in some special cases, the β-function continues to vanish exactly at all orders in perturbation theory. Such fields are called *truly* (or *exactly*) *marginal* operators. When a theory possesses a truly marginal operator, varying its continuous coupling does not destroy conformal invariance but rather smoothly deforms the theory into a new, physically distinct CFT. In this scenario, instead of an isolated critical point, one finds a continuous manifold (a line or surface) of fixed points, continuously parameterized by the marginal coupling.

The "good" theories are the renormalizable ones, where the dominating field in the potential is relevant, driving the theory out of the fixed point of RG (which corresponds to a critical point) towards a RG flow to some *"infrared" destiny* (critical or trivial). So what field theorists call renormalizability of a theory corresponds, in Statistical Mechanics, adopting the RG language, to relevant perturbations.

If $V(\phi)$ is a polynomial whose term with maximal power is ϕ^n, there exists a space-time dimension

$$D_{crit} = \frac{2n}{n-2}$$

called **upper critical dimension** such that, for $D > D_{crit}$, $[g_n]$ is irrelevant or marginal and the theory becomes non-renormalizable (indeed we derived the critical dimension by imposing $[g_n] = 0$, i.e. it is marginal). This puts a bound on which theories one can be renormalizable at a given spacetime dimension. For a single scalar theory like (1.2.3) we have:

Dimension	Max. field in the potential
$D = 4$	$n = 4 \mapsto \phi^4$
$D = 3$	$n = 6 \mapsto \phi^6$
$D = 2$	any power ($n = \infty$)

In $D = 2$ a scalar field admits any sort of potential, meaning we could add infinitely many relevant perturbations to the free massless boson theory, which is the simplest CFT in two dimensions.

As an example we cite the celebrated *Sine-Gordon theory*

$$S = \int \mathrm{d}^2x \left(\frac{1}{2}(\partial_\mu \phi)^2 - \frac{m^2}{\beta^2} \cos \beta\phi \right)$$

a very important theory in many aspects, ranging from soliton theory to being the prototypical example of integrable QFT. It is called Sine-Gordon because of the cosine potential: the equations of motion are very similar to Klein-Gordon equation, but with a sine non-linearity

$$\propto \sin \phi.$$

The Sine-Gordon theory is renormalizable in two dimensions, but it becomes non-renormalizable in higher dimensions due to the presence of the cosine potential, which can be expanded into an infinite series of powers of ϕ.

2 Conformal Symmetry in Field Theories

At a critical point, the correlation length diverges:

$$\xi \to \infty.$$

In the language of Quantum Field Theory (QFT), the inverse of the correlation length corresponds to the mass of the lightest excitation in the spectrum ($m \sim \xi^{-1}$), which defines the **energy gap** above the vacuum. When $\xi \to \infty$, this mass gap vanishes, resulting in a *gapless* or *massless* theory (analogous to Quantum Electrodynamics, where the fundamental photon is massless). Since a non-zero mass introduces a fundamental length or energy scale, a massless theory cannot distinguish between different observation scales. Therefore, a diverging correlation length inherently implies that the theory is *scale invariant*.

As a consequence, the system becomes invariant under global scale transformations of the spacetime coordinates (including time):

$$x^\mu \to \lambda x^\mu.$$

Under this rescaling, the fundamental objects of the theory (the local fields, regardless of their scalar, vector, or tensor nature) transform according to their **scaling dimension** Δ:

$$\phi(x) \to \lambda^{-\Delta}\phi(\lambda x).$$

In general, the space of all possible fields can be thought of as a vector space. It is always possible to select a preferred basis of fields such that every element has a definite scaling dimension. Any arbitrary field can then be expressed as a linear combination of these basis fields, a procedure conceptually identical to finding the eigenoperators of the Renormalization Group (RG) flow.

Away from criticality, correlation functions typically exhibit an exponential decay governed by the correlation length, $\sim e^{-|x|/\xi}$. However, at a critical point ($\xi \to \infty$), this exponential suppression factor evaluates to 1. Consequently, the scale invariance rigidly dictates that correlation functions must exhibit a pure power-law behavior:

$$\langle \phi(x)\phi(0)\rangle \sim \frac{1}{|x|^{2\Delta}}.$$

It is therefore very important to deepen our knowledge of scale transformations. As we shall see, a crucial result (the classical Polyakov theorem, a powerful extension of Noether's theorem) states that, under quite general assumptions, scale invariance in field theory is almost always enhanced to a much larger symmetry: *conformal invariance*. Thus, the first necessary step is to formally understand what conformal invariance is and how it governs the theory.

2.1 | Conformal Transformations

General coordinate transformations $x \to x'$ change the metric tensor as

$$g'_{\mu\nu}(x') = \frac{\partial x'^{\lambda}}{\partial x^{\mu}} \frac{\partial x'^{\sigma}}{\partial x^{\nu}} g_{\lambda\sigma}(x).$$

Conformal transformations are coordinate transformations of type

$$\frac{\partial x'^{\lambda}}{\partial x^{\mu}} = \Omega(x) g^{\lambda\mu},$$

preserving the metric up to a local rescaling:

$$g'_{\mu\nu}(x') = \Omega^2(x) g_{\mu\nu}(x), \quad \Omega^2(x) = e^{f(x)} > 0.$$

Physically, this represents a strictly *local* rescaling of the coordinates. Unlike a global scale transformation that stretches the entire space uniformly, a conformal transformation allows the scale factor $\Omega(x)$ to vary from point to point. Geometrically, if one imagines a coordinate grid, this operation allows you to fix a point and locally "inflate" or "deflate" the grid around it. The coordinates change scale in different ways at different positions in space, but without arbitrarily mixing the coordinate axes in a way that would alter the angles between intersecting lines.

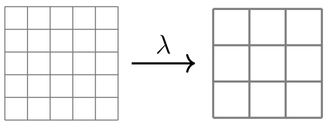


Figure 2.1: Global scale transformation: the grid is scaled uniformly at every point by a constant factor λ.

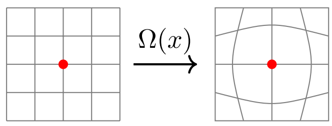


Figure 2.2: Local conformal transformation: the grid undergoes a local "inflation" determined by the position-dependent factor $\Omega(x)$.

This mechanism becomes particularly interesting when applied to a spacetime characterized by a **flat metric**, where the original $g_{\mu\nu}$ is a constant matrix independent of the position x. In a D-dimensional flat spacetime with metric

$$g_{\mu\nu} = \begin{cases} \delta_{\mu\nu} = & \text{diag}(1, 1, \ldots, 1) \text{ if Euclidean,} \\ \eta_{\mu\nu} = & \text{diag}(-1, 1, \ldots, 1) \text{ if Minkowskian,} \end{cases}$$

the **conformal condition** can be explicitly written as:

$$\partial_\mu x'^{\rho} \partial_\nu x'^{\sigma} g_{\rho\sigma} = \Omega^2(x) g_{\mu\nu}. \tag{2.1.1}$$

Thus if we have already Wick rotated to Euclidean signature, we will be working with the Euclidean metric $\delta_{\mu\nu}$, if not, we will be working with the Minkowski metric $\eta_{\mu\nu}$. In any case, the conformal condition can be expressed as shown above, where $g_{\mu\nu}$ is the appropriate metric for the signature we are working with.

2.1.1 | Conformal Killing Equation

Since we are interested in finding the Lie algebra of this transformations group, thus it is convenient to consider infinitesimal transformations. For an infinitesimal transformation we can write

$$x'^{\mu} = x^{\mu} + \epsilon^{\mu}(x), \quad \Omega^2(x) = 1 + f(x) + \ldots$$

and, since $\Omega^2(x)$ is a positive function, we chose to write it as the exponential of another function $f(x)$ (this is just a convenient parametrization, not a restriction). Thus we can write

$$\Omega^2(x) = e^{f(x)} \simeq 1 + f(x) + \dots$$

Taking the conformal condition (2.1.1) and inserting the infinitesimal transformation (recognizing that the r.h.s. is the metric in the new coordinates, i.e. $g'_{\mu\nu}$), we get

$$g'_{\mu\nu} = g_{\mu\nu} - (\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu),$$

and now expanding $g'_{\mu\nu} = e^{f(x)} g_{\mu\nu}$ to first order in ϵ, we get

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = f(x) g_{\mu\nu}.$$

We now take the trace of both members, multiplying by $g^{\mu\nu}$ (saturting the indices), getting to

$$f(x) = \frac{2}{D} \partial_\sigma \epsilon^\sigma(x),$$

which implies that the function $f(x)$ is determined by the divergence of $\epsilon^\mu(x)$. Inserting this back into the previous equation we get

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = \frac{2}{D} (\partial \cdot \epsilon) g_{\mu\nu}. \tag{2.1.2}$$

This is the conformal Killing equation, which is the fundamental equation defining the conformal Killing vectors $\epsilon^\mu(x)$ that generate the conformal transformations. The solutions to this equation will give us the explicit form of the infinitesimal conformal transformations in flat spacetime.

Solution of Conformal Killing Equation

To explicitly solve the conformal Killing equation, one can apply a standard tensorial manipulation. By taking a further derivative of both sides, one obtains an equation with three spacetime indices. By cyclically permuting these indices to generate three distinct equations (adding the first two and subtracting the third) one can isolate the second derivatives of the displacement vector $\epsilon(x)$.

After performing these algebraic steps, one finds that for spacetime dimensions $D > 2$, the second derivatives of the scalar function $f(x)$ must strictly vanish ($\partial_\mu \partial_\nu f = 0$). This implies that $f(x)$ can be at most a linear function of the coordinates. Since $f(x)$ is defined through the first derivative of $\epsilon(x)$, it naturally follows that the third derivatives of $\epsilon(x)$ must also vanish. Consequently, the components of $\epsilon^\mu(x)$ are restricted to be at most *quadratic* polynomials in the coordinates.

Indeed, for $D > 2$ the general solution for $\epsilon(x)$ is precisely quadratic in x^μ, and it takes the exact form:

$$\epsilon^\mu(x) = a^\mu + \omega^\mu{}_\nu x^\nu + \lambda x^\mu + (b \cdot x) x^\mu - \frac{1}{2} x^2 b^\mu.$$

If we now restrict the form of the solution, we can get some informations about the structure of the conformal group. In particular, we can identify four different types of transformations, based on the parameters appearing in the solution:

- **Translations** $x'^\mu = x^\mu + a^\mu$: these are the simplest transformations, corresponding to a uniform shift of the coordinates by a constant vector a^μ.
- **Rotations** $x'^\mu = x^\mu + \omega^\mu{}_\nu x^\nu$: this comes from the fact that we can always decompose the transformationmatrix into a symmetric and an antisymmetric part, and the latter corresponds to rotations.

- **Dilatations** $x'^{\mu} = (1+\lambda)x^{\mu}$: this corresponds to an infinitesimal scaling of the coordinates (inflation or deflation), and it is indeed proportional to the coordinates themselves. This is the symmetric part we cited before, which is not a rotation but a scaling. It is an important transformation, since it is related to the scale invariance of the theory, and it is peculiar among the conformal transformations.

- **Special conformal transformations** $x'^{\mu} = x^{\mu} + 2(b \cdot x)x^{\mu} - b^{\mu}x^2$: these are the most non-trivial transformations, featuring a quadratic dependence on the coordinates. It is a D-parameter family of transformations (which reduces to 4 parameters in $D = 4$ dimensions), parameterized by the constant vector b^{μ}. Geometrically, this complex non-linear transformation can always be gracefully decomposed into a sequence of three fundamental operations: an inversion ($x^{\mu} \to x^{\mu}/x^2$), followed by a constant translation by the vector b^{μ}, and finally a second inversion.

The first two types of transformations (translations and rotations) are the usual Poincaré transformations, which are symmetries of any relativistic field theory. The last two types (dilatations and special conformal transformations) are additional symmetries that arise in scale-invariant theories, and they are responsible for the enhancement from scale invariance to full conformal invariance.

We already know the form of the generators of translations and rotations, since they are the usual ones, and they give us meaningful information about the structure of the theory of a Poincaré invariant QFT. We are here considering systems at criticality, which are scale invariant, thus it seems natural to expect that the generators of dilatations will also give us important information about the structure of the theory at criticality. Finally, the generators of special conformal transformations are the most non-trivial ones, and they are responsible for the full conformal symmetry of the theory, even if now we do not have a clear physical intuition about their meaning, but they are crucial for the structure of the conformal group and for the constraints that conformal invariance imposes on the theory.

Considering the application of these transformation on generic functions of the coordinates, we can obtain the form of the **generators**

- **Translations**: $P_{\mu} = -i\partial_{\mu}$, which is the usual generator of translations in field theory, corresponding to the momentum operator.

- **Rotations**: $L_{\mu\nu} = i\epsilon_{\mu\nu\rho\sigma}x^{\rho}\partial^{\sigma}$, which is the usual generator of rotations (angular momentum) in field theory, corresponding to the angular momentum operator.

- **Dilatations**: $D = -ix_{\mu}\partial^{\mu}$, which is the generator of dilatations (scale transformations) in field theory, and it is related to the scaling dimension of fields and operators in the theory.

- **Special conformal transformations**: $K_{\mu} = -i(2x_{\mu}x^{\nu}\partial_{\nu} - x^2\partial_{\mu})$, which is the generator of special conformal transformations in field theory; we do not have a simple physical interpretation for this generator, but its importance will become clear when we discuss the structure of the conformal group.

2.1.2 | Conformal Algebra

The generators of the first two types satisfy the usual roto-translation (Poincaré) algebra:

$$\begin{cases} [P_\mu, P_\nu] = 0, \\ [P_\rho, L_{\mu\nu}] = \delta_{\rho\mu} P_\nu - \delta_{\rho\nu} P_\mu, \\ [L_{\mu\nu}, L_{\rho\sigma}] = \delta_{\nu\rho} L_{\mu\sigma} + \delta_{\mu\sigma} L_{\nu\rho} - \delta_{\mu\rho} L_{\nu\sigma} - \delta_{\nu\sigma} L_{\mu\rho}, \end{cases} \tag{2.1.3}$$

but now also the commutations with the dilatation operator D have to be added:

$$\begin{cases} [D, P_\mu] = P_\mu, \\ [D, L_{\mu\nu}] = 0, \end{cases} \tag{2.1.4}$$

and finally those involving the special conformal transformations K_μ:

$$\begin{cases} [K_\mu, K_\nu] = 0, \\ [K_\rho, L_{\mu\nu}] = \delta_{\rho\mu} K_\nu - \delta_{\rho\nu} K_\mu, \\ [K_\mu, P_\nu] = 2(\delta_{\mu\nu} D - L_{\mu\nu}), \\ [D, K_\mu] = -K_\mu. \end{cases} \tag{2.1.5}$$

Note, in particular, that the rotation (spin) operator and the dilatation one commute, which means that we can find a basis of local fields that diagonalizes both operators simultaneously (this should not come as a surprise, since this two transformations are built from the symmetryc and antisymmetric parts of the general infinitesimal transformations).

This algebra is isomorphic to $\mathfrak{so}(D+1,1)$ and the corresponding conformal group is $SO(D+1,1)$ in euclidean spacetime (it would become $SO(D,2)$ if a Minkowski metric is adopted instead). In any case, it is a group with $(D+1)(D+2)/2$ parameters. Physically, every parameter of the symmetry group translates into a mathematical constraint that the Lagrangian (or the correlation functions) of the theory must satisfy. As a general rule, the larger the symmetry group, the tighter the constraints imposed on the theory.

For example, in $D = 4$ dimensions, requiring standard Poincaré invariance imposes 10 constraints on the theory (corresponding to the 10 generators of spacetime translations and Lorentz rotations). However, if we upgrade our requirement to full conformal invariance, the parameters of the conformal group become 15. Therefore, the Lagrangian is subjected to 15 strict constraints: the original 10 from the Poincaré subgroup, plus 5 additional ones arising directly from the extended conformal symmetry (1 from dilatations and 4 from special conformal transformations). This highly constrained framework is precisely what restricts the possible forms of the theory and makes CFTs so mathematically powerful.

Little Group

Generators can be realized in various (irreducible) representations of the conformal group. The **little group** is the subgroup of the conformal group that leaves invariant a specific point, conventionally chosen as the origin of coordinates $x = 0$ (any point can be used as such, due to translational invariance).

We can classify the behavior of generic fields by studying how they transform exactly at this origin. For any other point in spacetime, the transformation rules can be deduced simply by applying a spatial translation to shift the field from 0 to x.

Let us illustrate this mechanism by considering rotations. When an infinitesimal Lorentz transformation $L_{\mu\nu}$ acts on a field evaluated at $x = 0$, it operates strictly on its internal indices:

$$L_{\mu\nu}\phi(0) \coloneqq S_{\mu\nu}\phi(0).$$

The operator $S_{\mu\nu}$ represents the **intrinsic spin** matrix of the field ϕ. Physically, a general coordinate transformation changes not only the spacetime coordinates themselves but also the internal structure of the object evaluated at those coordinates (the components in such basis). If the field is a simple scalar (a single function), this extra intrinsic term vanishes ($S_{\mu\nu} = 0$). However, if the field is a vector or a tensor (comprising multiple component functions), $S_{\mu\nu}$ actively mixes these components, representing the total internal angular momentum.

Analogously, we can define the action of the dilatation operator at the origin:

$$D\phi(0) \coloneqq \tilde{D}\phi(0).$$

To find out what these operators do to a field at an arbitrary point $x \neq 0$, we must conjugate them using the finite translation operator, $e^{ix^\rho P_\rho}$, which is obtained by exponentiating the infinitesimal translation generator P_ρ. Applying this transformation yields:

$$e^{ix^\rho P_\rho} L_{\mu\nu} e^{-ix^\rho P_\rho} = S_{\mu\nu} + i\epsilon_{\mu\nu\rho\sigma} x^\rho \partial^\sigma,$$

and

$$e^{ix^\rho P_\rho} D e^{-ix^\rho P_\rho} = \tilde{D} - ix_\mu \partial^\mu.$$

To explicitly compute these conjugations, we make use of the Baker-Campbell-Hausdorff formula:

$$e^{-A}Be^{A} = B + [B, A] + \frac{1}{2!}[[B, A], A] + \frac{1}{3!}[[[B, A], A], A] + \dots$$

These relations mathematically demonstrate the split of the total angular momentum into an orbital part $(-x_\mu P_\nu + x_\nu P_\mu)$ and an intrinsic spin part $(S_{\mu\nu})$, as well as the split of dilatations into a spacetime coordinate scaling $(x_\mu P^\mu)$ and an intrinsic scaling dimension $(\tilde{D})$:

$$\begin{aligned} L_{\mu\nu} &= S_{\mu\nu} + i\epsilon_{\mu\nu\rho\sigma} x^\rho \partial^\sigma, \\ D &= \tilde{D} - ix_\mu \partial^\mu. \end{aligned}$$

Of course, being strictly internal operators evaluated at the origin, $S_{\mu\nu}$ and $\tilde{D}$ also commute.

The same can also be done for the special conformal transformations, resulting in

$$K_\mu = -i(2x_\mu x^\nu \partial_\nu - x^2 \partial_\mu) + \tilde{K}_\mu,$$

where one could try to derive the explicit form of the internal operator $\tilde{K}_\mu$ by applying the same procedure, but it is not necessary for our purposes.

2.2 | Scale Invariance and Conformal Invariance

In this section we discuss the classical relation between scale invariance and conformal invariance, following the standard Noether approach.

2.2.1 | Noether Current for Scale Transformations

What are the Noether's currents when we apply the conformal group as the symmetry group of a classical field theory? To answer this, we apply these transformations to the Lagrangian and define an action.

Consider a local field theory in D Euclidean dimensions with action

$$S = \int d^D x \mathcal{L}(\phi_a, \partial_\mu \phi_a).$$

where the index a labels the various fields. Any continuous transformation involving coordinates and fields that leaves the action invariant ($\delta S = 0$) is a symmetry of the theory. Such transformations can include purely internal symmetries (which do not affect spacetime coordinates) or spacetime transformations like rotations, which can also induce an internal transformation of the field itself (its intrinsic spin).

Note that the Lagrangian density $\mathcal{L}$, under such a transformation, needs not to be strictly invariant; it can change by a total derivative

$$\mathcal{L} \to \mathcal{L} + \partial_\mu K^\mu,$$

of a quantity K_μ and still ensure that $\delta S = 0$. Under these circumstances, Noether's theorem dictates that for any continuous symmetry (parameterized by elements of a continuous group lying on a manifold whose dimension is independent of spacetime) there exists a corresponding current

$$j_\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta\phi_a - K_\mu.$$

This current turns out to be a conserved current, satisfying the continuity equation

$$\partial_\mu j^\mu = 0,$$

which in turn implies the conservation of the corresponding charge

$$Q = \int j^0 d^{D-1} x.$$

In the case of space-time translation invariance, we define the canonical **stress-energy tensor**. It carries two indices: one (μ) acting as the standard current index, and a second (ν) acting as a sort of internal index labeling the specific translation parameter. The canonical tensor is defined as:

$$T_{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial_\nu \phi_a - \delta_{\mu\nu} \mathcal{L}.$$

If the theory is invariant under space-time translations, $T_{\mu\nu}$ is conserved:

$$\partial^\mu T_{\mu\nu} = 0.$$

We now ask: what is the variation of the field when we apply a dilatation? Under an infinitesimal scale transformation

$$x^\mu \to x'^\mu = (1 + \epsilon) x^\mu,$$

a field $\phi_a(x)$ of classical scaling dimension Δ_a transforms as

$$\delta\phi_a(x) = -\epsilon(x^\mu\partial_\mu + \Delta_a)\phi_a(x).$$

The associated Noether current is the **dilatation current** (j_μ of the dilatation), given by:

$$D_\mu = x^\nu T_{\mu\nu} - V_\mu,$$

where V_μ is the so-called **virial current**. Its divergence is

$$\partial^\mu D_\mu = T^\mu_\mu - \partial^\mu V_\mu.$$

Therefore, requiring scale invariance ($\partial^\mu D_\mu = 0$) implies:

$$T^\mu_\mu = \partial^\mu V_\mu. \tag{2.2.1}$$

This is the general Noether condition for scale invariance: the trace of the energy-momentum tensor does not need to vanish identically, but only up to a total divergence.

While this analysis can be done in a huge variety of theories in any dimension, it is important to note that the most general variation of the action under an arbitrary conformal transformation will ultimately depend on this trace property.

2.2.2 | From Scale Invariance to Conformal Invariance

We now ask under which conditions scale invariance implies full conformal invariance. For an infinitesimal conformal transformation we may write

$$x^\mu \to x^\mu + \epsilon^\mu(x).$$

The variation of the action is controlled by the energy-momentum tensor:

$$\delta S = -\int d^Dx\ T^{\mu\nu}\partial_\mu\epsilon_\nu = -\frac{1}{2}\int d^Dx\ T^{\mu\nu}(\partial_\mu\epsilon_\nu + \partial_\nu\epsilon_\mu)$$

where we assume that the stress-energy tensor is symmetric or it has been symmetrized à la Belinfante (as shown in section 2.2.3). If $\epsilon^\mu(x)$ is a conformal Killing vector, then

$$\partial_\mu\epsilon_\nu + \partial_\nu\epsilon_\mu = \frac{2}{D}(\partial\cdot\epsilon)\delta_{\mu\nu},$$

and therefore

$$\delta S = -\frac{1}{D}\int d^Dx(\partial\cdot\epsilon)T^\mu_\mu.$$

Hence, if the trace vanishes

$$T_\mu{}^\mu = 0,$$

then the action is invariant under all conformal transformations, since its variation vanishes for any conformal Killing vector $\epsilon^\mu(x)$. We conclude that a sufficient condition for conformal invariance is the existence of a symmetric, conserved and traceless energy-momentum tensor.

2.2.3 | Belinfante Energy Momentum Tensor

Equation (2.2.1) shows that in a scale invariant theory the trace is a divergence. The next question is whether one can redefine the energy-momentum tensor so as to make the trace vanish identically. Suppose that the virial current is itself a divergence,

$$V_\mu = \partial^\nu \sigma_{\nu\mu},$$

or, more specifically, that it can be written in a form allowing an improvement term. Then one can define an improved (Belinfante) energy-momentum tensor

$$T^{Bel}_{\mu\nu} = T_{\mu\nu} + \partial^\rho B_{\rho\mu\nu}, \tag{2.2.2}$$

where $B_{\rho\mu\nu}$ is antisymmetric in the first two indices,

$$B_{\rho\mu\nu} = -B_{\mu\rho\nu}.$$

so that conservation is preserved:

$$\partial^\mu \Theta_{\mu\nu} = 0.$$

With a suitable choice of the improvement term (CCJ = Callan-Coleman-Jackiw), one may achieve

$$\Theta^\mu{}_\mu = 0.$$

In that case the theory is not only scale invariant, but conformally invariant. The CCJ choice cannot always be implementable, leading to the fact that for $D > 2$ there can be some (very special) cases of theories that are scale invariant but their energy-momentum tensor cannot be made traceless. In $D = 2$ however, this is always possible.

2.2.4 | Polyakov theorem

We can now state the classical result in the form relevant for our purposes.

Theorem 2.1 (Classical Polyakov theorem)**.** *Consider a local D-dimensional field theory admitting a symmetric conserved energy-momentum tensor. If the theory is scale invariant and if the trace of the energy-momentum tensor can be improved, so that there exists an improved tensor* $\Theta_{\mu\nu}$ *satisfying*

$$\partial^\mu \Theta_{\mu\nu} = 0, \quad \Theta_{\mu\nu} = \Theta_{\nu\mu}, \quad \Theta^\mu{}_\mu = 0,$$

then the theory is conformally invariant.

Proof. For an infinitesimal conformal transformation generated by a conformal Killing vector $\epsilon^\mu(x)$, the variation of the action is

$$\delta S = -\frac{1}{2}\int d^D x \Theta^{\mu\nu}(\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu).$$

Since

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = \frac{2}{D}(\partial \cdot \epsilon) g_{\mu\nu}$$

one gets

$$\delta S = -\frac{1}{D}\int d^D x (\partial \cdot \epsilon)\Theta^\mu_\mu$$

But $\Theta^\mu_\mu = 0$, hence

$$\delta S = 0.$$

Therefore the action is invariant under all infinitesimal conformal transformations. This proves classical conformal invariance of any QFT having a traceless improved energy-momentum tensor.

□

Comments

The theorem should be understood as follows:

- scale invariance implies that the trace is a total divergence;
- if this divergence can be removed by improving the energy-momentum tensor, then the improved tensor is traceless;
- tracelessness plus conservation is exactly what is needed to generate the full local conformal symmetry.

Thus the passage from scale invariance to conformal invariance is controlled by the trace of the energy-momentum tensor.

Remark (Quantum comment). *At the quantum level the situation is subtler. Even when the classical theory is conformally invariant, quantization may spoil the tracelessness of the energy-momentum tensor through an anomaly. In two dimensions this anomaly is encoded in the central extension of the symmetry algebra and, equivalently, in the non-vanishing central charge appearing in the* TT *operator product expansion. We postpone the quantum discussion to later chapters, where the Virasoro algebra and the central charge will emerge naturally.*

2.3 | Algebra of local fields

Local fields $A(x) \in \mathcal{A}$ are operators in a QFT depending on only one space-time point. This set includes fundamental fields, but also their powers and derivatives. However, they are not freely generated: the algebra of local fields is subject to constraints (for example, equations of motion like $\partial^2\phi = \phi^5$). The set $\mathcal{A}$ is called the algebra of local fields. Indeed:

- Local fields add up and can be multiplied by scalars, so $\mathcal{A}$ has the structure of a vector space. We can select a preferred basis $\{\varphi_n(x)\}$ for this space, such that all fields can be expressed as a linear combination of this basis.
- Usually, we choose a basis of local fields that diagonalizes two commuting operators (since they commute, they can be simultaneously diagonalized):
  - **Dilatation**: $\tilde{D}\varphi_n(0) = \lambda^{d_n}\varphi_n(0)$. The eigenvalue d_n (or Δ_n) is the scale dimension. This is the crucial number we are interested in, as it dictates how the field rescales and ultimately determines the critical exponents that characterize specific universality classes.
  - **Spin**: $S\varphi_n(0) = e^{is_n\theta}\varphi_n(0)$. Local fields can have various spins ($s_n = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$).
- $\varphi_n(x)$ is therefore uniquely characterized by this pair of numbers $\{d_n, s_n\}$.
- All other fields are linear combinations

  $$A(x) = \sum_{n=0}^{\infty} a_n\varphi_n(x), \quad \varphi_0 \equiv \mathbb{I},$$

  where the equality here has to be understood in the *weak sense*. This means that two operators are considered equal if they yield the exact same result when inserted into *any* correlation function of the theory. You can have two formally different expressions, but if they do the same job on the full algebra of fields, they represent the same physical field. Formally, for any string of local fields $X = A_1(x_1)\ldots A_n(x_n) \in \mathcal{A}^{\otimes n}$, we have

  $$A(x) = B(x) \Leftrightarrow \langle A(x)X(x_1,\ldots,x_n)\rangle = \langle B(x)X(x_1,\ldots,x_n)\rangle.$$

2-point functions of local fields, at criticality, are fixed by roto-translation and scale invariance:

$$\mathcal{D}_{ij}(x_1 - x_2) = \langle\varphi_i(x_1)\varphi_j(x_2)\rangle = \frac{A_{ij}}{|x_1 - x_2|^{d_i+d_j}},$$

where $A_{ij} = A\delta_{ij}$. So they are all zero if $\phi_j \neq \phi_i$. The constant A can be fixed by normalizing the field ϕ_i:

$$\mathcal{D}_{ij}(x_1 - x_2) = \langle\varphi_i(x_1)\varphi_j(x_2)\rangle = \frac{\delta_{ij}}{|x_1 - x_2|^{2d_i}}.$$

Near criticality we have a mass scale $m \sim \xi^{-1}$, i.e. the inverse of the correlation length. The expectation values of local fields (think e.g. to the order parameters) rescale as:

$$\langle\varphi_i(x)\rangle = \mu_i m^{d_i}.$$

At criticality, i.e. in presence of scale invariance and $\xi \to \infty$:

$$\langle\phi_0\rangle = \langle\mathbb{I}\rangle = 1, \quad \langle\phi_i(x)\rangle = 0 \quad (i = 1, 2, \ldots).$$

The vector space of local fields is raised to an algebra if a suitable notion of product between fields is established: the **Operator Product Expansion (OPE)**. Any product of fields that are separated by some distance can be expanded as a sum of individual fields. Consider a composite (non-local) field $A(x_1)B(x_2)$. In the limit $x_1 \to x_2$, we expect that it reproduces the properties of a local field $C(x_2)$, which can then be expanded in the basis of local fields. This is conceptually vital to understand objects like ϕ^2, which is rigorously defined as $\phi(x_1)\phi(x_2)$ evaluated as $x_1 \to x_2$. Note that x_1 and x_2 do not need to be infinitesimally close; the OPE holds as long as there are no other field singularities in the middle (i.e., within a certain radius of convergence).

Wilson assumed that for any pair of local fields $A(x_1)$ and $B(x_2)$ the OPE holds:

$$A(x_1)B(x_2) = \sum_{k=0}^{\infty} c_k(x_1, x_2)\varphi_k(x_2).$$

In particular, for our chosen basis:

$$\varphi_p(x_1)\varphi_q(x_2) = \sum_r c^r_{pq}(x_1, x_2)\varphi_r(x_2).$$

The functions $c^r_{pq}(x_1, x_2)$ are called structure functions of the OPE algebra. Because the theory is *translationally invariant*, these functions cannot depend on the absolute coordinates x_1 and x_2, but only on their difference $x_1 - x_2$. Furthermore, *rotational invariance* (Poincaré invariance in the Wick rotated Euclidean description) implies they depend only on the distance between the two points (the modulus, not the full vector):

$$c^r_{pq}(|x_1 - x_2|) = c^r_{pq}(|x_{12}|).$$

Scale invariance at criticality further restricts the function:

$$c^r_{pq}(x_1, x_2) = \frac{C^r_{pq}}{|x_{12}|^{d_p+d_q-d_r}}.$$

The numbers C^r_{pq} are called structure constants.

Here and in the following we denote, for short,

$$x_{ij} = x_i - x_j.$$

The set of all scaling dimensions d_n (the spectrum of the theory) together with the structure constants C^r_{pq} form the so-called **CFT data**. These quantities completely define the dynamical content of the theory. If they are known exactly, the theory is considered exactly solved. This is because, by recursively applying the Operator Product Expansion, the product of two fields inside any correlation function can be systematically replaced by a sum over single fields. Through this contraction procedure, any N-point correlation function can be reduced to a combination of $(N-1)$-point functions, and eventually completely decomposed down to known 2-point or 1-point functions.

However, in a generic conformal field theory, these structure constants and scaling dimensions are not known *a priori*. One must rely on sophisticated non-perturbative methods (such as the conformal bootstrap mentioned in the previous chapter) to constrain and compute these numbers.

Furthermore, the physical importance of the OPE is most evident in the short-distance limit, when two fields are brought arbitrarily close to each other ($x_1 \to x_2$). In this limit, the expansion typically exhibits a divergence dictated by the power-law denominator $|x_{12}|^{d_p+d_q-d_r}$. While the

OPE naturally contains both regular and singular terms, it is precisely this *singular part* that encodes the most vital algebraic information about the correlation functions and the symmetries of the theory. To fully appreciate how these short-distance singularities dictate the entire structure of the theory, it is highly advantageous to restrict our attention to the incredibly rich case of two dimensions.

2.3.1 | The Conformal Group in Two Dimensions

If the dimension of spacetime is larger than two ($D > 2$), the conformal group is a finite-dimensional Lie group. While it is larger than the standard Poincaré group and therefore provides more constraints on the theory, the number of these constraints remains strictly finite. However, in exactly two dimensions, the situation changes drastically and the conformal symmetry becomes infinite-dimensional.

To see this, let us work in 2D Euclidean space (if we start from a Minkowski metric, we can always perform a Wick rotation). In a 2D Minkowski spacetime, the natural variables would be the *lightcone coordinates*; in Euclidean space, the analogous operation is to map the 2D Cartesian plane into the complex plane. We take a new basis by changing coordinates from the real Cartesian coordinates x^1 and x^2 to the complex coordinates z and $\overline{z}$:

$$\begin{cases} z = x^1 + ix^2, \\ \overline{z} = x^1 - ix^2, \end{cases} \implies \begin{cases} x^1 = \dfrac{z+\overline{z}}{2}, \\ x^2 = \dfrac{z-\overline{z}}{2i}. \end{cases}$$

While physically one coordinate is the complex conjugate of the other, it is mathematically very convenient to treat z and $\overline{z}$ as two formally independent variables characterizing a point in $\mathbb{C}^2$.

In these new coordinates, the flat Euclidean line element becomes:

$$\mathrm{d}s^2 = (\mathrm{d}x^1)^2 + (\mathrm{d}x^2)^2 = \mathrm{d}z\mathrm{d}\overline{z},$$

where $\mathrm{d}z\mathrm{d}\overline{z}$ also acts as the area element (d^2x) up to a constant factor. The corresponding derivatives with respect to the complex coordinates are:

$$\begin{cases} \partial_z = \dfrac{1}{2}(\partial_1 - i\partial_2), \\ \partial_{\overline{z}} = \dfrac{1}{2}(\partial_1 + i\partial_2), \end{cases} \implies \partial_z z = 1, \quad \partial_z \overline{z} = 0,$$

and similarly for $\partial_{\overline{z}}$. From the line element $\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu$, we can read off the metric tensor in the $(z, \overline{z})$ basis, which takes the off-diagonal form:

$$g = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}.$$

Consequently, any field in our theory can be redefined as a function of these two independent variables:

$$\varphi(x) \to \varphi(z,\, \overline{z}).$$

Now, let us consider conformal transformations. A coordinate transformation is conformal if it simply rescales the line element by a strictly positive local factor Ω:

$$(\mathrm{d}s')^2 = \Omega(z,\, \overline{z})\mathrm{d}s^2.$$

In complex coordinates, we can achieve this by applying independent transformations to z and $\overline{z}$. Let us map them to new variables w and $\overline{w}$:

$$\begin{cases} z \mapsto w = f(z), \\ \overline{z} \mapsto \overline{w} = \overline{f}(\overline{z}), \end{cases} \implies \begin{cases} \mathrm{d}w = f'(z)\mathrm{d}z, \\ \mathrm{d}\overline{w} = \overline{f}'(\overline{z})\mathrm{d}\overline{z}. \end{cases}$$

Here, $f(z)$ is an analytic (holomorphic) function, meaning its derivative $f'(z)$ exists, and $\overline{f}(\overline{z})$ is an anti-holomorphic function. The new line element becomes:

$$(\mathrm{d}s')^2 = \mathrm{d}w\mathrm{d}\overline{w} = f'(z)\overline{f}'(\overline{z})\mathrm{d}z\mathrm{d}\overline{z},$$

which perfectly matches the conformal condition with the scale factor completely factorized into a holomorphic and an anti-holomorphic part:

$$\Omega(z, \overline{z}) = f'(z)\overline{f}'(\overline{z}).$$

This reveals a profound result: *any* analytic transformation of the complex plane is a conformal transformation. Geometrically, analytic functions preserve local angles, maintaining the shape of infinitesimally small figures, even though globally they can stretch and severely distort areas (a classic example is the Mercator projection of a sphere, where the North Pole is stretched into an entire infinite line).

Since there are infinitely many analytic functions, the conformal group in two dimensions is **infinite-dimensional**. This implies we have infinitely many symmetries and, consequently, infinitely many constraints on the algebra of local fields. When a theory is subjected to infinitely many constraints, it often becomes exactly solvable, allowing us to compute quantities far beyond standard perturbation theory.

To rigorously link this back to the conformal Killing equation, consider an infinitesimal conformal transformation:

$$\begin{aligned} z &\mapsto z + \epsilon(z), \\ \overline{z} &\mapsto \overline{z} + \overline{\epsilon}(\overline{z}), \end{aligned}$$

where we define the displacement vector components as:

$$\epsilon(z) = \epsilon_1 + i\epsilon_2, \quad \overline{\epsilon}(\overline{z}) = \epsilon_1 - i\epsilon_2.$$

By definition, an infinitesimal transformation is conformal if it satisfies the conformal Killing equation. In $D = 2$ dimensions, this equation reads:

$$\partial_\mu\epsilon_\nu + \partial_\nu\epsilon_\mu = \frac{2}{2}(\partial \cdot \epsilon)\delta_{\mu\nu} = (\partial_1\epsilon_1 + \partial_2\epsilon_2)\delta_{\mu\nu}.$$

Evaluating this tensor equation component by component ($\mu = 1, \nu = 1$ and $\mu = 1, \nu = 2$), we obtain exactly the celebrated **Cauchy-Riemann equations**:

$$\begin{cases} \partial_1\epsilon_1 = \partial_2\epsilon_2, \\ \partial_1\epsilon_2 = -\partial_2\epsilon_1. \end{cases}$$

These equations are the necessary and sufficient condition for $\epsilon(x^1, x^2) = \epsilon_1 + i\epsilon_2$ to be a complex analytic function $\epsilon(z)$. This confirms that in two dimensions, the set of all conformal transformations coincides precisely with the set of all holomorphic and anti-holomorphic mappings.

[...]

$$\epsilon(z) = \sum_{n=-\infty}^{\infty} \epsilon_n z^{n+1},$$

unless you have a constant function, if it is analytic there must be somewhere in the domain of analiticity in which the function is greater or higher than some value, and outside the domain of analiticity the function can diverge. This laurent series is the most general form of an analytic function, and it is convergent in the annulus defined by the inner and outer radius of convergence.

We can consider $\epsilon_n(z)$ our basis for the expansion

$$\epsilon_n(z) = -\epsilon z^{n+1},$$

then the generators of the transformations on scalar fields $\varphi(x)$ as

$$\hat{\ell}_n = -z^{n+1}\partial_z,$$

such that

1. $\hat{\ell}_0 = -z\partial_z$ is the generator of **dilatations**,
2. $\hat{\ell}_{-1} = -\partial_z$ is the generator of **translations**,
3. $\hat{\ell}_1 = -z^2\partial_z$ is the generator of **special conformal transformations**.

The dilatation or translation is in z, but we need the $\overline{z}$ direction, such that we need to define

$$\overline{\ell}_n = -\overline{z}^{n+1}\partial_{\overline{z}}.$$

So we have

1. $\overline{\ell}_0 = -\overline{z}\partial_{\overline{z}}$ is the generator of **dilatations**,
2. $\overline{\ell}_{-1} = -\partial_{\overline{z}}$ is the generator of **translations**,
3. $\overline{\ell}_1 = -\overline{z}^2\partial_{\overline{z}}$ is the generator of **special conformal transformations**.

Thus if we were to write the four momentum operator in the complex coordinates, we would have

$$P = \begin{pmatrix} E \\ p \end{pmatrix} = \begin{pmatrix} \ell_{-1} + \overline{\ell}_{-1} \\ i(\ell_{-1} - \overline{\ell}_{-1}) \end{pmatrix},$$

where E is the energy and p is the momentum. The dilatation operator would be

$$D = \hat{\ell}_0 + \overline{\ell}_0,$$

[...]

If you reason in higher dimensions, the special conformal group is six-dimensional, since

$$D = 2 \implies \frac{(D+1)(D+2)}{2} = 6.$$

Imagine to apply the commutator $[\ell_n, \ell_m]$ on a test function $f(z)$, we can find

$$[\ell_n, \ell_m] = \left((n-m)z^{n+m+1}\partial_z\right) = (n-m)\ell_{n+m},$$

which is the Witt algebra, the algebra of the generators of conformal transformations in two dimensions. The same holds for the barred generators $\bar{\ell}_n$. The Witt algebra is an infinite-dimensional Lie algebra, since n and m can take any integer value and be applied on any test function $f(z)$ in the domain of analyticity. The Witt algebra is the classical version of the Virasoro algebra, which is the quantum version of the conformal algebra in two dimensions. Visaroro was a young phd in Turin when he discovered the algebra, after is supervisor (which already had found the Mobius transformations we are going to introduce later) told him to find the algebra of the generators of conformal transformations in two dimensions, and he found this infinite-dimensional algebra.

This generators, with $n = -1, 0, 1$ (barred and not), are six generators which form the algebra of the 2×2 special linear group $\mathfrak{sl}(2, \mathbb{C})$, which is the double cover of the Lorentz group $SO(3, 1)$. This is the conformal group in two dimensions. It is isomorphic to $SU(2) \oplus SU(2)$.

We have an infinite number of conformal transformations, since they are all the analytic functions, so how can it be six dimensional? We can map the coordintes of the plane to $z = e^w$, which inverse gives the logarithm of z: we are mapping the plane to a strip (given by the riemann surface of the logarithm), and with periodic boundary conditions we get a cylinder. The conformal transformations on the plane are mapped to conformal transformations on the cylinder, and the six generators of the conformal group are mapped to the six generators of the conformal group on the cylinder. very important (it is a way of compactifying the plane).

Now we can ask which are the transformations starting from a plane and ending in a plane, i.e. which are the transformations that do not change the point at infinity. These are the Mobius transformations, which are given by

$$w = \frac{az + b}{cz + d}, \quad ad - bc = 1,$$

which has six parameters. Thus global conformal transformations are given by the Mobius transformations, which are a subgroup of the infinite-dimensional group of local conformal transformations.

[...]

We can now infer someting about the fields: after the remapping

$$\begin{cases} z \\ \bar{z} \end{cases} \mapsto \begin{cases} w = f(z) \\ \bar{w} = \bar{f}(\bar{z}) \end{cases},$$

we can find fields that transforms as

$$\dots$$

which are called **primary fields**, and they are the building blocks of the theory. We can also understand quasi-primary fields, which are fields that transforms as primary fields under the global conformal transformations (Mobius transformations), but not under the local conformal transformations. There are also other fields which do not have tensorial stucture under the conformal transformations, but they are not important for us. Some of the secondary fields happens to be quasi-primary, but not all of them. The primary fields are the most important fields in the theory and we will distiguish them from the secondary fields, which are obtained by applying the generators of the conformal transformations on the primary fields.

For example, the energy-momentum tensor $T(z)$ is secondary field, and it would seem lika a bad news since it is the most important field in the theory, but we will see the consequences.

If we compute tge derivatives of w and $\bar{w}$ with respect to z and $\bar{z}$, we can find

$$\dots$$

and now expanding

$$\phi(z,\,\overline{z}) = \phi(w-\epsilon,\,\overline{w}-\overline{\epsilon}) = \phi(w,\,\overline{w}) - \epsilon\partial_w\phi(w,\,\overline{w}) - \overline{\epsilon}\partial_{\overline{w}}\phi(w,\,\overline{w}) + \ldots$$

such that we can find the variation of the field ϕ under the conformal transformation

$$\delta\phi = -\epsilon\partial_z\phi - \overline{\epsilon}\partial_{\overline{z}}\phi - h\partial_z\epsilon\phi - \overline{h}\partial_{\overline{z}}\overline{\epsilon}\phi,$$

where we can see how the two variables are completely decoupled, and the transformation of the field is given by the sum of the transformation in z and the transformation in $\overline{z}$.

dilatations…

$$D = \ell_0 + \overline{\ell}_0 = h + \overline{h} = \Delta,$$

rotations…

$$L = \ell_0 - \overline{\ell}_0 = h - \overline{h} = s,$$

such that we can characterize the fields by their scaling dimension Δ and their spin s (or equivalently by h and $\overline{h}$). A scalar field is spinless, so $s = 0$ and $h = \overline{h}$. A chiral field is a field that depends only on z or only on $\overline{z}$, so it has either $h = 0$ or $\overline{h} = 0$.

Appendices